

16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
DRAWING LIST															
#	DRAWING NUMBER	DRAWING TITLE	REVISION												
1	3041-PID-0000-1	DRAWING LIST	A												
2	3041-PID-0001-1	LEGEND SHEET	A												
3	3041-PID-0002-1	LEGEND SHEET	A												
4	3041-PID-0200-1	PREGASIFIER	A												
5	3041-PID-0200-2	PREGASIFIER O2, LS, N2	A												
6	3041-PID-0200-3	PREGASIFIER COOLING	A												
7	3041-PID-0300-1	PEM	A												
8	3041-PID-0300-2	PEM LS, O2, & N2	A												
9	3041-PID-0300-3	PEM DRAINS	A												
10	3041-PID-0301-1	TRC SG LINE	A												
11	3041-PID-0310-1	PEM SKID N2 HEADERS	A												
12	3041-PID-0310-2	PEM SKID CW HEADERS	A												
13	3041-PID-0301-1	TRC SG LINE	A												
14	3041-PID-0400-1	DC POWER SUPPLIES	A												
15	3041-PID-0400-2	AC POWER SUPPLY	A												
16	3041-PID-0400-3	DC POWER SUPPLY	A												
17	3041-PID-0500-1	AC ELECTRODE HOLDERS	A												
18	3041-PID-0500-2	DC ELECTRODE FEEDERS	A												
19	3041-PID-0500-3	DC RETURN ELECTRODE HOLDERS	A												
20	3041-PID-0600-1	TRC	A												
21	3041-PID-0600-2	TRC GAD INJECTION	A												
22	3041-PID-0601-1	TRC GAD FUEL TRAIN	A												
23	3041-PID-0900-1	UTILITY SKID N2 HEADERS	A												
24	3041-PID-0900-2	UTILITY SKID O2 HEADER	A												
25	3041-PID-0900-3	UTILITY SKID CW HEADERS	A												
26	3041-PID-0900-4	UTILITY SKID COOLING PUMPS	A												
27	3041-PID-0900-5	UTILITY SKID LS HEADER	A												
28	3041-PID-0900-6	UTILITY SKID PREHEATERS	A												
29	3041-PID-0900-7	UTILITY SKID IA HEADER	A												
30	3041-PID-0900-8	GAS ANALYZER	A												
31	3041-PID-1000-1	FEED HANDLING 1 & SHREDDING	A												
32	3041-PID-1000-2	FEED SHREDDER VENTILATION SYSTEM	A												
33	3041-PID-1000-3	FEED HANDING 2	A												
34	3041-PID-2000-1	GLASS HANDLING SYSTEM	A												
35	3041-PID-2100-1	METAL HANDLING SYSTEM	A												
36	3041-PID-2200-1	GLASS ADDITIVES CONVEYOR	A												
37	3041-PID-2200-2	GLASS ADDITIVES FEEDER	A												
38	3041-PID-2300-1	WASTEWATER COLLECTION	A												
39	3041-PID-3000-1	PRIMARY QUENCH	A												
40	3041-PID-3000-2	PRIMARY QUENCH TANK & PUMPS	A												
41	3041-PID-3100-1	BAGHOUSE	A												
42	3041-PID-3100-2	BAGHOUSE	A												
43	3041-PID-3100-3	BAGHOUSE HEAT TRACE	A												
44	3041-PID-3200-1	SCRUBBER	A												
45	3041-PID-3200-2	SCRUBBER PUMP AND PIPING	A												
46	3041-PID-3200-3	CAUSTIC SUPPLY	A												
47	3041-PID-3300-1	SYNGAS BLOWER 1	A												
48	3041-PID-3300-2	SYNGAS BLOWER 2	A												
49	3041-PID-3500-1	SULFUR ADSORBER SYSTEM	A												
50	3041-PID-3800-1	HG REMOVAL & FILTRATION	A												
51	3041-PID-3950-1	GAS ANALYZER	A												
52	3041-PID-4100-1	FLARE	A												
53	3041-PID-5000-1	SYNGAS COMPRESSOR	A												
54	3041-PID-5001-1	SYNGAS SURGE TANK	A												
55	3041-PID-5200-1	PRESSURE SWING ADSORPTION	A												
56	3041-PID-5200-2	PRESSURE SWING ADSORPTION	A												

DRAWING LIST			
#	DRAWING NUMBER	DRAWING TITLE	REVISION
57	3041-PID-5201-1	TAIL GAS SURGE TANK	A
58	3041-PID-5300-1	HYDROGEN COMPRESSOR	A
59	3041-PID-5400-1	H2 TANK FILLING STATIONS	A
60	3041-PID-5500-1	H2 ANALYZER	A
61	3041-PID-5700-1	THROX WATER TRAP ARRESTOR	A
62	3041-PID-5701-1	THERMAL OXIDIZER HEADERS	A
63	3041-PID-5701-2	THERMAL OXIDIZER STACK	A
64	3041-PID-8400-1	BUILDING VENTILATION SYSTEM	A
65	3041-PID-9000-1	FUEL GAS DISTRIBUTION	A
66	3041-PID-9100-1	PROCESS WATER SYSTEM	A
67	3041-PID-9100-2	PROCESS WATER DISTRIBUTION	A
68	3041-PID-9200-1	COOLING WATER SYSTEM	A
69	3041-PID-9201-1	PEM COOLING WATER SYSTEM	A
70	3041-PID-9202-1	H2 COOLING WATER SYSTEM	A
71	3041-PID-9300-1	NITROGEN DISTRIBUTION	A
72	3041-PID-9301-1	OXYGEN DISTRIBUTION	A
73	3041-PID-9302-1	CARBON DIOXIDE SUPPLY	A
74	3041-PID-9500-1	INSTRUMENT AIR SYSTEM	A
75	3041-PID-9500-2	INSTRUMENT AIR DISTRIBUTION	A
76	3041-PID-9800-1	BFW SYSTEM	A
77	3041-PID-9801-1	BFW PUMPS	A
78	3041-PID-9802-1	STEAM SYSTEM	A
79	3041-PID-9803-1	STEAM SYSTEM 2	A

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- NOTES:
- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

PLOT DATE:
5/1/2024

DRAWN: A BENNETT
DATE:

CHECKED:
DATE:

ENGR:
DATE:

APVD:
DATE:

REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041
P&ID
DRAWING LIST

SCALE	NONE	SHEET NO	1	SHEETS	1
DWG NO	3041-PID-0000-1	REV	B		
CADFILE	3041-PID-0000-1.DWG	4/20/2023			

MISC. PIPING SYMBOLS

PRIMARY LINE

SECONDARY LINE

EXISTING PRIMARY LINE

EXISTING SECONDARY LINE

VENDOR PACKAGE BOUNDARY

HEAT TRACED LINE

TIE POINT

LINE CONNECTOR

SIGNAL CONNECTOR

STEAM TRAP

SPECIALITY ITEM

PERMANENT Y-TYPE STRAINER
(W/BLOW-OFF VALVE & CAP)

CONE STRAINER

TEMPORARY STRAINER

FILTER

BASKET STRAINER

MIXING NOZZLE
(SPARGER OR EDUCTOR)

DAMPER / LOUVER

EXPANSION JOINT

ELECTRIC MOTOR

CLOSED DRAIN

OPEN DRAIN

PIPING SPEC. # CHANGE

FLANGE

SPECTACLE BLIND (NORMALLY OPEN)

SPECTACLE BLIND (NORMALLY CLOSED)

BLIND FLANGE

CAP

PLUG

LINE SIZE CHANGE OR CONCENTRIC REDUCER

ECCENTRIC REDUCER

FLEXIBLE HOSE

HOSE CONNECTOR

COMPLETE PACKAGE
FURNISHED BY VENDOR

HOLD, REVISED

AIR FILTER WITH HOOD

ABBREVIATIONS

(E) EXISTING

(F) FURNISHED

(R) RELOCATE

AG ABOVE GROUND

ATM ATMOSPHERE

BFP BACK FLOW PREVENTER

BL BATTERY LIMIT

BTL BOTTOM TANGENT LINE

BYP BYPASS

CC CHEMICAL CLEANOUT

CL CENTERLINE

CO CLEANOUT

CONN CONNECTION

CSC CAR SEAL CLOSED

CSO CAR SEAL OPEN

CTR CENTER

CTRL CONTROL

D/P DIFFERENTIAL PRESSURE

DCS DISTRIBUTED CONTROL SYSTEM

DES DESIGN

DIA DIAMETER

DP DESIGN PRESSURE

DRN DRAIN

DT DESIGN TEMPERATURE

DWVG DRAWING

EL ELEVATION

ESD EMERGENCY SHUTDOWN

FHB FEED HANDLING BUILDING

FC FAIL CLOSED

FI FAIL INDETERMINATE

FL FAIL LOCKED (LAST POSITION)

FLG FLANGE

FO FAIL OPEN

FOP FACE OF FLANGE

FOR FORWARD/OFF/REVERSE

FP FULL PORT

FV FULL VACUUM

GO GEAR OPERATED

GR GRADE

HC HOSE CONNECTION

HDR HEADER

HH HAND HOLE

HOA HAND/OFF/AUTOMATIC

HOB HAND/OFF/BACKUP

HOR HAND/OFF/REMOTE

HP HIGH PRESSURE

HPT HIGH POINT

IAS INSTRUMENT AIR SUPPLY

ISBL INSIDE BATTERY LIMITS

LC LOCKED CLOSE

LO LOCKED OPEN

LP LOW PRESSURE

LPT LOW POINT

MAX MAXIMUM

MIN MINIMUM

MOV MOTOR OPERATED VALVE

MTL MATERIAL

MW MANWAY

NC NORMALLY CLOSED

NNF NORMALLY NO FLOW

NO NORMALLY OPEN

NOZ NOZZLE

O/C OPEN/CLOSE

O/O ON/OFF

OP OUTPUT

OSBL OUTSIDE BATTERY LIMITS

OVHD OVERHEAD

PLC PROGRAMMABLE LOGIC CONTROLLER

PRESS PRESSURE

PV PROCESS VARIABLE

REQD REQUIRED

RO REVERSE OSMOSIS

RTD RESISTANCE TEMPERATURE DETECTOR

S/S START/STOP

SC SAMPLE CONNECTION

SCH SCHEDULE

SD SHUTDOWN

SG SPECIFIC GRAVITY

SIS SAFETY INSTRUMENT SYSTEM

SO STEAM OUT

SP SET POINT

SS STAINLESS STEEL

STD STANDARD

T/C THERMOCOUPLE

T/T TANGENT TO TANGENT

TDH TOTAL DIFFERENTIAL HEAD

TEMP TEMPERATURE

THRD THREADED

TL TANGENT LINE

TSO TIGHT SUT-OFF

TYP TYPICAL

UG UNDERGROUND

VAC VACUUM

VB VORTEX BREAKER

VNT VENT

VTA VENT TO ATMOSPHERE

W/ WITH

W/O WITHOUT

EQUIPMENT ID.

FA-0730
(1) (2) (3)

1. EQUIPMENT TYPE (SEE BELOW)

2. AREA (2 DIGITS)

3. SEQUENCE (2 DIGITS)

A MIXING EQUIPMENT

B BLOWERS

C COMPRESSORS

D MECHANICAL DRIVERS

E HEAT EXCHANGERS

EE ELECTRICAL EQUIPMENT

F FURNACES

P PUMPS

R REACTORS

T TOWERS / COLUMNS

TK TANKS

U MISCELLANEOUS

V EQUIPMENT

Y VESSELS

ELECTRODE HOLDERS / FEEDERS

LINE IDENTIFICATION

(1) (2) (3) (4)

XX"-XXXX-XXXX-XXXX

XX - XX" - XX

(5) (6) (7)

(1) (2) (3) (4) (6,5,7)

XX"-XXXX-XXXXXX-XXXX-XX"XX

1 LINE SIZE

2 LINE SERVICE CODE

3 LINE NUMBER

4 PIPE SPECIFICATION

5 INSULATION TYPE

6 INSULATION THICKNESS

7 HEAT TRACING (HT)

SERVICE IDENTIFICATION

AV ATMOSPHERIC VENT

BA BREATHING AIR

BD BLOWDOWN

BFW BOILER FEED WATER

BRR BRINE RETURN

BRS BRINE SUPPLY

CC CONTAMINABLE CONDENSATE

CE COMBUSTION EXHAUST

CF COLD FLARE

CHS CHEMICAL SEWER

CHWR CHILLED WATER RETURN

CHWS CHILLED WATER SUPPLY

CO2 CARBON DIOXIDE

CTWR COOLING TOWER WATER RETURN

CTWS COOLING TOWER WATER SUPPLY

CV COLD VENT

CWR COOLING WATER RETURN

CWS COOLING WATER SUPPLY

DMV DEMINERALIZED WATER

DR DRAIN

DS DILUTION STEAM

DW DRINKING WATER

ER ETHYLENE REFRIGERANT

FF FLUSHING FLUID

FG FUEL GAS

FO FUEL OIL

FW FIRE WATER

GL GLASS

GLR GLYCOL RETURN

GLS GLYCOL SUPPLY

H HYDROGEN

HS HIGH PRESSURE STEAM

IA INSTRUMENT AIR

IS INTERMEDIATE PRESSURE STEAM

LNG LIQUID NATURAL GAS

LO LUBE OIL

LPG LIQUIFIED PETROLEUM GAS

LS LOW PRESSURE STEAM

ME METHANOL

MR MIXED REFRIGERANT

MS MEDIUM PRESSURE STEAM

MTL METAL

N NITROGEN

NA CAUSTIC

NAS CAUSTIC SEWER

NG NATURAL GAS

NH AMMONIA

OWS OILY WATER SEWER

OX OXYGEN

P GENERAL PROCESS

PA PLANT AIR

PC PROCESS CONDENSATE

PR PROPYLENE REFRIGERANT

PW PROCESS WATER

OO QUENCH OIL

RS RECOVERED SLOPS

RV RELIEF VENT

RW RAW WATER

SB SCRUB WATER

SC STEAM CONDENSATE

SG SYNTHESIS GAS

SO SEAL OIL

SS SANITARY SEWER

ST STEAM

STS STORM SEWER

SW SERVICE WATER

SWR SEA WATER RETURN

SWS SEA WATER SUPPLY

TWR TEMPERED WATER RETURN

TWS TEMPERED WATER SUPPLY

UR UREA

VC VACUUM CONDENSATE

VE VACUUM EXHAUST

WF WARM FLARE OIL

WO WASH OIL

WW WASTEWATER

DRAWING IDENTIFICATION

G100P-PID-0620-1
(1) (2) (3) (4) (5)

1 JOB NUMBER

2 DISCIPLINE (SEE BELOW)

3 AREA (2 DIGITS)

4 SUB AREA (2 DIGITS)

5 SHEET NUMBER

DISCIPLINE

A ARCHITECTURAL

C CIVIL

CS CONTROL SYSTEM

E ELECTRICAL

GA GENERAL ARRANGEMENT

M MECHANICAL

P PIPING

PEM MELTER SHELL

PFD PROCESS FLOW DIAGRAM

PID PIPING AND INSTRUMENTATION DIAGRAM

R REFRACTORY

S STRUCTURAL

SCH SCHEDULE

AREA

02 PREGASIFIER

03 PEM®

04 INDUCTION, AC, & DC POWER

05 AC & DC ELECTRODE SYSTEMS

06 TRC

09 PEM® UTILITIES

10 SOLIDS FEED SYSTEM

20 GLASS DRAIN

21 METAL DRAIN

22 GLASS ADDITIVES SYSTEM

23 WASTEWATER COLLECTION

30 PRIMARY QUENCH

31 BAGHOUSE

32 SCRUBBER & CAUSTIC SUPPLY

33 SYNGAS BLOWER

35 SULFUR ADSORBER

38 MERCURY ADSORBER & FILTRATION

39 SYNGAS ANALYSIS

41 E-FLARE

50 SYNGAS COMPRESSION

52 PSA

53 H2 COMPRESSION

54 H2 FILLING STATION

57 THERMAL OXIDIZER

84 VENTILATION

90 FUEL GAS UTILITIES & DISTRIBUTION

91 PROCESS WATER UTILITES & DISTRIBUTION

92 COOLING WATER SYSTEMS & DISTRIBUTION

93 N2 & O2 UTILITES & DISTRIBUTION

95 IA SYSTEMS & DISTRIBUTION

98 BFW & STEAM SYSTEMS & DISTRIBUTION

PIPING IDENTIFICATION

PIPE SPEC.NO.

CLASS RATING (#)

DESCRIPTION

1AA1 150# 304L STAINLESS STEEL PIPE

1AA2 150# 304L STAINLESS STEEL PIPE

UAB1 N/A 304L STAINLESS STEEL TUBING

1AC1 150# 316L STAINLESS STEEL PIPE

1AC2 150# 316L STAINLESS STEEL PIPE

UAD1 N/A 316L STAINLESS STEEL TUBING

1AE1 150# HASTELLOY C-276 PIPE

UAF1 N/A HASTELLOY C-276 TUBING

1AG1 150# AL6XN PIPE

UAH1 N/A AL6XN TUBING

1CA1 150# CARBON STEEL PIPE

1CA2 150# CARBON STEEL PIPE

1CA3 150# CARBON STEEL PIPE

1CB1 150# GALVANIZED CARBON STEEL PIPE

1CC1 150# 1-1/4 CHROME-1/2 MOLY PIPE

UKA1 N/A COPPER PIPE

1LA1 150# TEFLON LINED CARBON STEEL PIPE

UPA1 N/A PVC PIPE

UPB1 N/A PVC TUBING

UPC1 N/A CPVC PIPE

UPD1 N/A PP PIPE

UPE1 N/A PP TUBING

UPF1 N/A LDPE TUBING

UPG1 N/A PVDF PIPE

UPH1 N/A PVDF TUBING

UPI1 N/A PTFE PIPE

UPJ1 N/A PTFE TUBING

UPK1 N/A EPDM RUBBER HOSE

UPL1 N/A PVC HOSE

UPM1 N/A CARBON REINFORCED PVC TUBING

UPN1 N/A NONCONDUCTIVE HOSE

UPO1 250# STEAM HOSE

INSULATION TYPE

H NONE

IC HEAT CONSERVATION

IH COLD

PP HOT

PERSONNEL PROTECTION

HEAT TRACE TYPE

E NONE

ET ELECTRIC

S ELECTRIC TRACE

SJ STEAM

STEAM JACKET

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DRAWN: A BENNETT

DATE:

CHECKED:

DATE:

ENGR:

DATE:

APVD:

DATE:

REVISIONS

REV NO

DCR/DESCRIPTION

INITIAL ISSUE

3041
P&ID
LEGEND SHEET

SCALE NONE

DWG NO 3041-PID-0001-1

CADFILE 3041-PID-0001-1.DWG

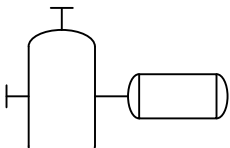
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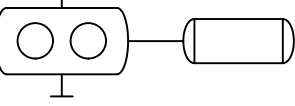
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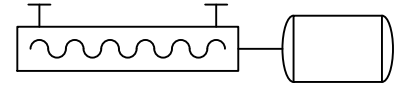
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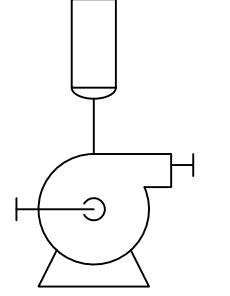
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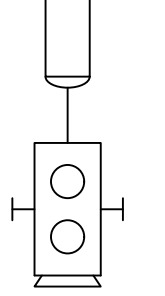
PUMPS & BLOWERS

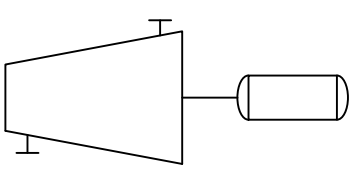
CENTRIFUGAL PUMP

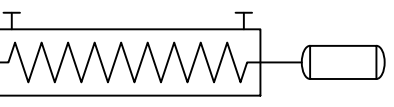
POSITIVE DISPLACEMENT PUMP

PROGRESSIVE CAVITY PUMP

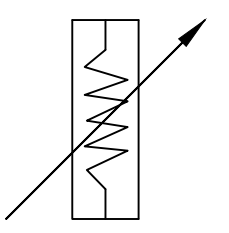
CENTRIFUGAL BLOWER

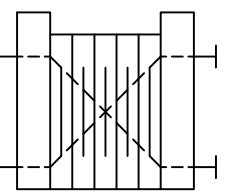
POSITIVE DISPLACEMENT BLOWER

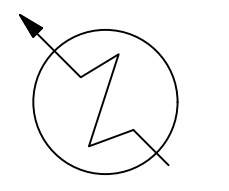
CENTRIFUGAL COMPRESSOR

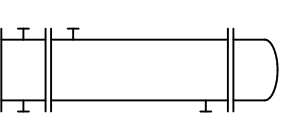
SCREW COMPRESSOR

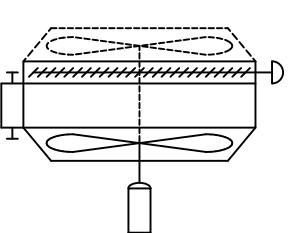
EXCHANGERS

PLATE EXCHANGER (PFD)

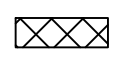
PLATE EXCHANGER (P&ID)

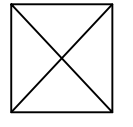
SHELL & TUBE EXCHANGER (PFD)

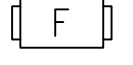
SHELL & TUBE EXCHANGER (P&ID)

AIR COOLER

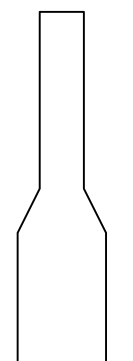
MISCELLANEOUS

FILTER

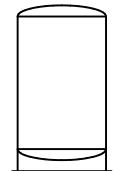
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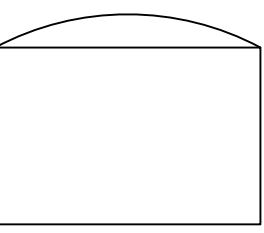
FLAME ARRESTOR

DETONATION ARRESTOR

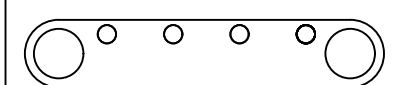
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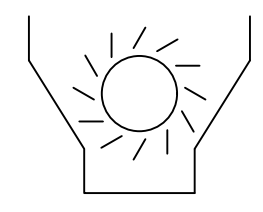
TANKS & VESSELS

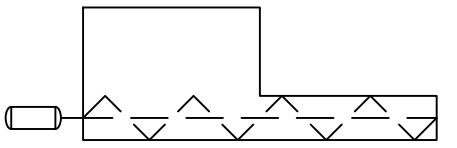
VERTICAL VESSEL

TANK

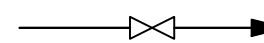
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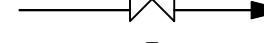
CONVEYOR

SHREDDER

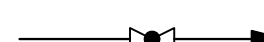
AUGER FEEDER

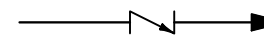
VALVES

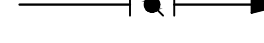
GATE

DIAPHRAGM

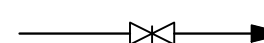
KNIFE GATE

PINCH

GLOBE

CHECK

BUTTERFLY

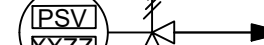
BALL

PLUG

NEEDLE

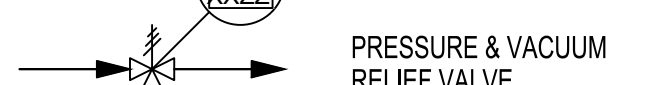
3-WAY

4-WAY

ANGLE

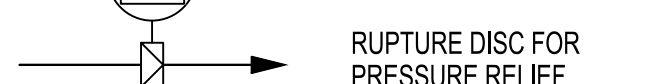
PRESSURE RELIEF VALVE (PSV)

VACUUM RELIEF VALVE

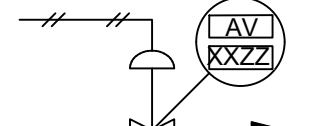
PRESSURE & VACUUM RELIEF VALVE

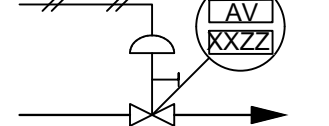
PILOT OPERATED RELIEF VALVE

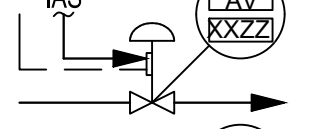
RUPTURE DISC FOR PRESSURE RELIEF

RUPTURE DISC FOR VACUUM RELIEF

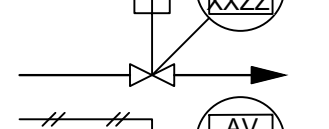
VALVE ACTUATORS

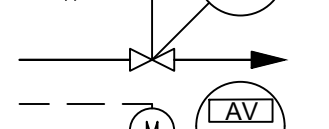
DIAPHRAGM

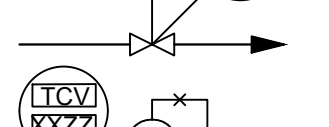
DIAPHRAGM W/ MANUAL OVERRIDE

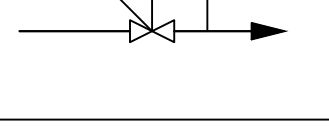
DIAPHRAGM W/ POSITIONER

SELF-CONTAINED PRESSURE REGULATOR

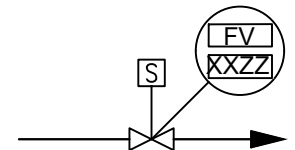
PISTON, SINGLE-ACTION

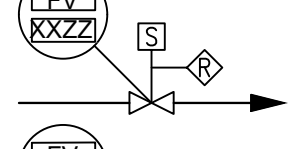
PISTON, DOUBLE-ACTION

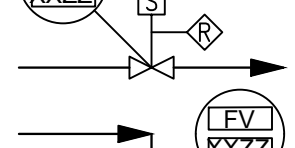
MOTOR

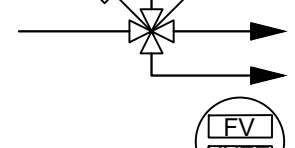
TEMPERATURE REGULATOR (FILLED SYSTEM)

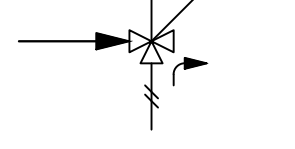
SOLENOID VALVES

SOLENOID

SOLENOID W/ AUTOMATIC RESET

SOLENOID W/ MANUAL RESET

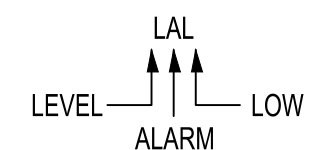
FOUR-WAY SOLENOID

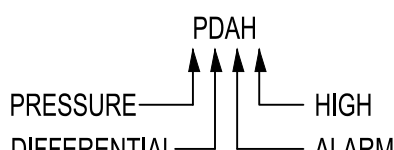
THREE-WAY SOLENOID FLOW DIRECTION IN DE-ENERGIZED CONDITION

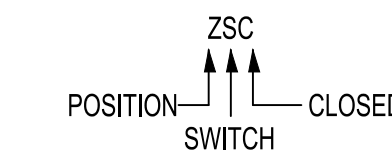
INSTRUMENT IDENTIFICATION

FIRST LETTER			SUCCEEDING LETTERS			
MEASURED OR INITIATING VARIABLE		MODIFIER	READOUT OR PASSIVE FUNCTION	OUTPUT FUNCTION	MODIFIER	
A	ANALYSIS, ANALYZER	DIFFERENTIAL	ALARM	USERS CHOICE	USERS CHOICE	
B	BURNER, FLAME, COMBUSTION, BELT					
C	CONDUCTIVITY					
D	DENSITY					
E	VOLTAGE (EMF)					
F	FLOW RATE	RATIO (FRACTION)	SENSOR (PRIMARY ELEMENT)	CONTROL	CLOSE DIVERT	
G	USER'S CHOICE					
H	HAND					
I	CURRENT (ELECTRICAL)	SCAN	INDICATE			
J	POWER					
K	TIME, TIME SCHEDULE					
L	LEVEL	TIME RATE OF CHANGE	CONTROL STATION	HIGH		
M	USER'S CHOICE					
N	USER'S CHOICE					
O	USER'S CHOICE	MOMENTARY			LIGHT	LOW MIDDLE OR INTERMEDIATE
P	PRESSURE, VACUUM					
Q	QUANTITY OR HEAT DUTY					
R	RADIATION	INTEGRATE, TOTALIZE	USER'S CHOICE ORIFICE, RESTRICTION POINT (TEST) CONNECTION	USERS CHOICE		
S	SPEED, FREQUENCY					
T	TEMPERATURE					
U	MULTIVARIABLE	SAFETY	RECORD		USERS CHOICE	
V	VIBRATION, MECHANICAL ANALYSIS					
W	WEIGHT OR FORCE					
X	UNCLASSIFIED	X AXIS	MULTIFUNCTION	REGULATING SWITCH		
Y	EVENT, STATE, OR PRESENCE					
Z	POSITION, DIMENSION					
		Y AXIS	UNCLASSIFIED		TRANSMIT	
		Z AXIS	WELL UNCLASSIFIED	MULTIFUNCTION		
		X AXIS	UNCLASSIFIED		VALVE, DAMPER OR LOUVER	
		Y AXIS	UNCLASSIFIED	THROUGH MULTIFUNCTION		
		Z AXIS	WELL UNCLASSIFIED		DRIVE, ACTUATE OR UNCLASSIFIED FINAL CONTROL ELEMENT	

EXAMPLES:

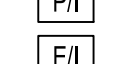
LEVEL $\uparrow\uparrow\uparrow$ LOW ALARM

PRESSURE DIFFERENTIAL $\uparrow\uparrow\uparrow$ HIGH ALARM

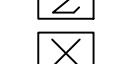
POSITION $\uparrow\uparrow\uparrow$ CLOSED SWITCH

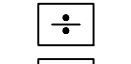
SIGNAL CONDITIONING DEVICE SYMBOLS

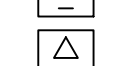
CURRENT-TO-PRESSURE

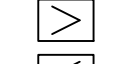
PRESSURE-TO-CURRENT

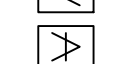
VOLTAGE-TO-CURRENT

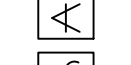
SQUARE ROOT

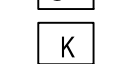
SUMMING

MULTIPLYING

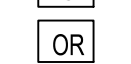
DIVIDING

BIAS

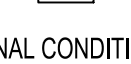
DIFFERENCE

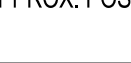
HIGH SELECTING

LOW SELECTING

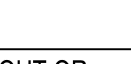
HIGH LIMITING

LOW LIMITING

INTEGRAL

PROPORTIONAL

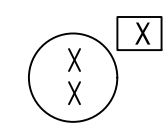
DERIVATIVE

CONSTANT

LOGICAL OR

LOGICAL AND

RELAY OR GATE



INSTRUMENTATION

LOCAL

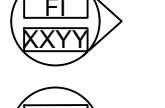
MOUNTED ON MAIN PANEL

MOUNTED BEHIND MAIN PANEL

MOUNTED ON LOCAL PANEL

MOUNTED BEHIND LOCAL PANEL

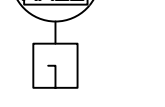
ROTAMETER WITH INTEGRAL VALVE

FLOW ELEMENT ROTAMETER

FLOW ELEMENT ULTRASONIC

FLOW ELEMENT VENTURI

FLOW ELEMENT TURBINE

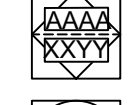
FLOW ELEMENT PITOT

FLOW ELEMENT THERMAL MASS

DIAPHRAGM SEAL

N = STREAM NUMBER ON PFD

PLC FUNCTION OPERATOR ACCESSIBLE

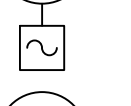
PLC FUNCTION OPERATOR INACCESSIBLE

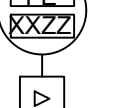
DCS FUNCTION OPERATOR ACCESSIBLE

DCS FUNCTION OPERATOR INACCESSIBLE

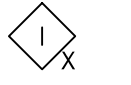
FLOW ELEMENT MAGMETER

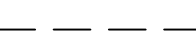
FLOW ELEMENT ULTRASONIC

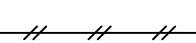
FLOW ELEMENT VORTEX

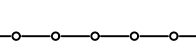
FLOW ELEMENT ORIFICE

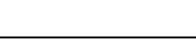
FLOW ELEMENT TARGET

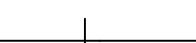
INTERLOCK/LOGIC FUNCTION

ELECTRIC SIGNAL

PNEUMATIC SIGNAL

SOFTWARE LINK

LEADER

HYDRAULIC SIGNAL

ELECTROMAGNETIC SIGNAL

MECHANICAL LINK

CAPILLARY TUBE

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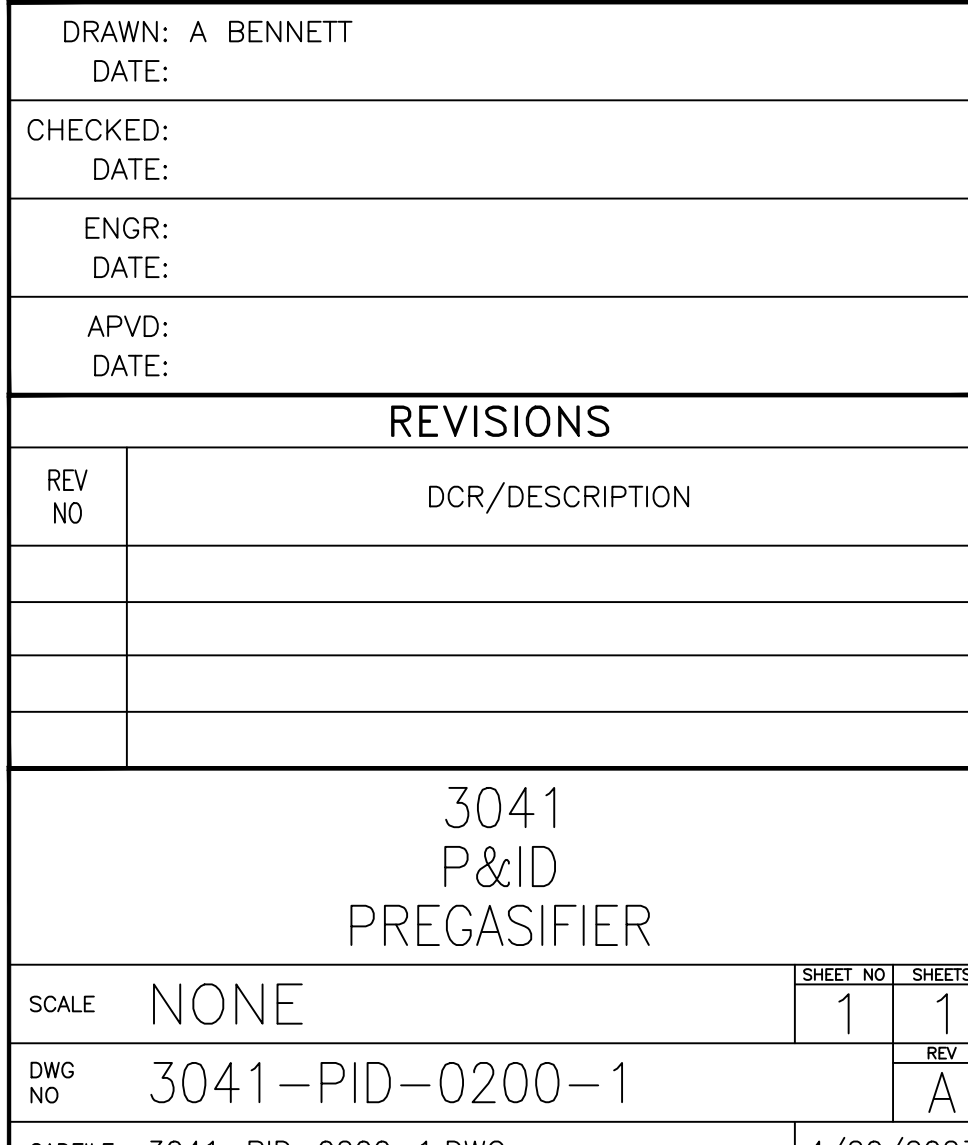
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DRAWN: A BENNETT DATE:	
CHECKED: DATE:	
ENGR: DATE:	
APVD: DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION

3041 P&ID LEGEND SHEET	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-0002-1	REV A
CADFILE 3041-PID-0002-1.DWG	4/20/2023

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1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. PIPE SIZE SHOWN IS APPROXIMATE INSIDE / OUTSIDE DIAMETER REQUIRED. THE DIFFERENCE INDICATES THE ALLOWANCE FOR REFRACTORY.
3. PREGASIFIER LIFTING MECHANISM FOR GRATE INSTALLATION AND REMOVAL.
4. PREHEATER MAYBE REMOVED BEFORE PROCESSING AND REQUIRES SUPPORT TO REMOVE. AIR AND GAS LINES NEED TO BE EASILY DISCONNECTED FROM PREHEATER. ALSO THE PREHEATER REQUIRES A MOUNTING PAD FOR THE IGNITER TRANSFORMER BOX NEAR THE PREHEATER.
5. AN ARRAY OF SEVEN THERMOCOUPLES DOWN THE SIDE OF THE PREGASIFIER.
6. PREGASIFIER PREHEATER BLOWER, VALVES, AND CONTROLS ON 3004-PID-0900-6
7. TEMPORARY.



V-0200

PREGASIFIER VESSEL

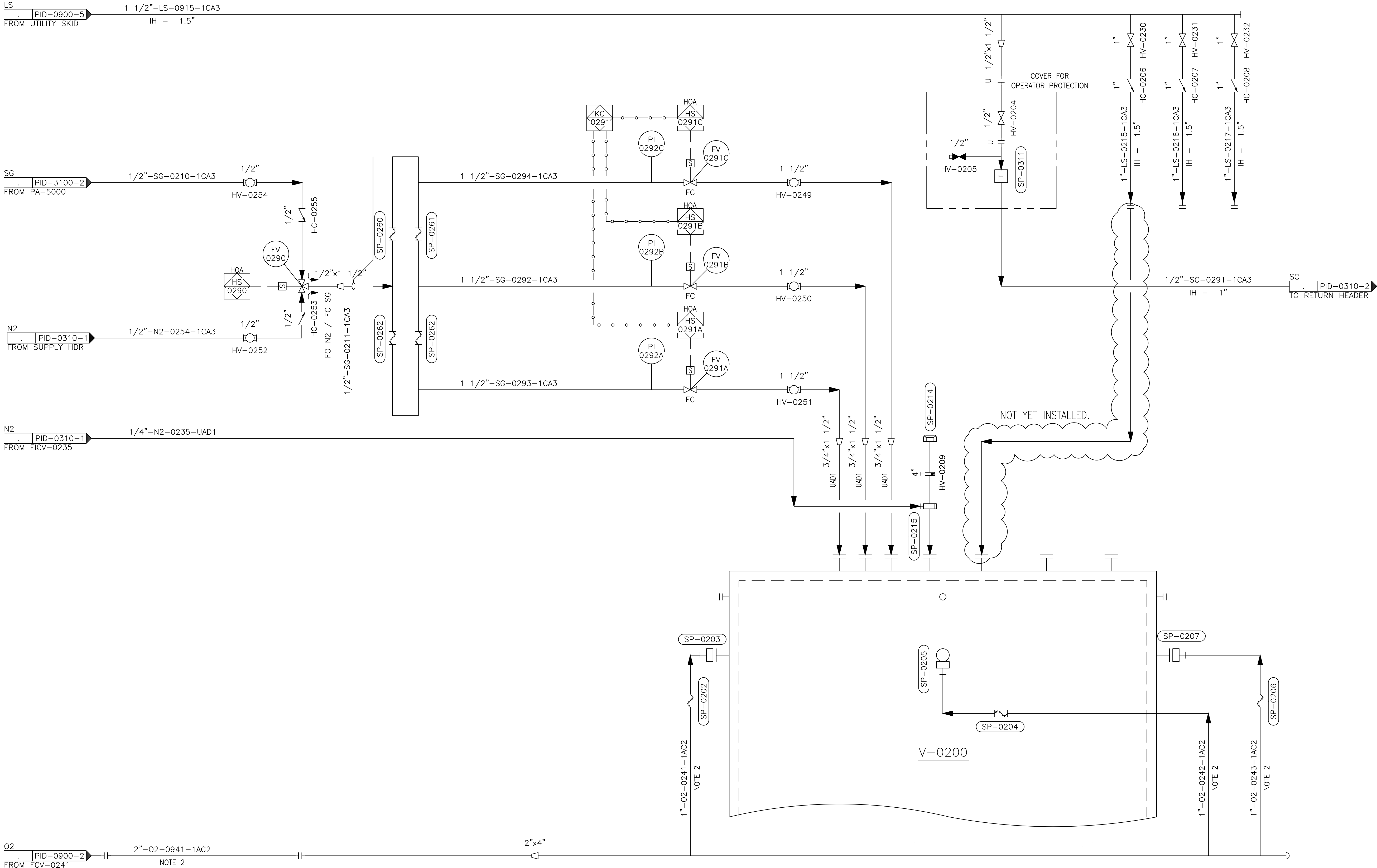
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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. ALL O2 SERVICE VALVES AND INSTRUMENTATION TO BE OXYGEN CLEANED AND COMPATIBLE.
3. A MOUNTING PAD MAY BE REQUIRED NEAR EACH GAD FOR THE IGNITER TRANSFORMER.
4. TEMPORARY



DRAWN: A BENNETT
DATE:

CHECKED: P. PEREZ
DATE: 6/28/2020

ENGR:
DATE:

APVD:
DATE:

REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN
C	REVISED AND REDRAWN

3041
P&ID
PREGASIFIER O2, LS, N2

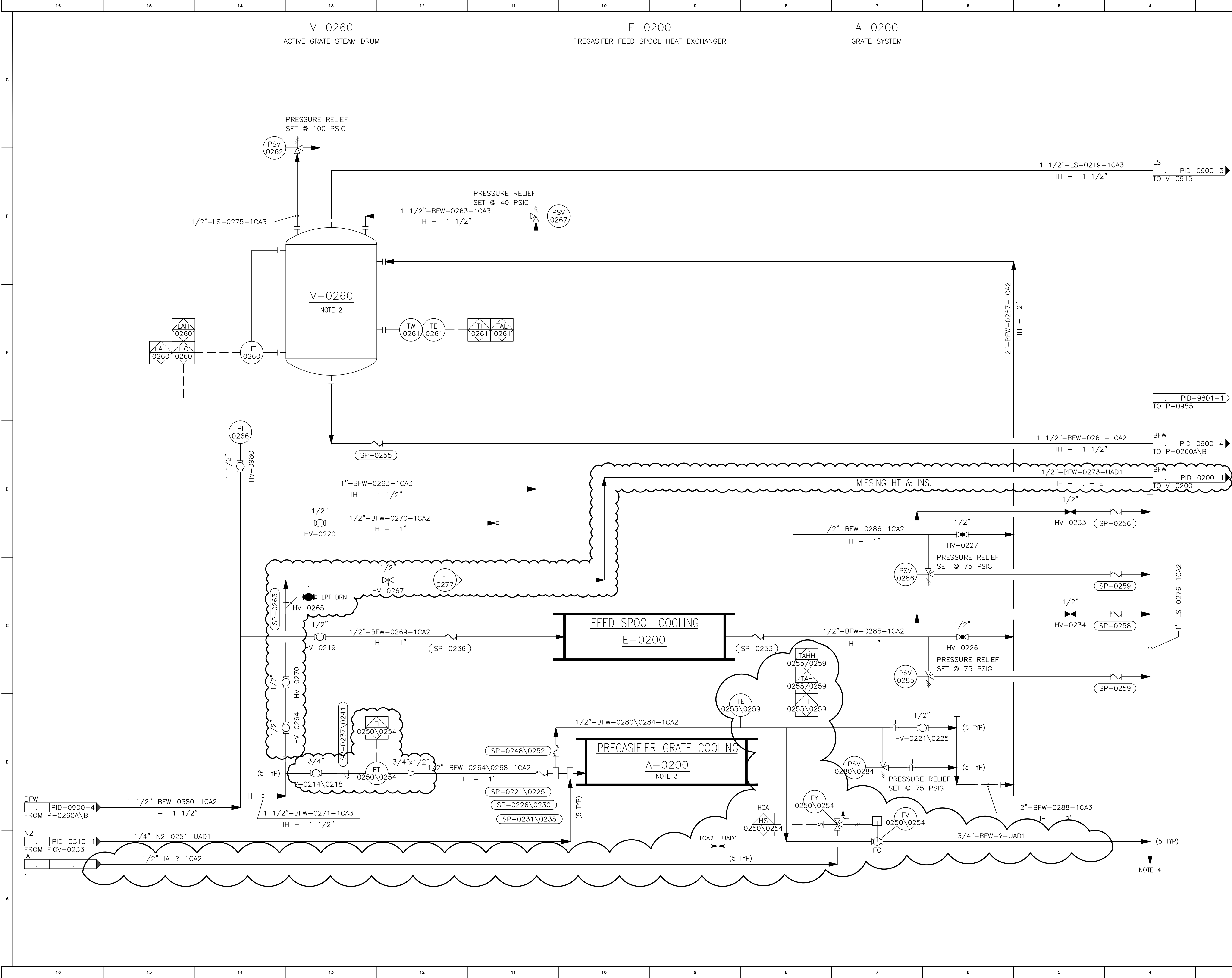
SCALE NONE

DWG NO 3041-PID-0200-2

CADFILE 3041-PID-0200-2.DWG

SHEET NO	SHEETS
1	1
REV	
C	

9/27/2023



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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. STEAM DRUM MUST BE LOCATED ABOVE THE T-SPINNER AND THE STEAM DRUM HAS 3" OF INSULATION.
3. THERE ARE FIVE GRATE BARS TO BE COOLED.
4. VENT AT GROUND LEVEL TO ALLOW WATER TO DRAIN OUT FROM LINE.

DRAWN: A BENNETT	
DATE:	
CHECKED: P. PEREZ	
DATE: 6/28/2020	
ENGR:	
DATE:	
APVD:	
DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN
C	REVISED AND REDRAWN

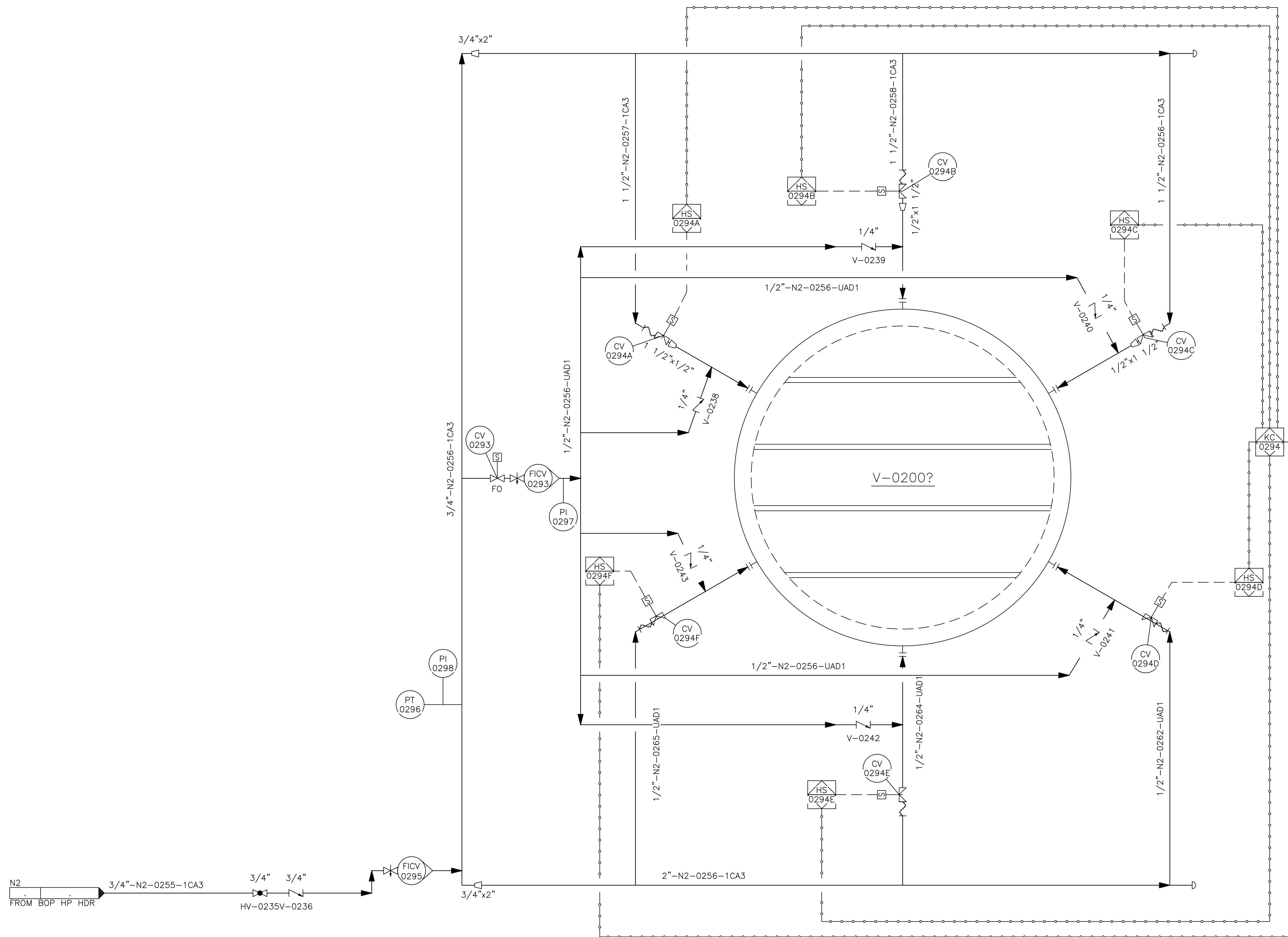
3041	
P&ID	
PREGASIFIER COOLING	
SCALE	NONE
DWG NO	3041-PID-0200-3
CADFILE	3041-PID-0200-3.DWG
SHEET NO	1
SHEETS	1
REV	C
9/27/2023	

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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. VIEWPORT LOCATED ON 14 IN FLANGE LOOKING AT DRAIN.
3. ACCESS PORT FOR WORKING ON DRAIN.



DRAWN: P SALAS
DATE: 9/11/23

CHECKED:
DATE:

ENGR:
DATE:

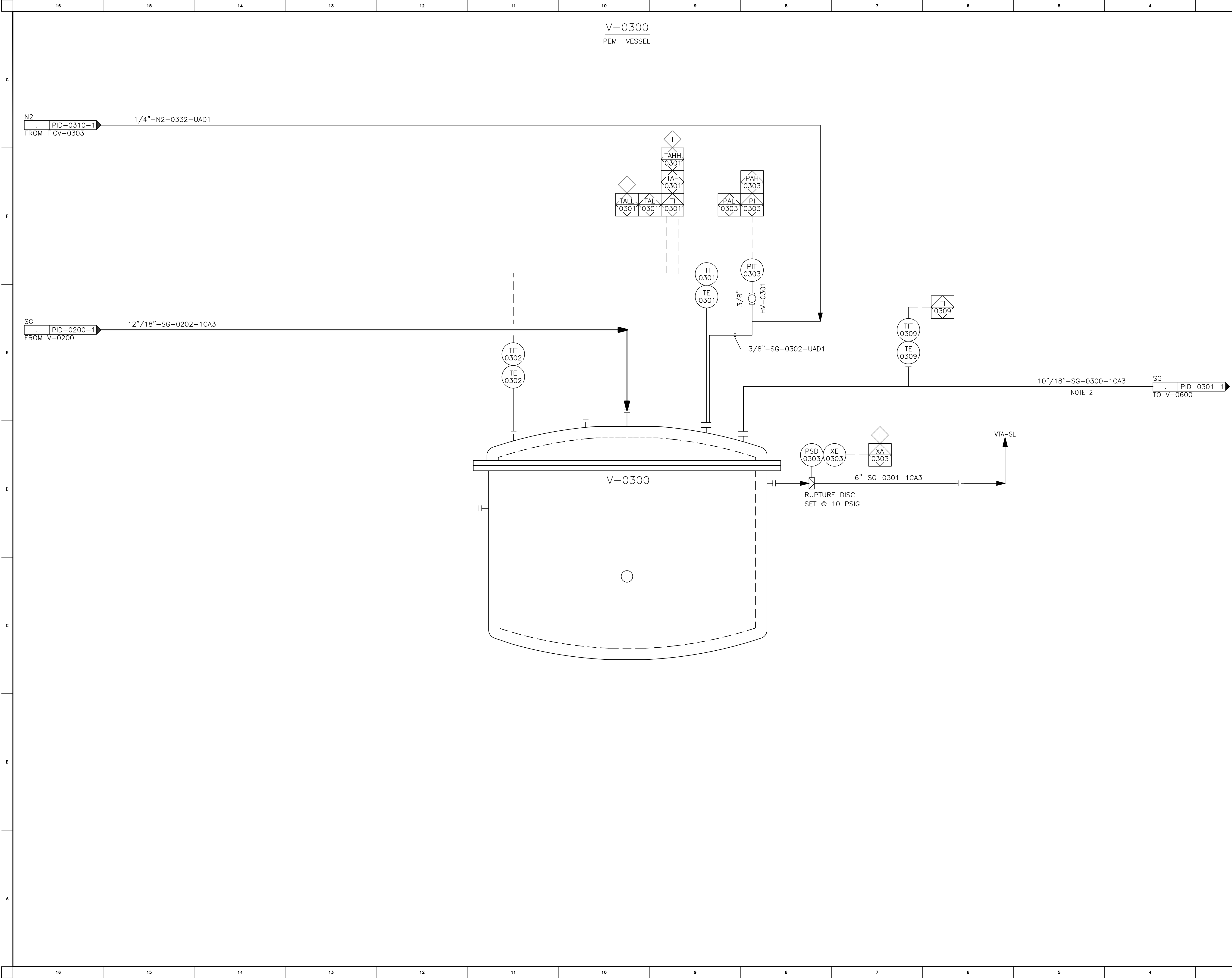
APVD:
DATE:

REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	ADDED FILL LINE ROTAMETER AND PRESSURE GAUGES

3041
P&ID
DDG GRATE PULSERS

SCALE	NONE	SHEET NO	1	SHEETS	1
DWG NO	3041-PID-0200-4			REV	B
CADFILE	3041-PID-0200-4.DWG			3/20/2024	



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- NOTES:
1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 2. PIPE SIZE SHOWN IS APPROXIMATE INSIDE / OUTSIDE DIAMETER REQUIRED. THE DIFFERENCE INDICATES THE ALLOWANCE FOR REFRACTORY.

DRAWN: A BENNETT
DATE:

CHECKED: P PEREZ
DATE: 6/28/2020

ENGR:
DATE:

APVD:
DATE:

REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN
C	REVISED AND REDRAWN

3041
P&ID
PEM

SCALE	NONE	SHEET NO	1	SHEETS	1
DWG NO	3041-PID-0300-1	REV	C		
CADFILE	3041-PID-0300-1.DWG				9/27/2023

V-0300
PEM VESSEL

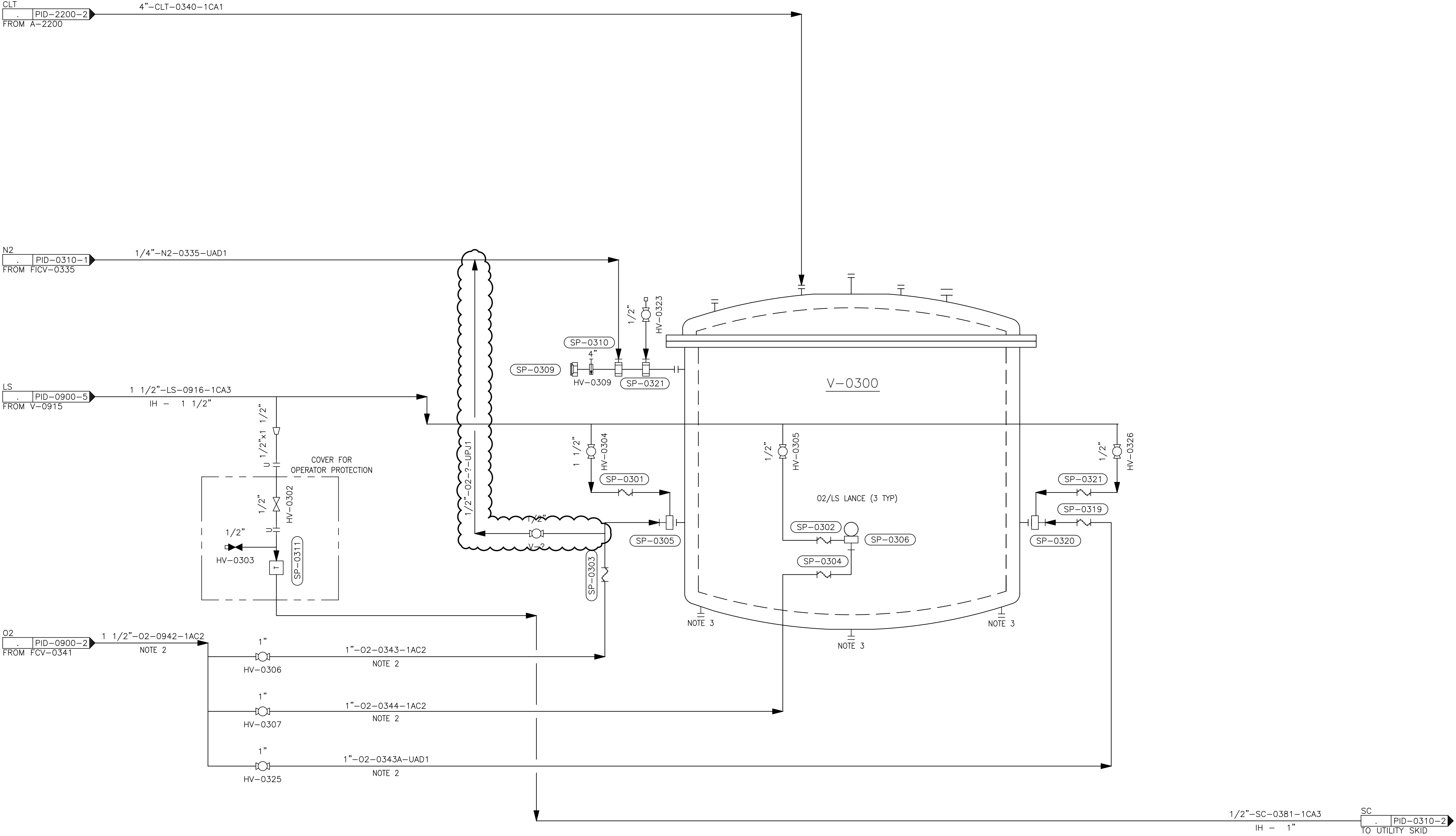
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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. ALL O2 SERVICE VALVES AND INSTRUMENTATION TO BE OXYGEN CLEANED AND COMPATIBLE.
3. DRAIN PORT FOR GLASS AND METAL REMOVAL.



DRAWN: A BENNETT DATE:	
CHECKED: P PEREZ DATE: 6/28/2020	
ENGR: DATE:	
APVD: DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN
C	REVISED AND REDRAWN

3041 P&ID PEM LS, O2, & N2	
SCALE	NONE
DWG NO	3041-PID-0300-2
CADFILE	3041-PID-0300-2.DWG
3/21/2024	

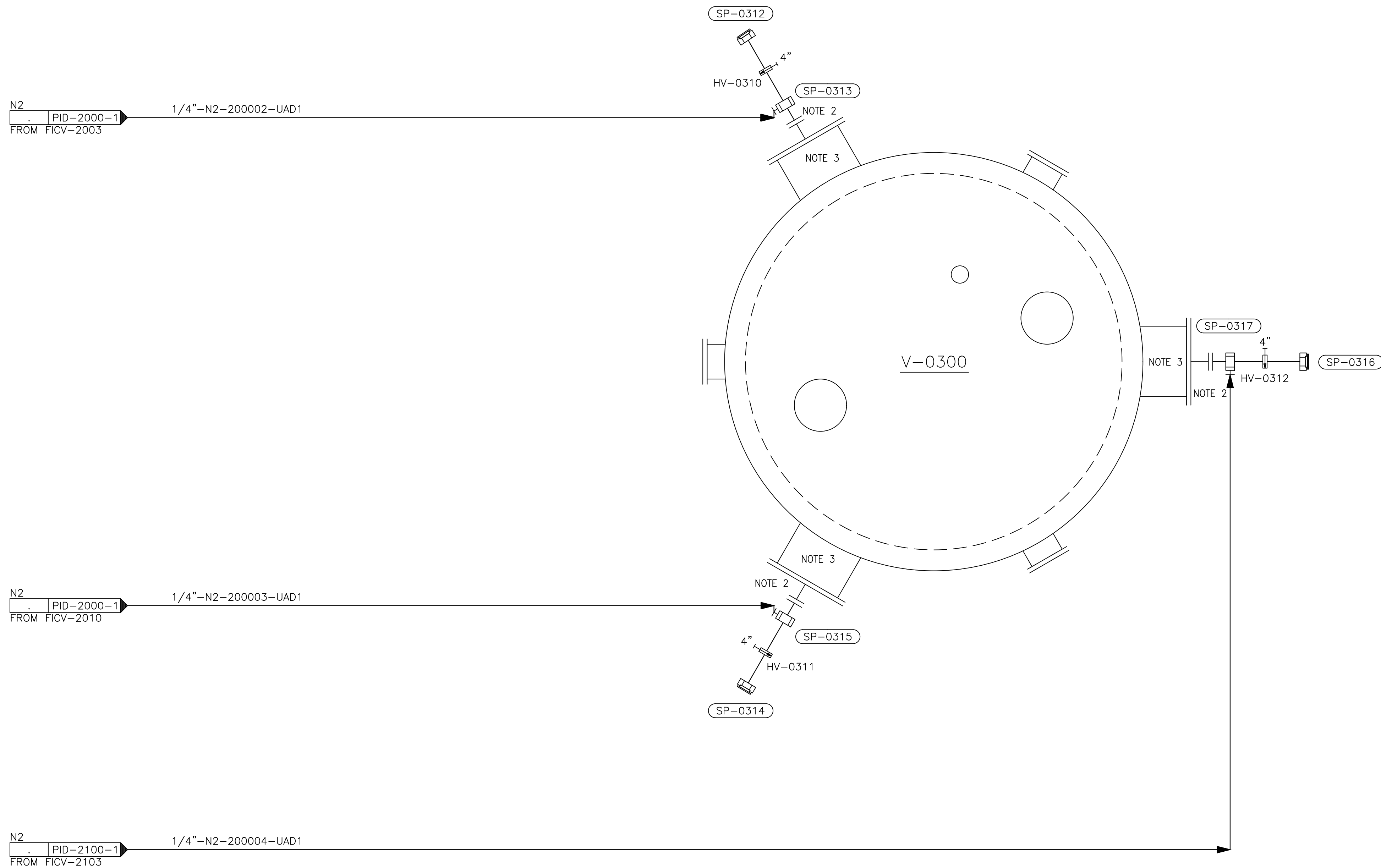
InfoToc Info

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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. VIEWPORT LOCATED ON 14 IN FLANGE LOOKING AT DRAIN.
3. ACCESS PORT FOR WORKING ON DRAIN.



DRAWN: A BENNETT
DATE:

CHECKED: P PEREZ
DATE: 6/28/2020

ENGR
DATE

APVD
DATE

REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN
C	REVISED AND REDRAWN

3041
P&ID
PEM DRAINS

SCALE	NONE	SHEET NO	1	SHEETS	1
DWG NO	3041-PID-0300-3	REV	C		
CADFILE	3041-PID-0300-3.DWG			9/11/2023	

PA-0610F-0610
TRC PREHEATER PACKAGE

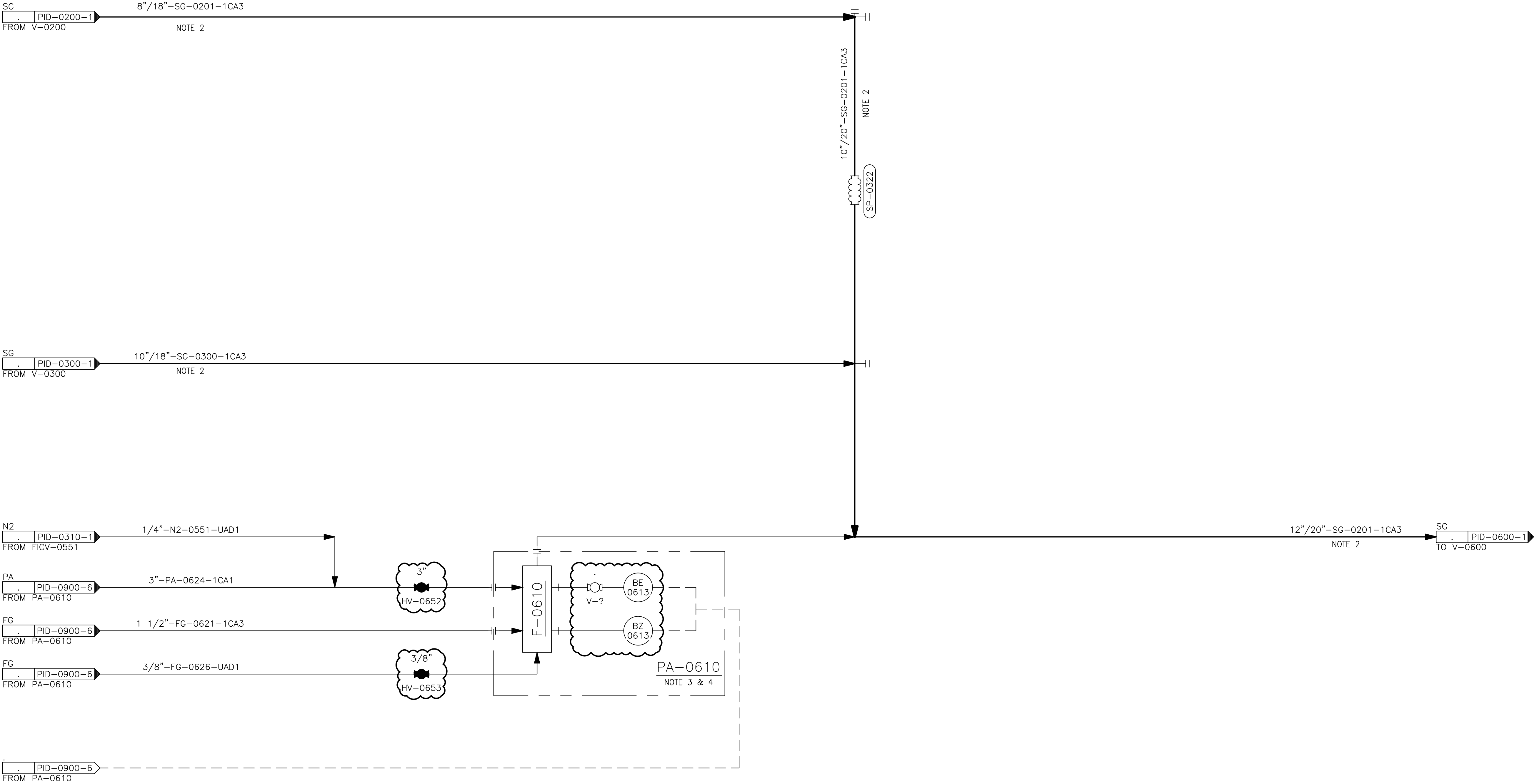
InEnTec Inc.

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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. PIPE SIZE SHOWN IS APPROXIMATE INSIDE / OUTSIDE DIAMETER REQUIRED. THE DIFFERENCE INDICATES THE ALLOWANCE FOR REFRACTORY.
3. PREHEATER MAYBE REMOVED BEFORE PROCESSING AND REQUIRES SUPPORT TO REMOVE. AIR AND GAS LINES NEED TO BE EASILY DISCONNECTED FROM PREHEATER. ALSO THE PREHEATER REQUIRES A MOUNTING PAD FOR THE IGNITER TRANSFORMER BOX NEAR THE PREHEATER.
4. TRC PREHEATER BLOWER, VALVES, AND CONTROLS ON 3041-PID-0900-6.



DRAWN: A BENNETT
DATE:
CHECKED: P PEREZ
DATE: 6/28/2020

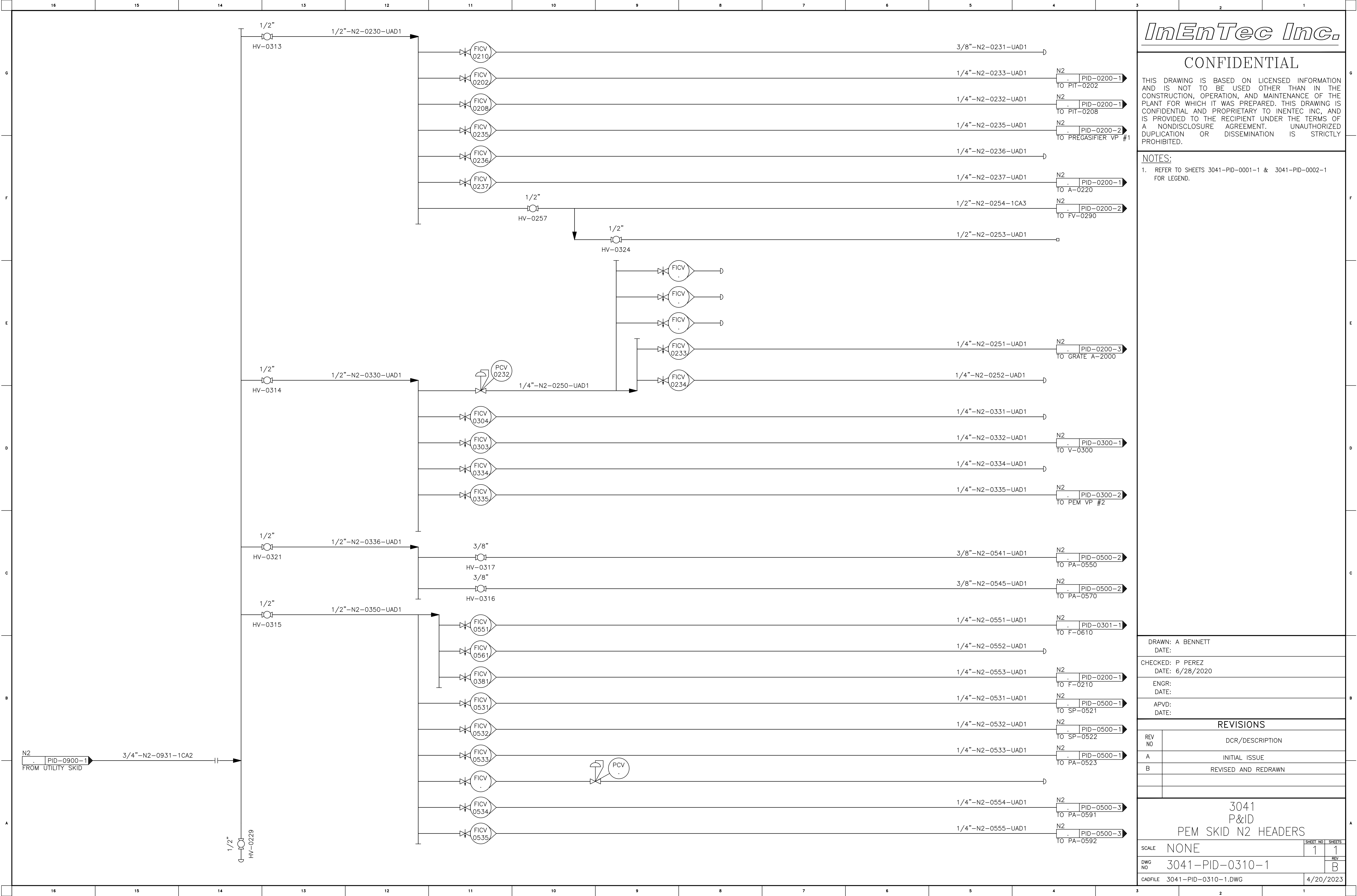
ENGR:
DATE:

APVD:
DATE:

REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041
P&ID
PEM-TRC SG DUCT

SCALE	NONE	SHEET NO	1	SHEETS	1
DWG NO	3041-PID-0301-1	REV	A		
CADFILE	3041-PID-0301-1.DWG				



InEnTec Inc.

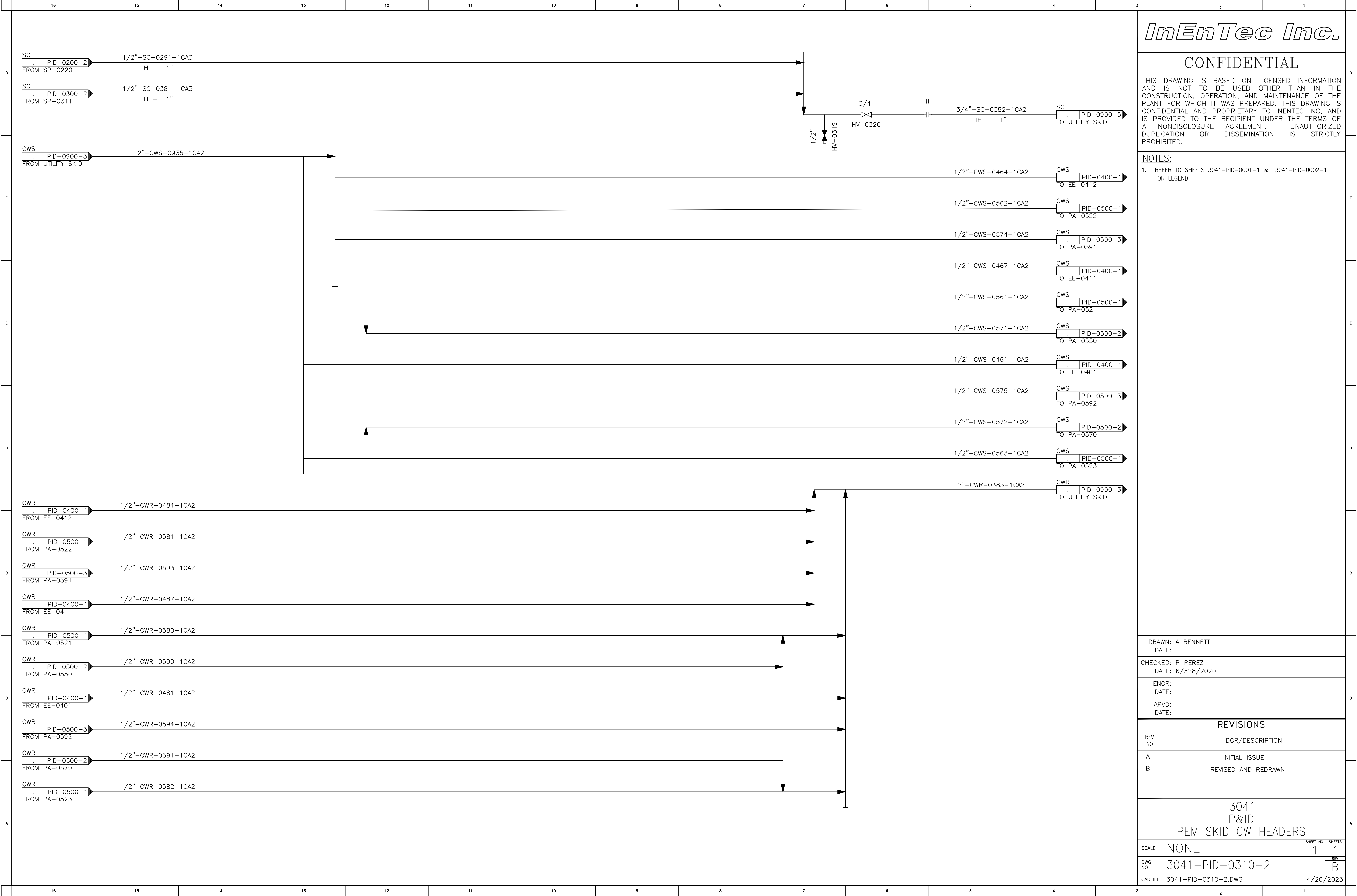
CONFIDENTIAL

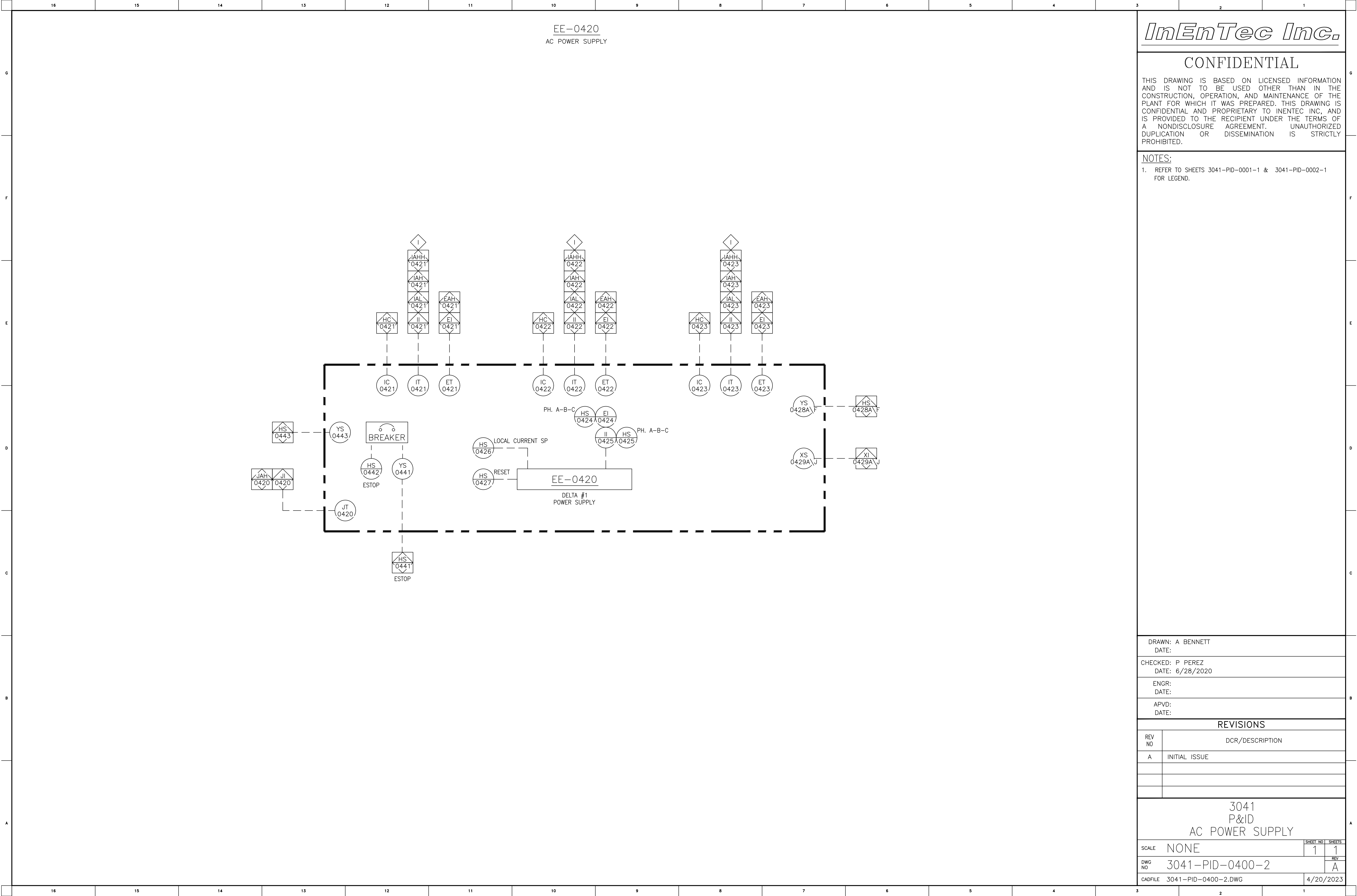
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NOTES:
1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT DATE:		
CHECKED: P PEREZ DATE: 6/28/2020		
ENGR: DATE:		
APVD: DATE:		
REVISIONS		
REV NO	DCR/DESCRIPTION	
A	INITIAL ISSUE	
B	REVISED AND REDRAWN	

3041 P&ID PEM SKID N2 HEADERS		
SCALE	NONE	SHEET NO 1 SHEETS 1
DWG NO	3041-PID-0310-1	REV B
CADFILE	3041-PID-0310-1.DWG	4/20/2023





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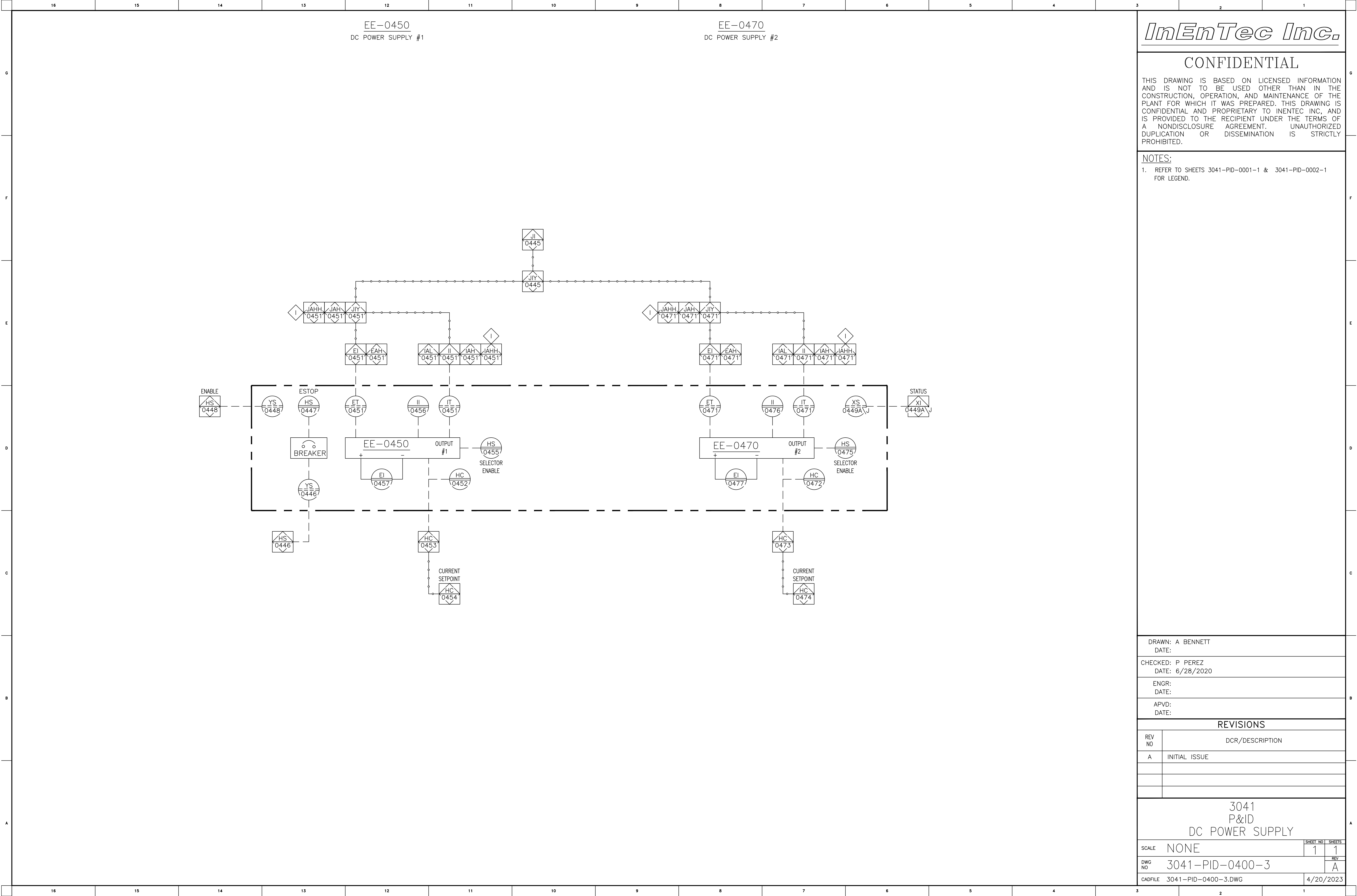
CONFIDENTIAL

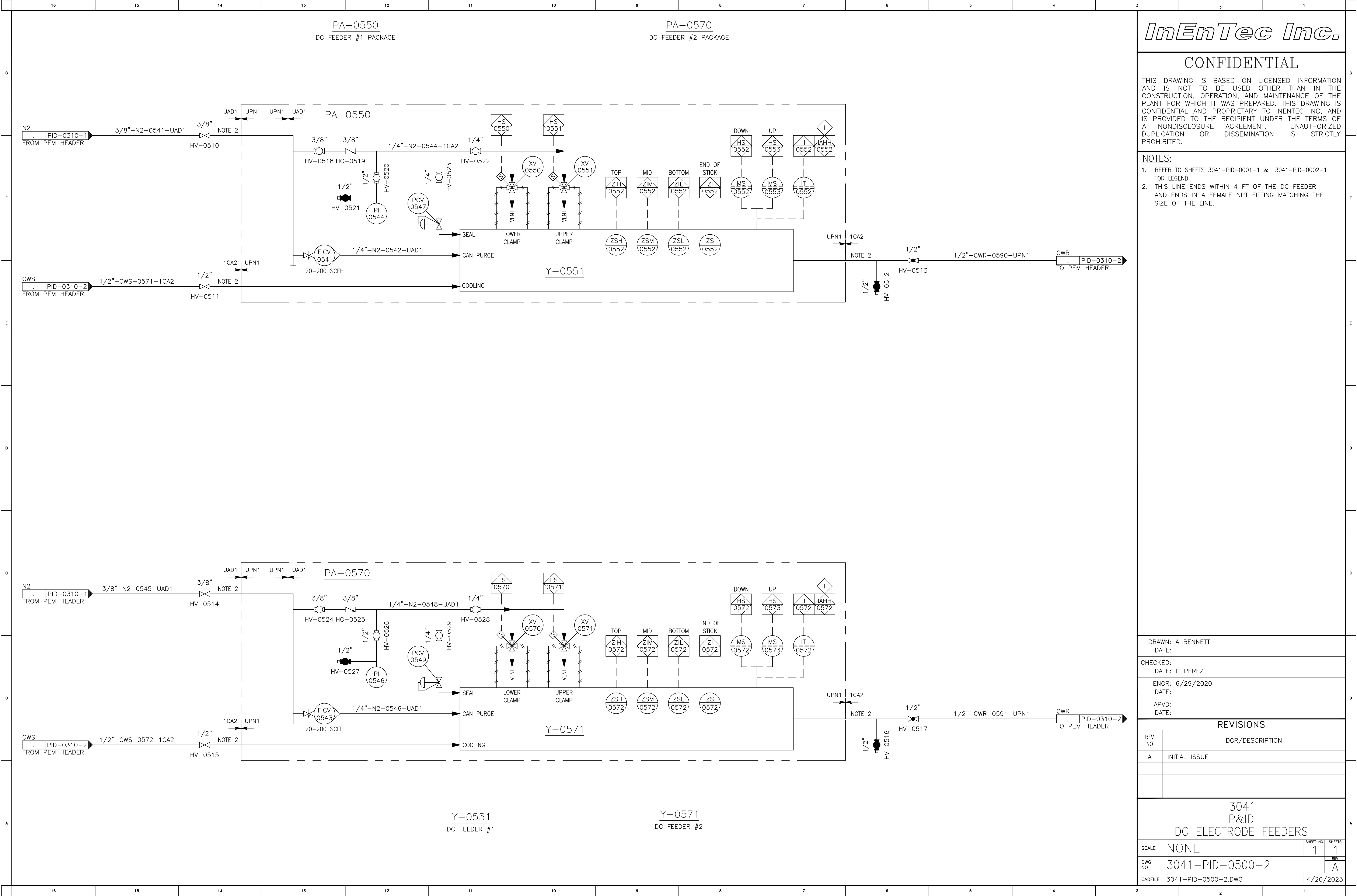
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- NOTES:
1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT DATE:	
CHECKED: P PEREZ DATE: 6/28/2020	
ENGR: DATE:	
APVD: DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID AC POWER SUPPLY		
SCALE	NONE	SHEET NO 1 SHEETS 1
DWG NO	3041-PID-0400-2	REV A
CADFILE	3041-PID-0400-2.DWG	4/20/2023





InEnTec Inc.

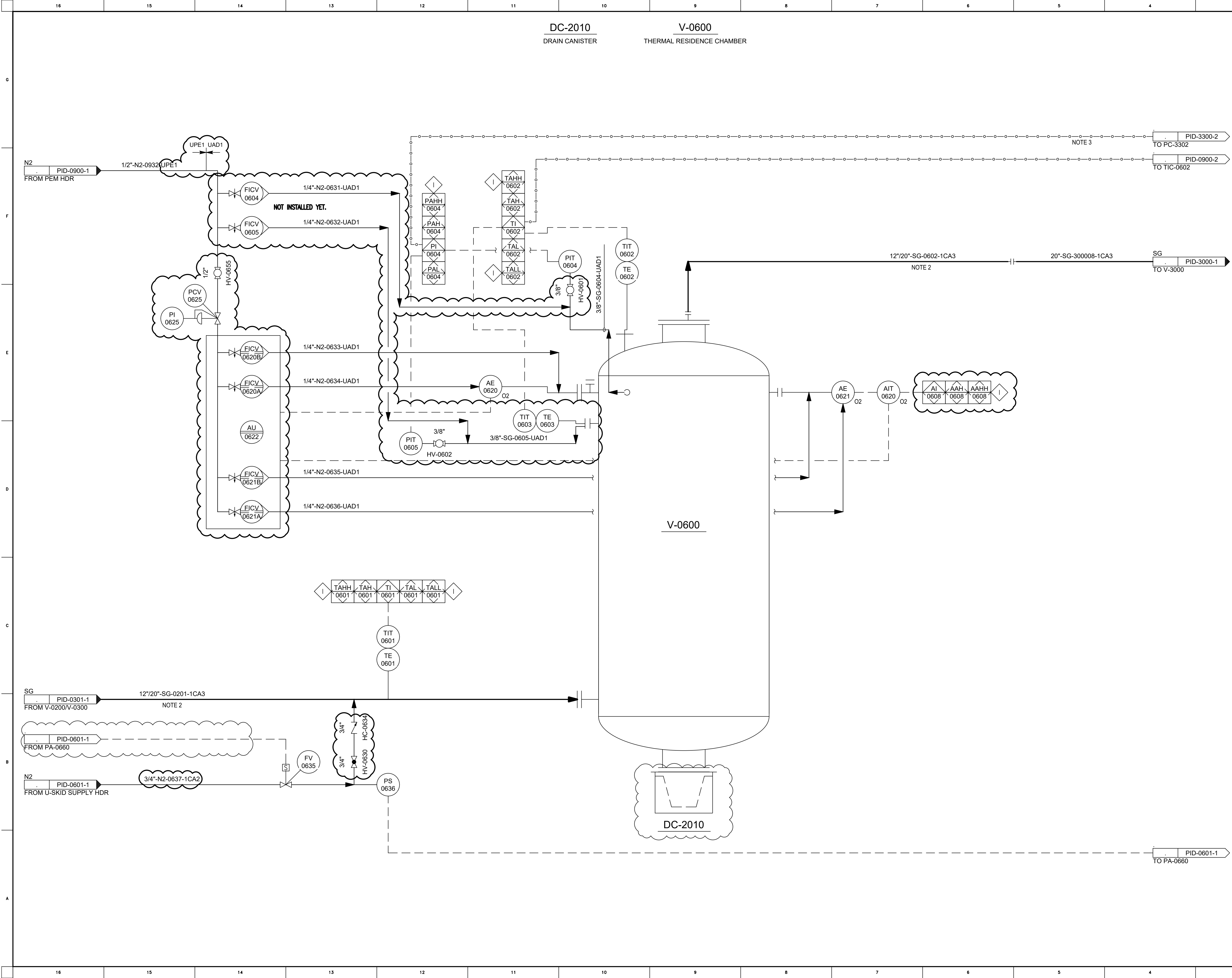
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- NOTES:**
- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 - THIS LINE ENDS WITHIN 4 FT OF THE DC FEEDER AND ENDS IN A FEMALE NPT FITTING MATCHING THE SIZE OF THE LINE.

DRAWN: A BENNETT DATE:	
CHECKED: DATE: P PEREZ	
ENGR: 6/29/2020 DATE:	
APVD: DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID DC ELECTRODE FEEDERS	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-0500-2	REV A
CADFILE 3041-PID-0500-2.DWG	4/20/2023



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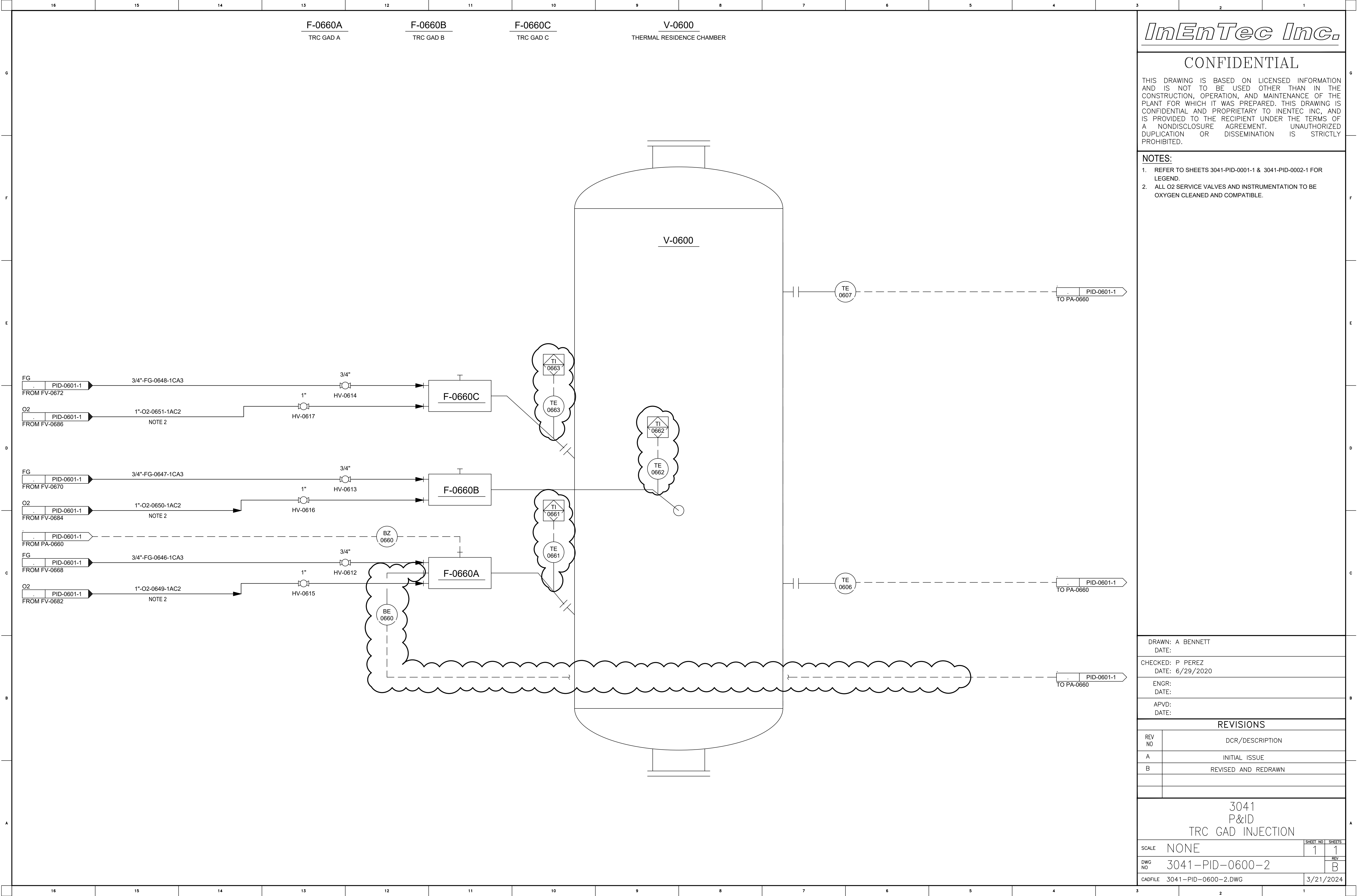
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- NOTES:
1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 2. PIPE SIZE SHOWN IS APPROXIMATE INSIDE / OUTSIDE DIAMETER REQUIRED. THE DIFFERENCE INDICATES THE ALLOWANCE FOR REFRACTORY.
 3. PRESSURE CONTROL BY SYNGAS CLEANING TRAIN.

DRAWN: A BENNETT DATE:	
CHECKED: P PEREZ DATE: 6/29/2020	
ENGR: DATE:	
APVD: DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN
C	REVISED AND REDRAWN

3041 P&ID TRC	
SCALE	NONE
DWG NO	3041-PID-0600-1
CADFILE	3041-PID-0600-1.DWG
SHEET NO	1
SHEETS	1
REV	C
3/21/2024	



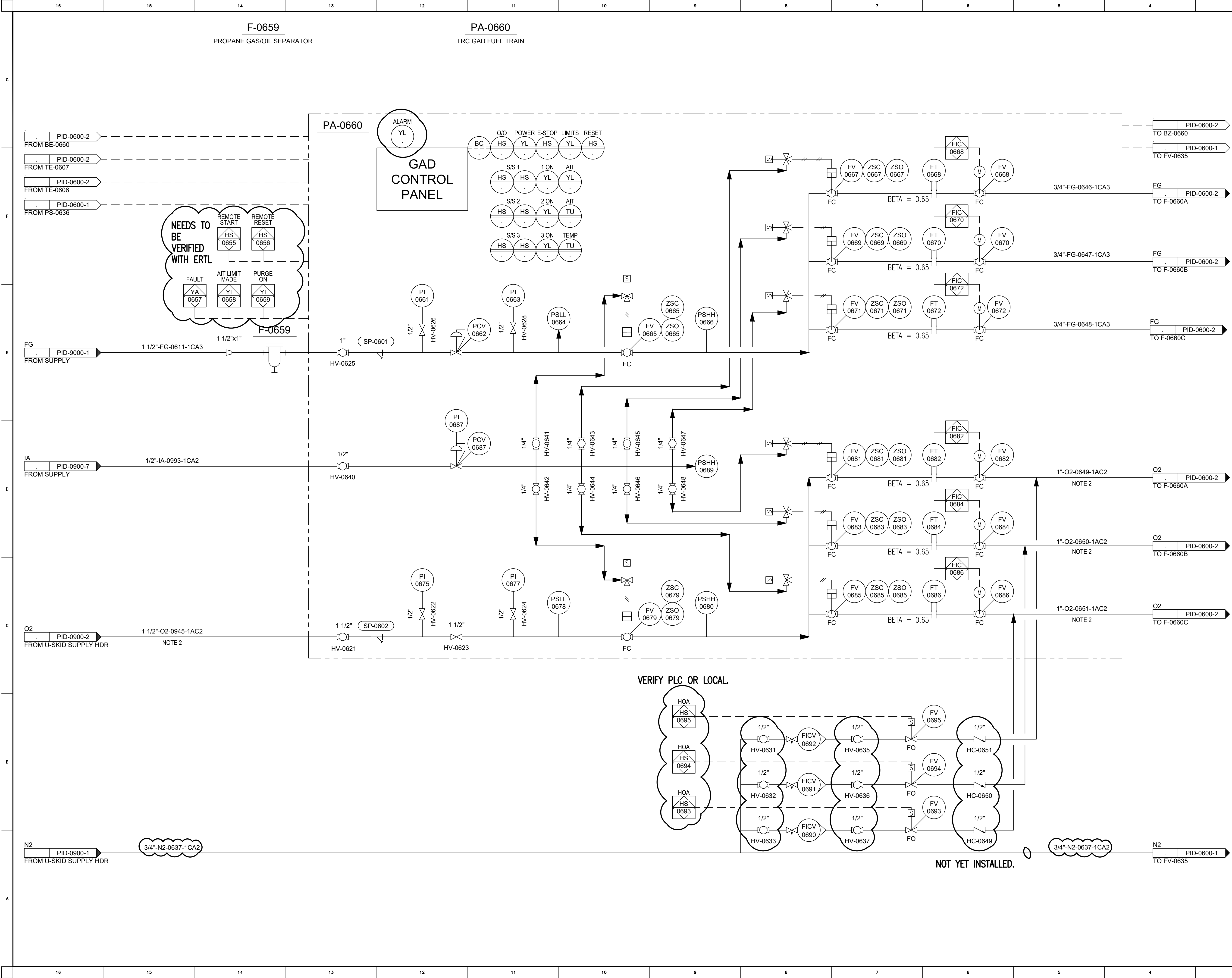
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NOTES:

- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
- ALL O2 SERVICE VALVES AND INSTRUMENTATION TO BE OXYGEN CLEANED AND COMPATIBLE.



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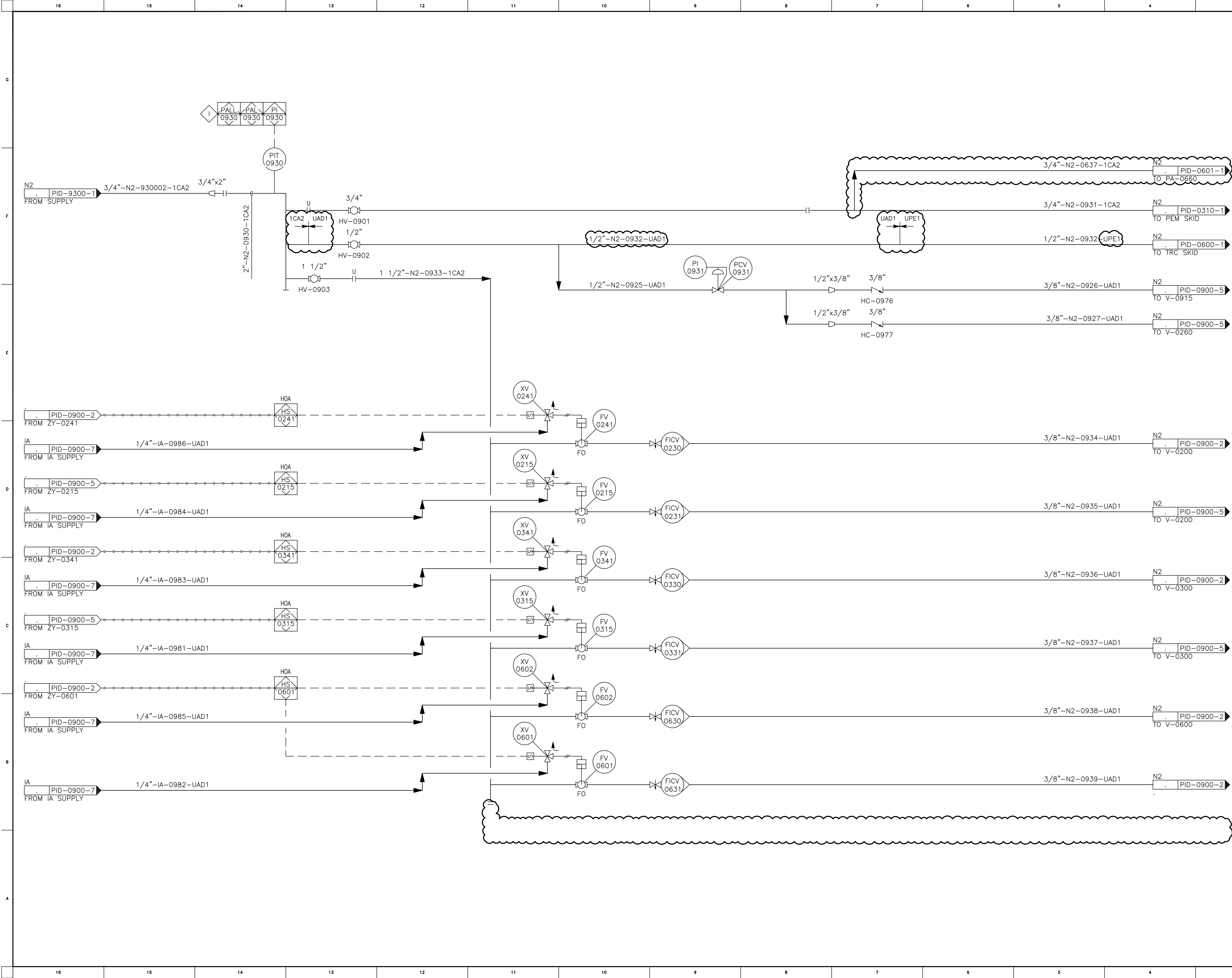
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- NOTES:
- REFER TO SHEETS 3041-PID-0001 & 3041-PID-0002 FOR LEGEND.
 - ALL O2 SERVICE VALVES AND INSTRUMENTATION TO BE OXYGEN CLEANED AND COMPATIBLE.

DRAWN: A BENNETT DATE:	
CHECKED: P PEREZ DATE: 6/29/2020	
ENGR: DATE:	
APVD: DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID TRC GAD FUEL TRAIN	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-0601-1	REV A
CADFILE 3041-PID-0601-1.DWG	4/20/2023



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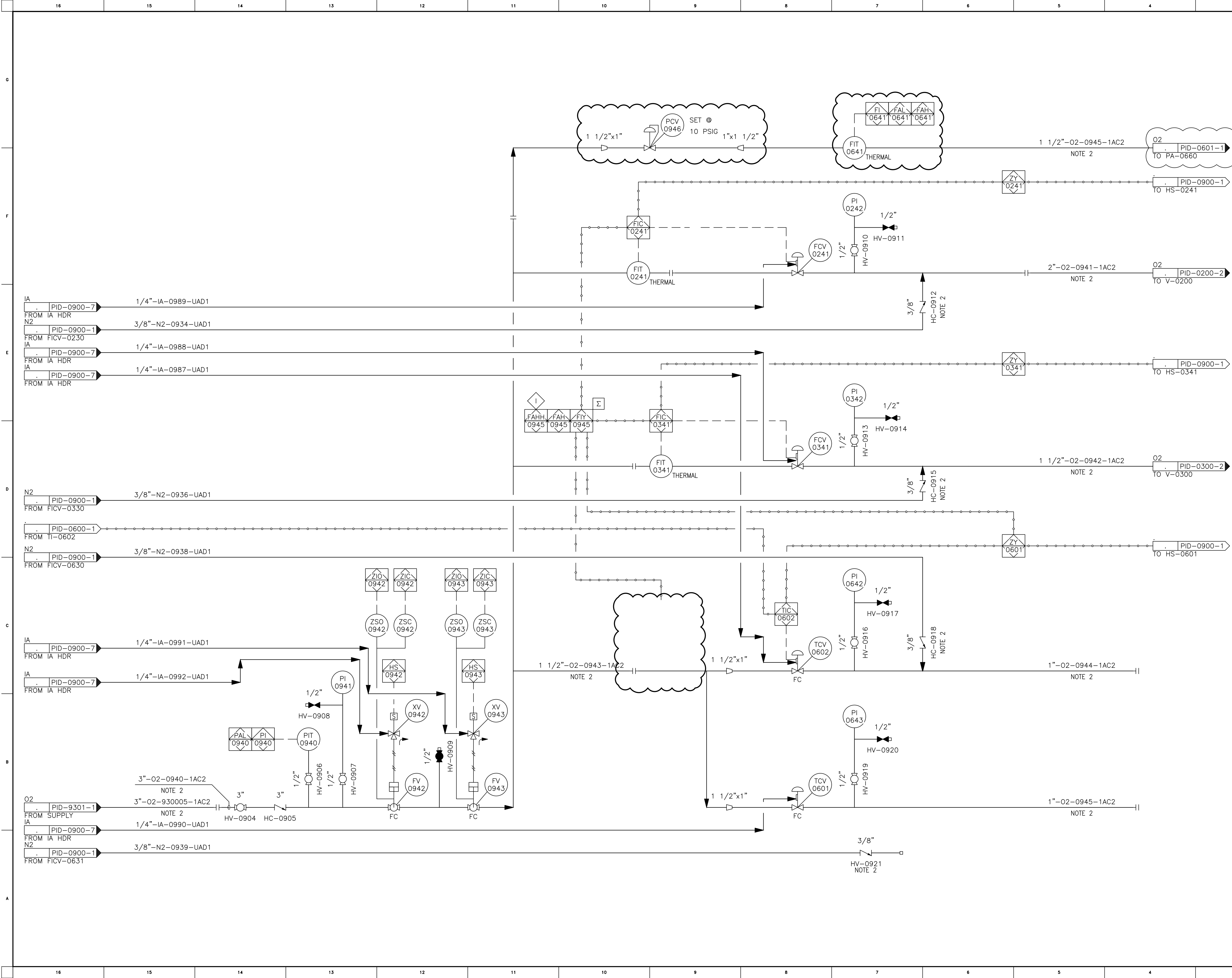
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- NOTES:
- 1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT	
DATE:	
CHECKED: P PEREZ	
DATE: 6/29/2020	
ENGR:	
DATE:	
APVD:	
DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041	
P&ID	
UTILITY SKID N2 HEADERS	
SCALE	NONE
DWG NO	3041-PID-0900-1
CADFILE	3041-PID-0900-1.DWG
SHEET NO	1
SHEETS	1
REV	A
4/20/2023	



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- NOTES:
1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 2. ALL O2 SERVICE VALVES AND INSTRUMENTATION TO BE OXYGEN CLEANED AND COMPATIBLE.

DRAWN: R RAMOS DATE: 12-4-19	
CHECKED: P PEREZ DATE: 6/29/2020	
ENGR: P SALAS DATE: 12-4-19	
APVD: DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID UTILITY SKID O2 HEADER	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-0900-2	REV B
CADFILE 3041-PID-0900-2.DWG	4/20/2023

EE-0415
HEAT STATION SWITCH #2

EE-0400
DRAIN POWER SUPPLY #1

EE-0410
DRAIN POWER SUPPLY #2

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CONFIDENTIAL

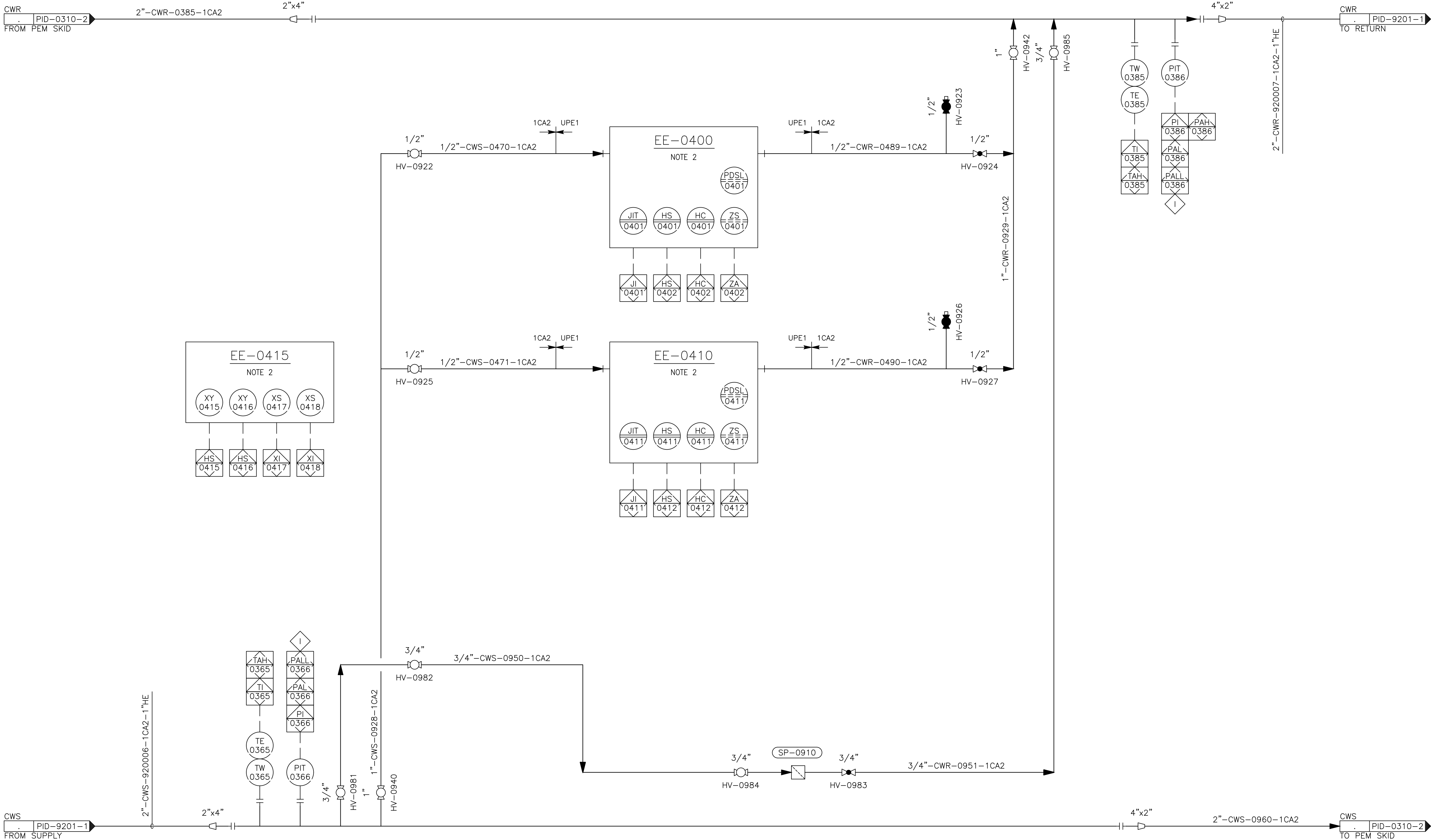
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NOTES:

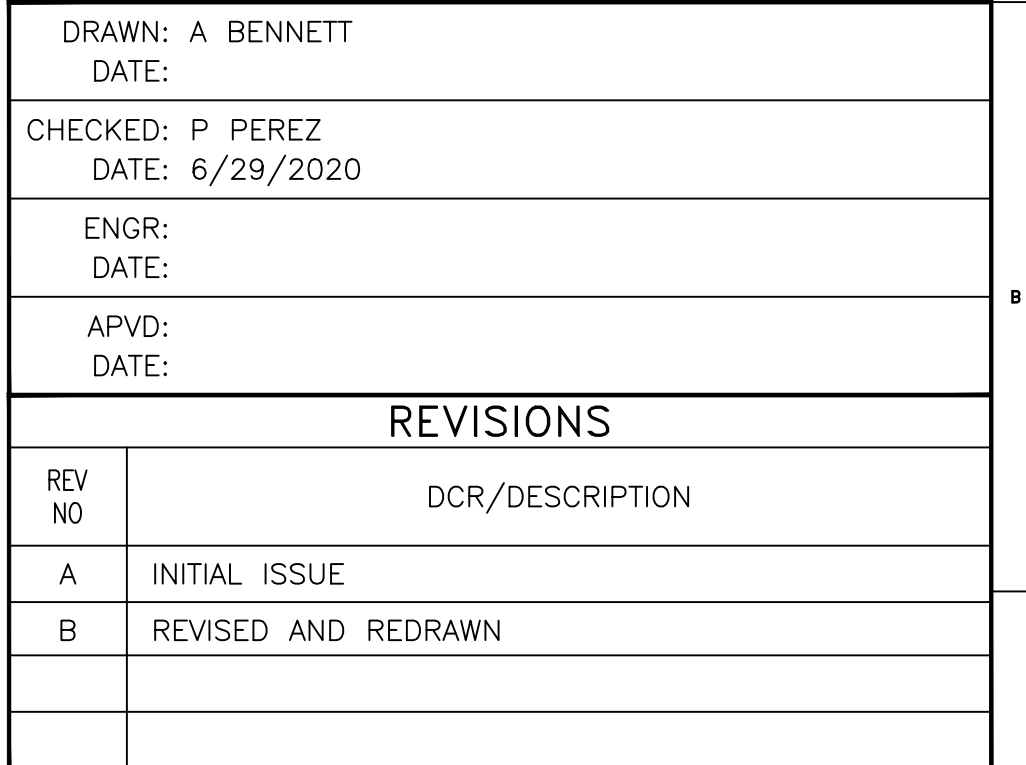
1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. HEAT STATIONS AND DRAINS LOCATED ON 3041-PID-0400-1.

DRAWN: A BENNETT DATE:	
CHECKED: P PEREZ DATE: 6/29/2020	
ENGR: DATE:	
APVD: DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID UTILITY SKID CW HEADERS			
SCALE	NONE	SHEET NO	1
DWG NO	3041-PID-0900-3	REV	A
CADFILE	3041-PID-0900-3.DWG	4/20/2023	



1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.



SCALE	NONE	SHEET NO	1	SHEETS	1
DWG NO	3041-PID-0900-4			REV	B
CADFILE	3041-PID-0900-4.DWG			4/20/2023	

V-0915
STEAM SUPPLY VESSEL

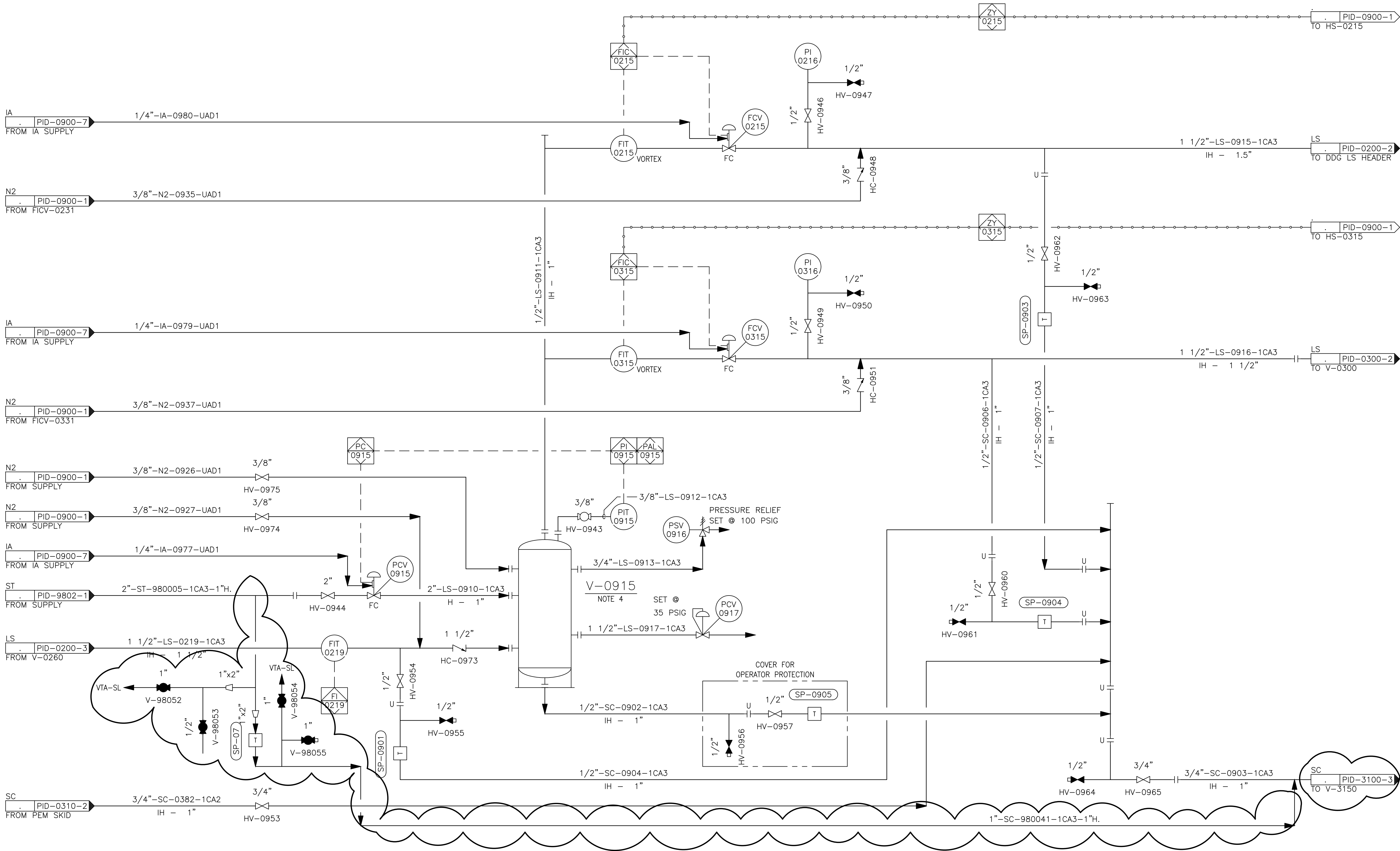
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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. PCV-0915 HAS 3" OF INSULATION.



DRAWN: A BENNETT
DATE:

CHECKED: P PEREZ
DATE: 6/29/2020

ENGR:
DATE:

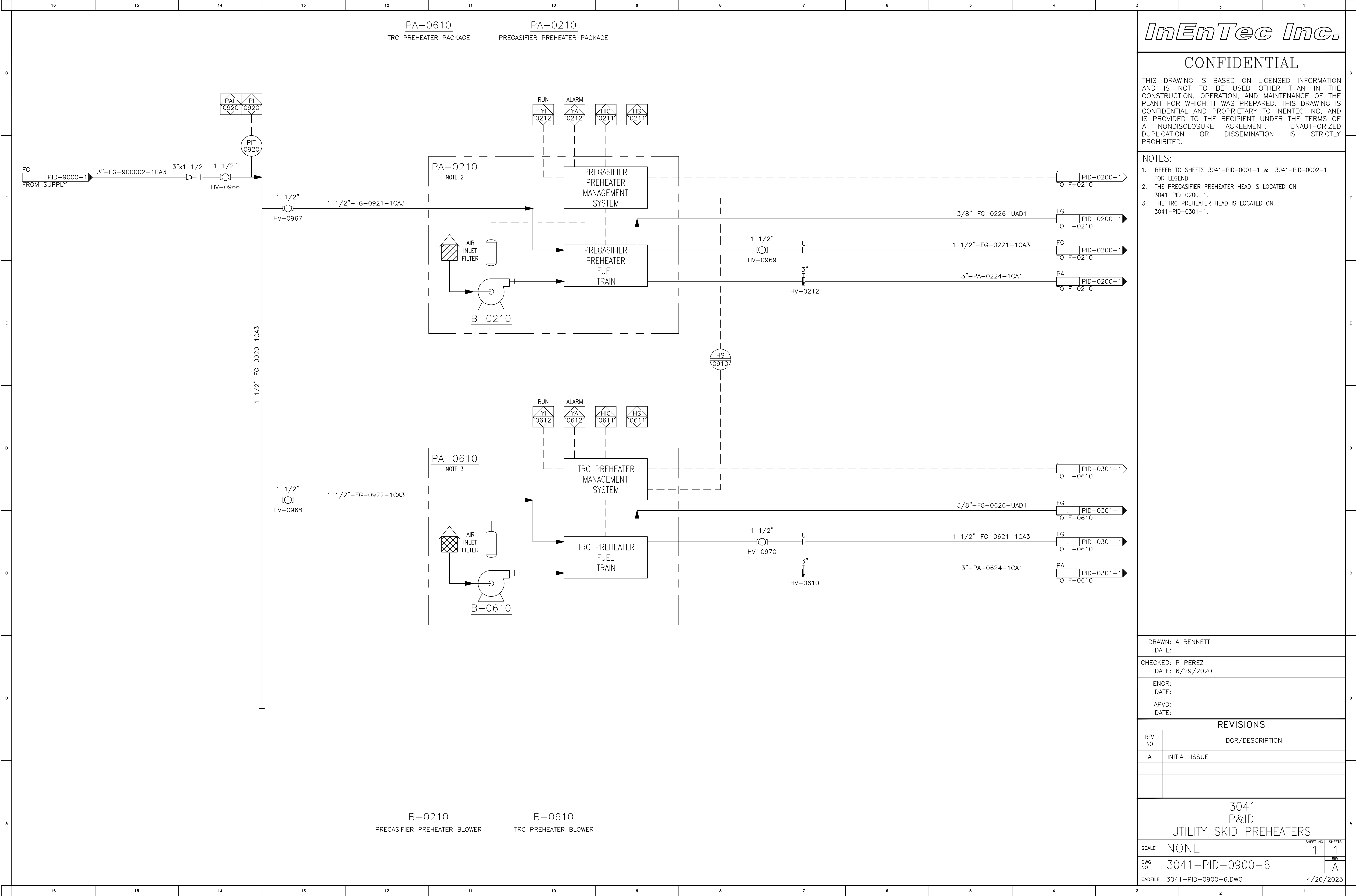
APVD:
DATE:

REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041
P&ID
UTILITY SKID LS HEADER

SCALE	NONE	SHEET NO	1	SHEETS	1
DWG NO	3041-PID-0900-5	REV	A		
CADFILE	3041-PID-0900-5.DWG				4/20/2023



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- NOTES:
- 1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 - 2. THE PREGASIFIER PREHEATER HEAD IS LOCATED ON 3041-PID-0200-1.
 - 3. THE TRC PREHEATER HEAD IS LOCATED ON 3041-PID-0301-1.

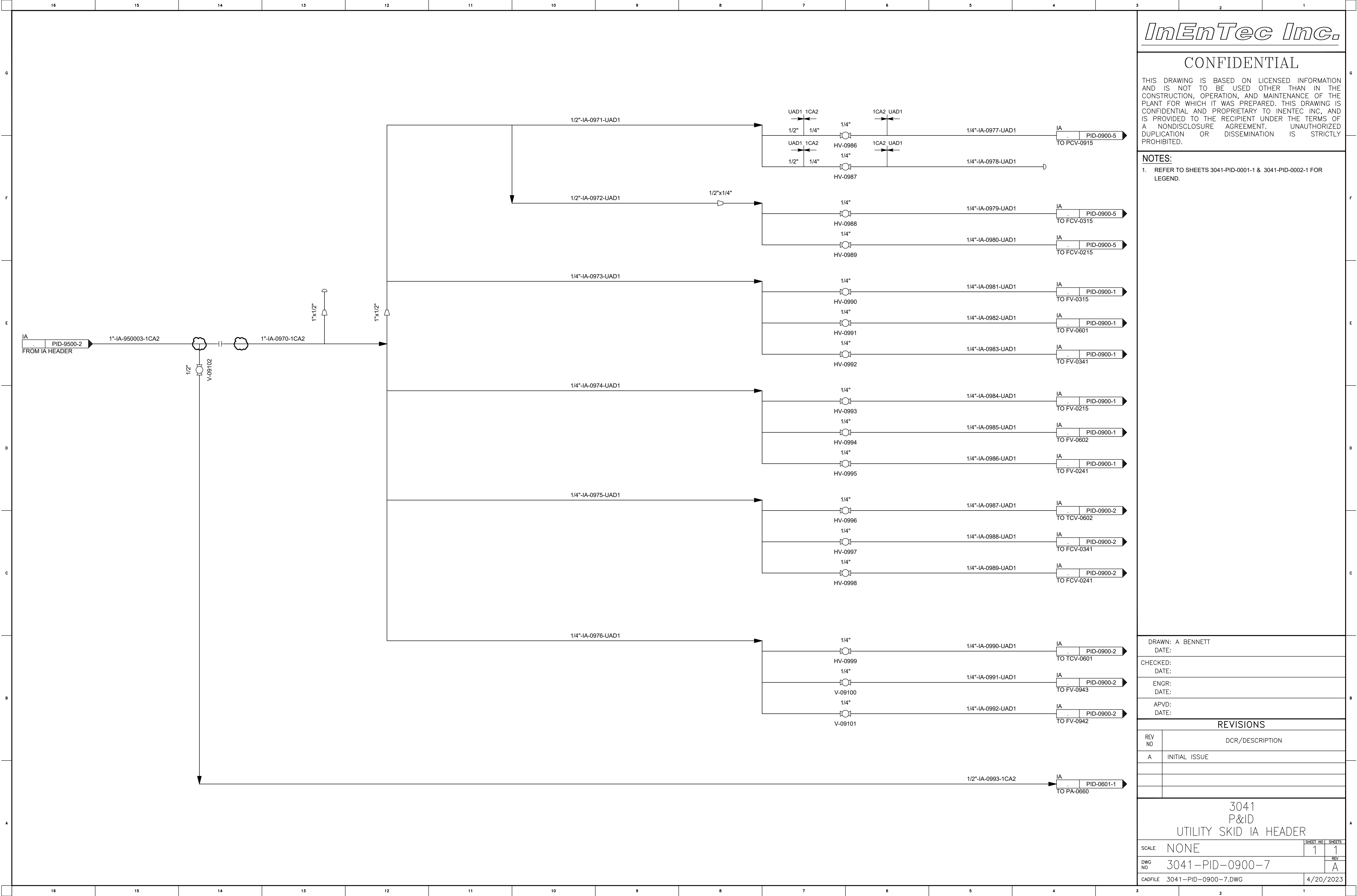
DRAWN: A BENNETT	
DATE:	
CHECKED: P PEREZ	
DATE: 6/29/2020	
ENGR:	
DATE:	
APVD:	
DATE:	

REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID UTILITY SKID PREHEATERS	
SCALE	NONE
DWG NO	3041-PID-0900-6
CADFILE	3041-PID-0900-6.DWG

SHEET NO	1	SHEETS	1
REV	A		

4/20/2023



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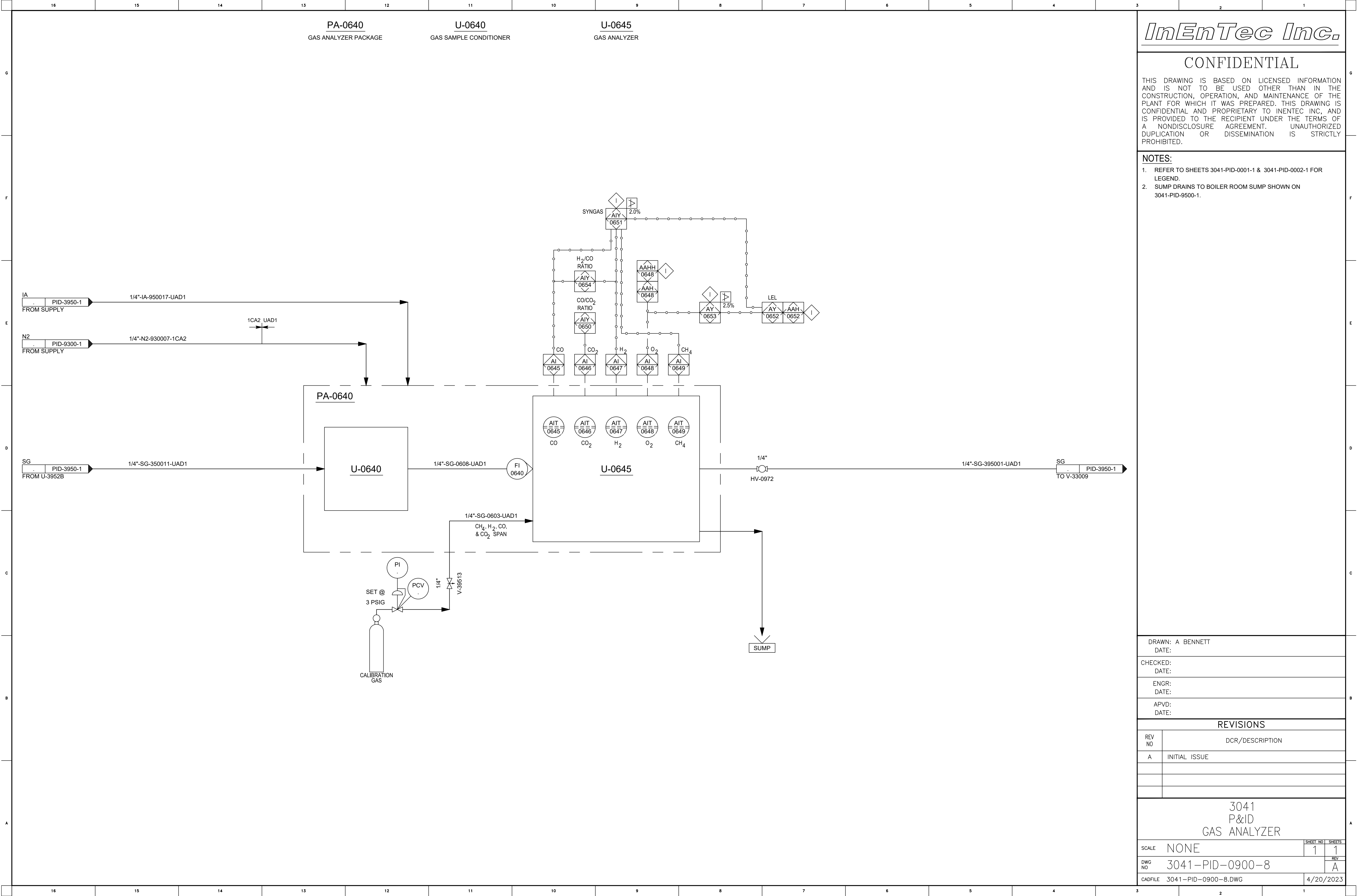
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- NOTES:
- 1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT	
DATE:	
CHECKED:	
DATE:	
ENGR:	
DATE:	
APVD:	
DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID UTILITY SKID IA HEADER			
SCALE	NONE	SHEET NO	SHEETS
DWG NO	3041-PID-0900-7	REV	A
CADFILE	3041-PID-0900-7.DWG	4/20/2023	



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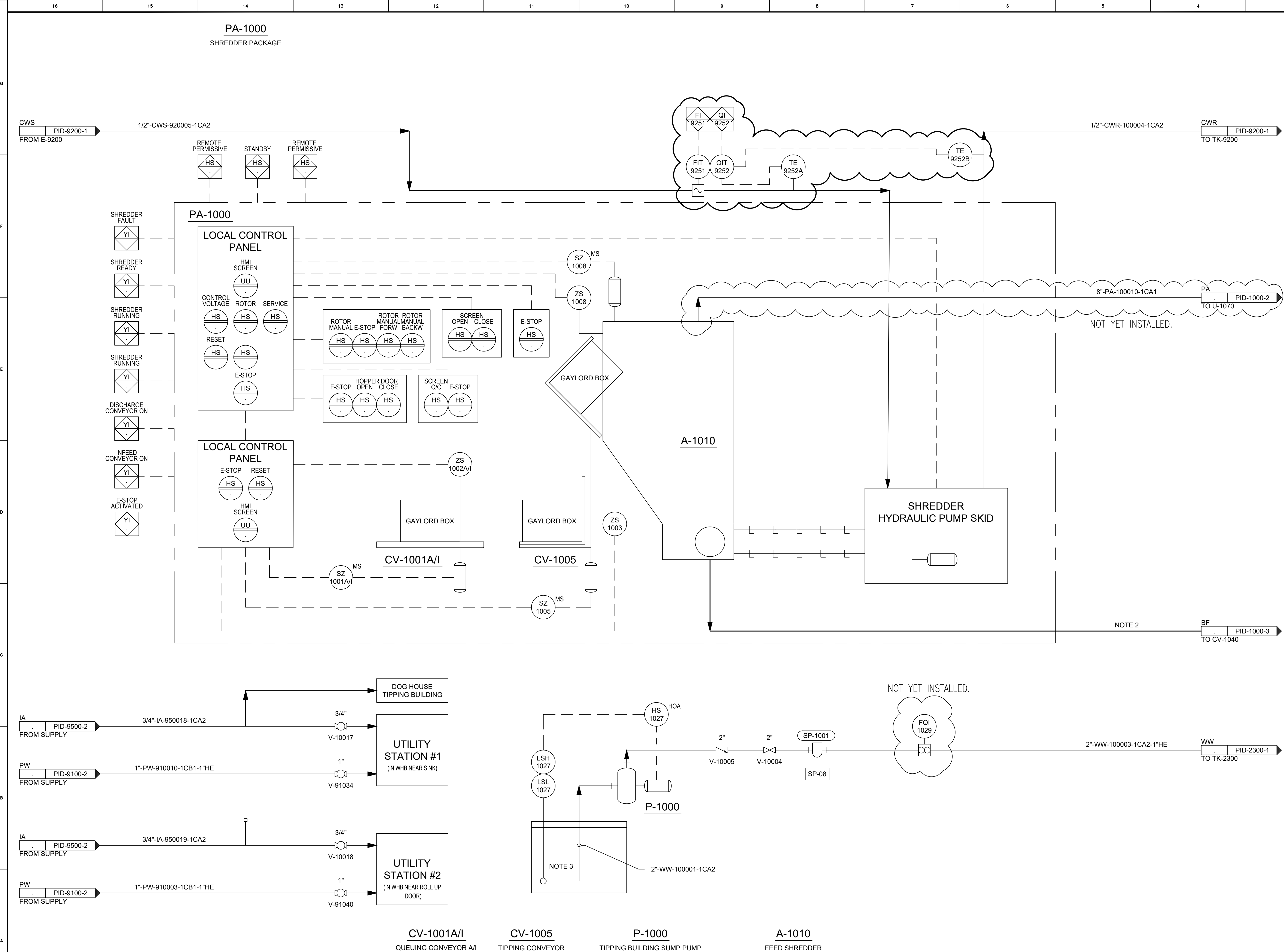
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- NOTES:
- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 - SUMP DRAINS TO BOILER ROOM SUMP SHOWN ON 3041-PID-9500-1.

DRAWN: A BENNETT	
DATE:	
CHECKED:	
DATE:	
ENGR:	
DATE:	
APVD:	
DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID GAS ANALYZER	
SCALE	NONE
DWG NO	3041-PID-0900-8
CADFILE	3041-PID-0900-8.DWG
SHEET NO	1
SHEETS	1
REV	A
DATE	4/20/2023



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- NOTES:
- 1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 - 2. THIS LINE IS NOT A PIPELINE. THE SHREDDER A-1010 IS CONNECT TO CV-1040 WITH A FABRICATED SHEET METAL PIECE.
 - 3. FEED HANDLING BUILDING SUMP.

DRAWN: P SALAS	
DATE:	
CHECKED:	
DATE:	
ENGR:	
DATE:	
APVD:	
DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN

3041	
P&ID	
FEED HANDLING 1 & SHREDDING	
SCALE	NONE
DWG NO	3041-PID-1000-1
CADFILE	3041-PID-1000-1.DWG
SHEET NO	1
SHEETS	1
REV	B
3/21/2024	

U-1070
CYCLONE

U-1075A
PREFILTER HOUSING

U-1075B
PREFILTER HOUSING

InEnTec Inc.

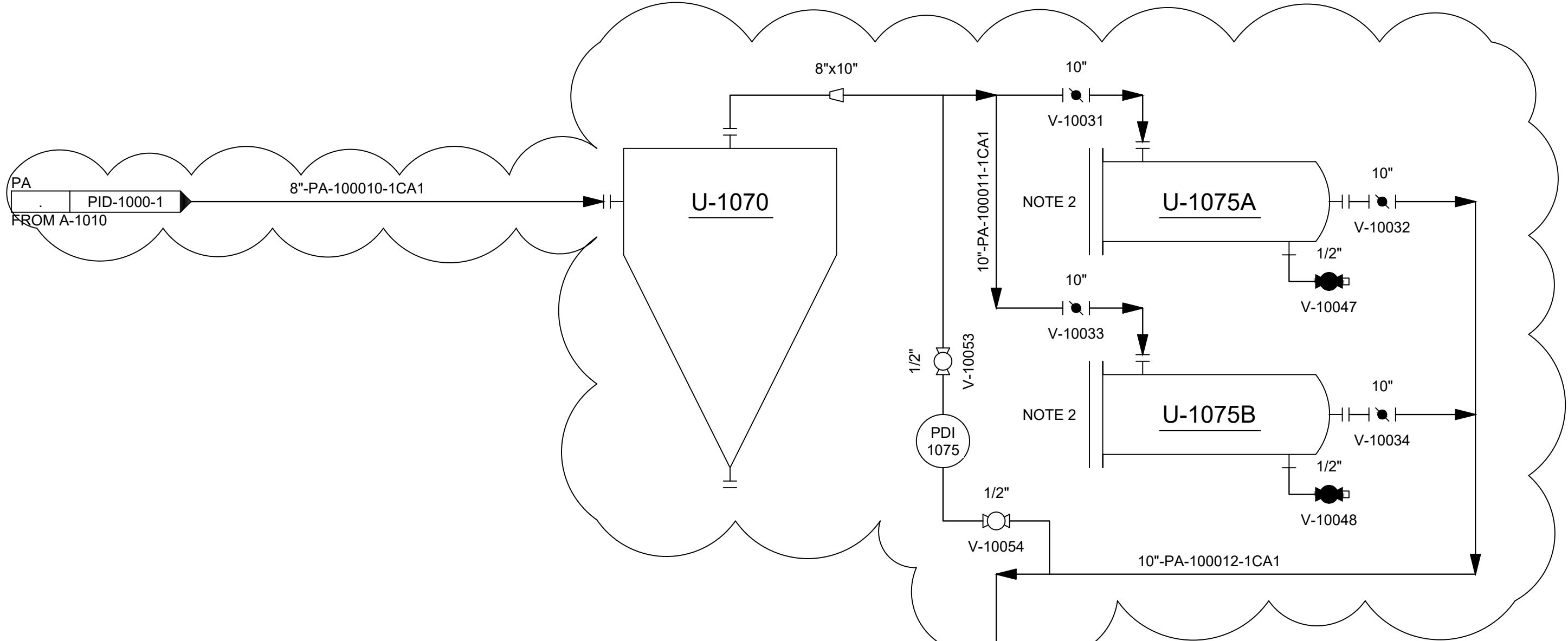
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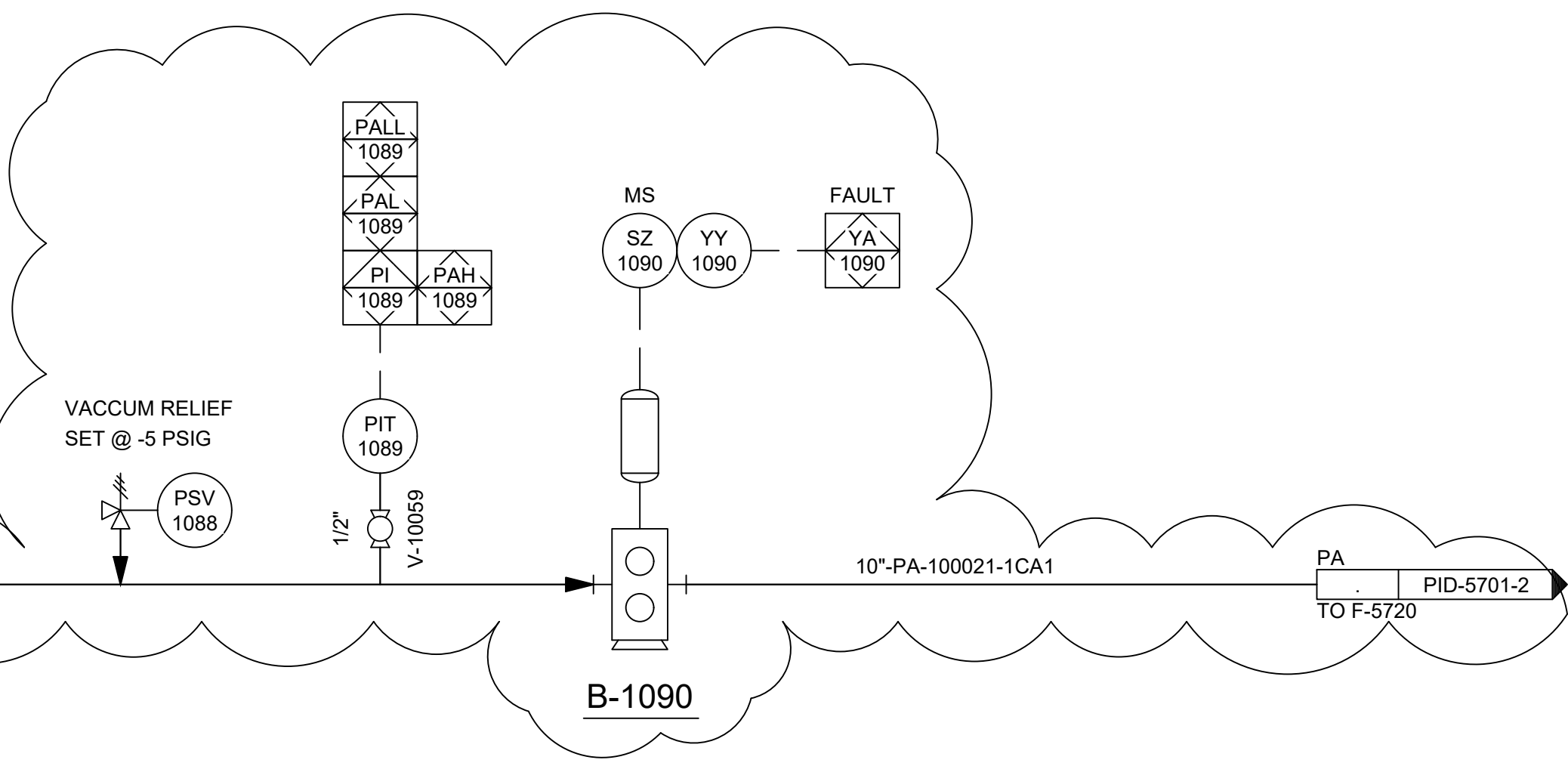
NOTES:

- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
- KEEP END CLEAR FOR DISASSEMBLY AND FILTER ELEMENT REPLACEMENT.

NEEDS TO BE INSTALLED.



NEEDS TO BE INSTALLED.



B-1090

POSITIVE DISPLACEMENT BLOWER

DRAWN: A BENNETT
DATE:

CHECKED:
DATE:

ENGR:
DATE:

APVD:
DATE:

REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041
P&ID
FEED SHREDDER VENTILATION SYSTEM

SCALE NONE

DWG NO 3041-PID-1000-2

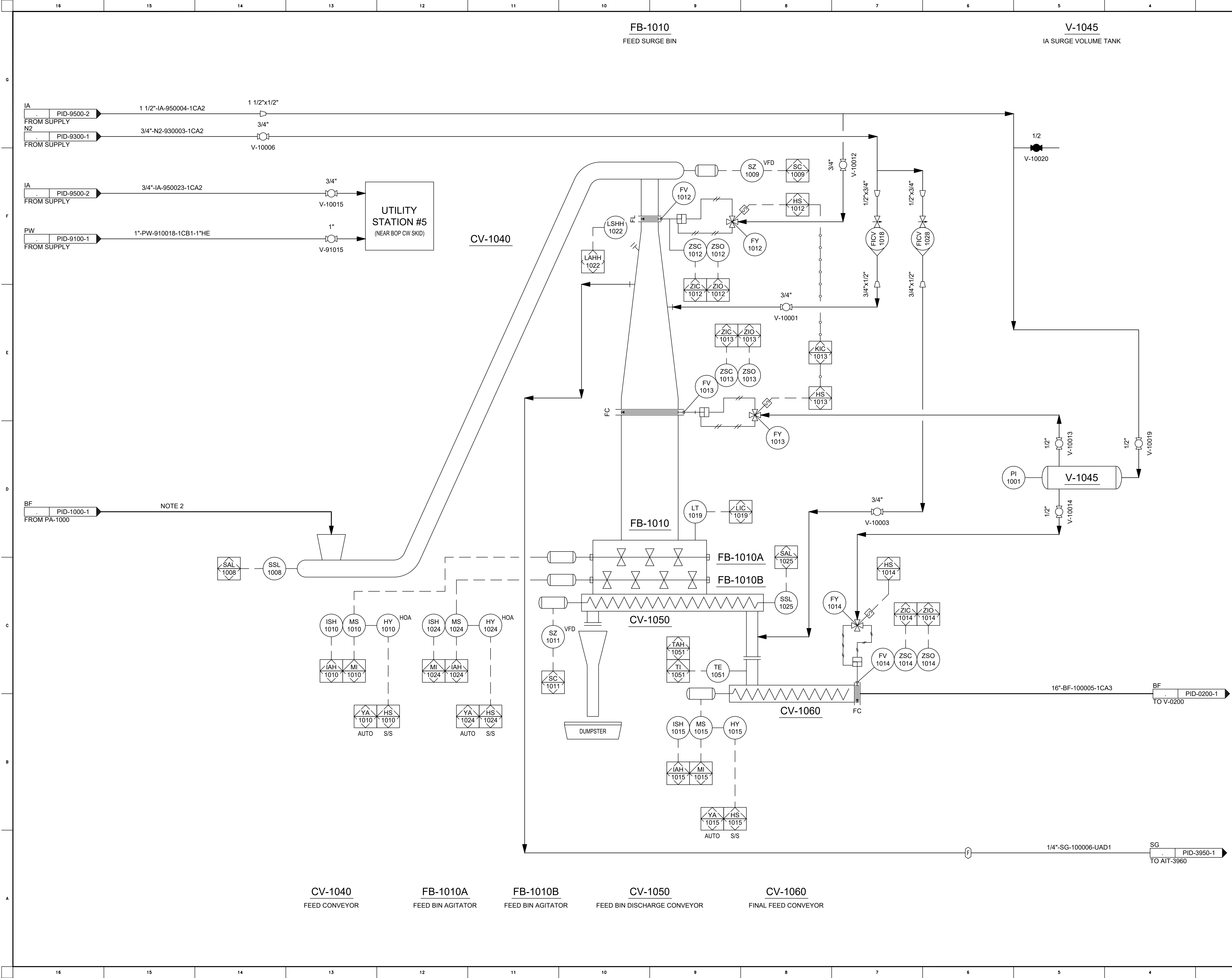
CADFILE 3041-PID-1000-2.DWG

SHEET NO 1

SHEETS 1

REV A

4/20/2023



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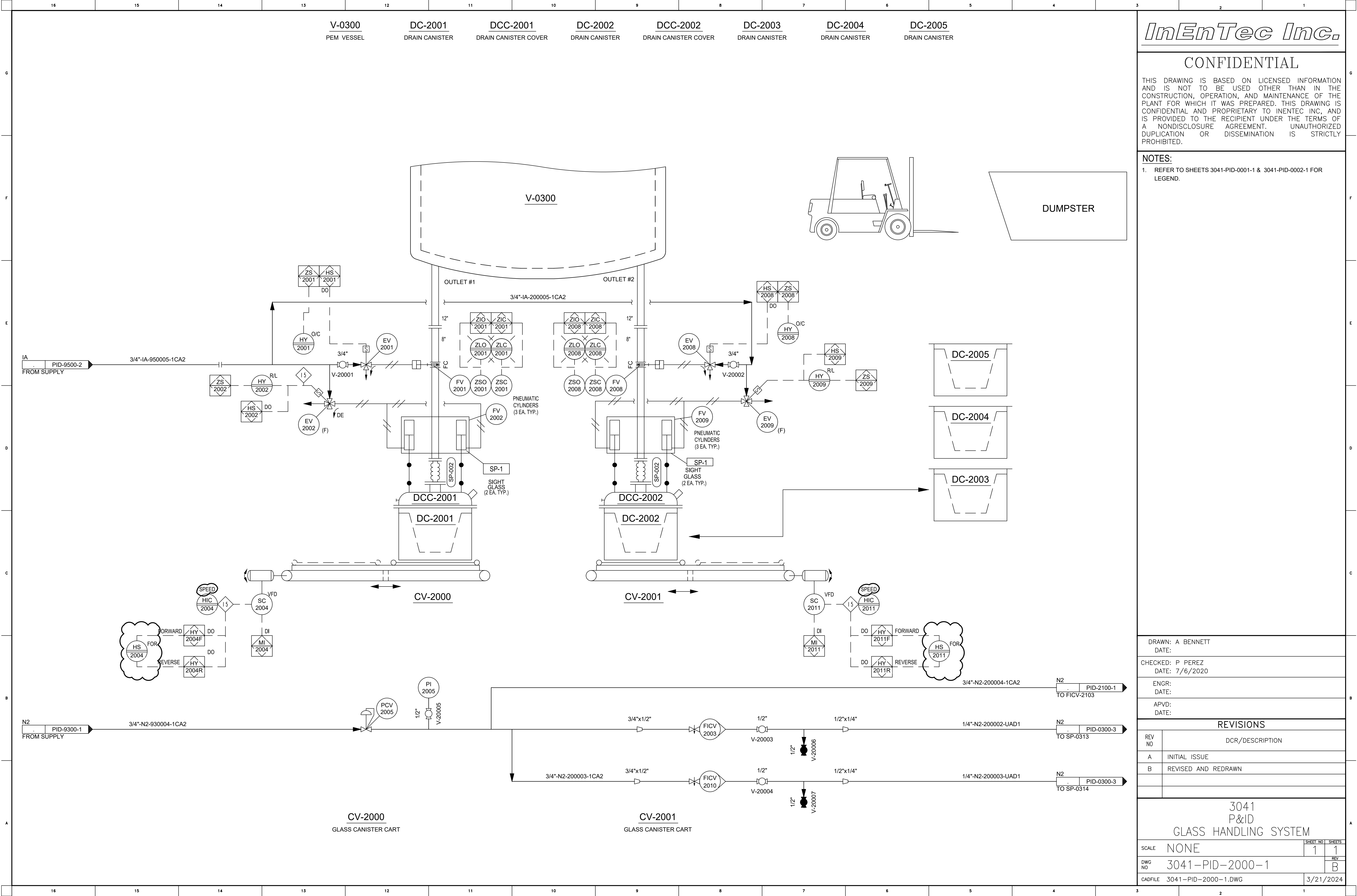
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- NOTES:
- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 - THIS LINE IS NOT A PIPELINE. THE SHREDDER A-1010 CONNECTS TO CV-1040 WITH A FABRICATED SHEET METAL TRANSITION PIECE.

DRAWN: A BENNETT	
DATE:	
CHECKED: P PEREZ	
DATE: 6/30/2020	
ENGR:	
DATE:	
APVD:	
DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID FEED HANDLING 2	
SCALE	NONE
DWG NO	3041-PID-1000-3
CADFILE	3041-PID-1000-3.DWG
SHEET NO	1
SHEETS	1
REV	A
DATE	3/21/2024



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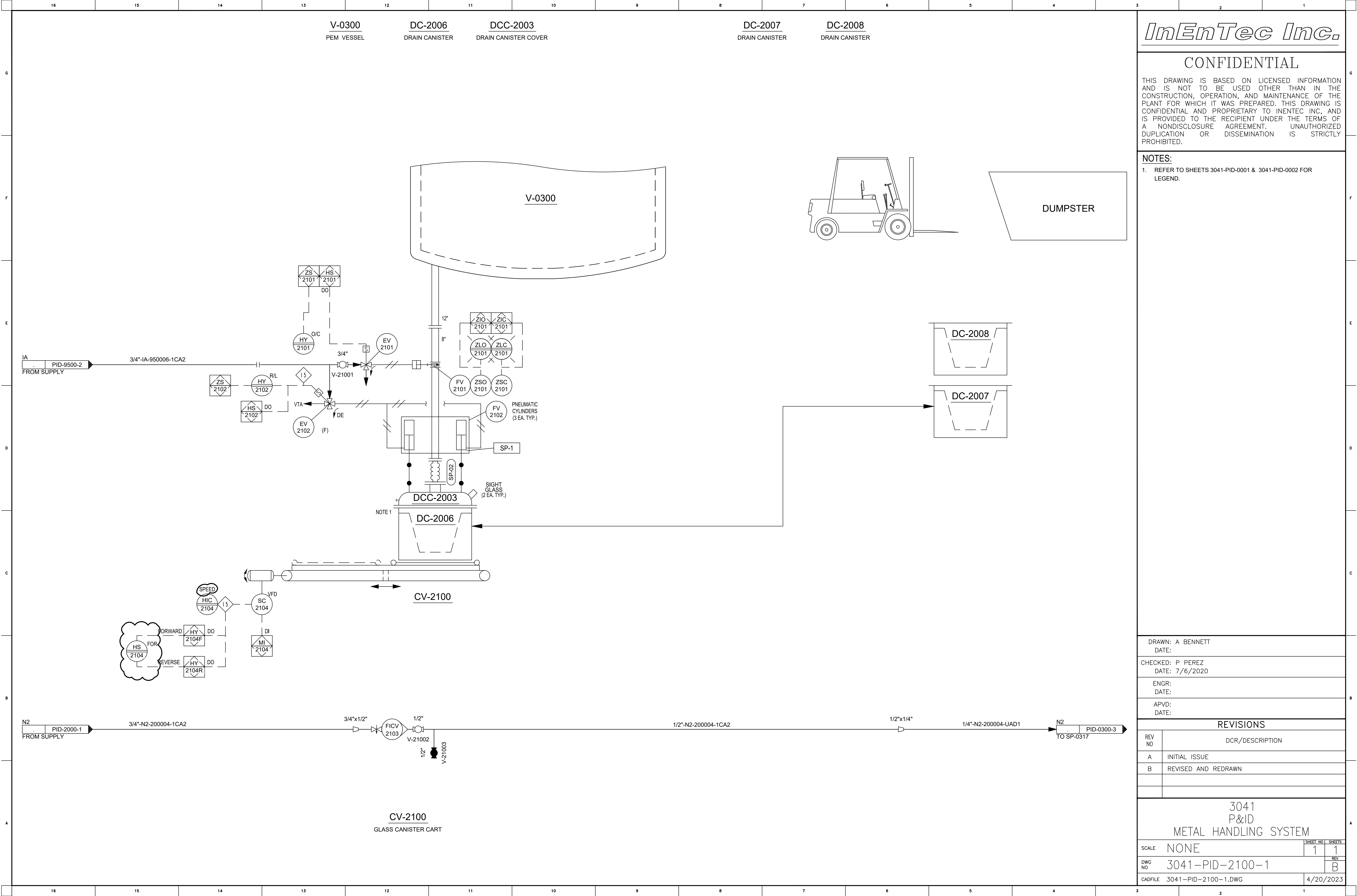
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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT DATE:	
CHECKED: P PEREZ DATE: 7/6/2020	
ENGR: DATE:	
APVD: DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN

3041 P&ID GLASS HANDLING SYSTEM	
SCALE	NONE
DWG NO	3041-PID-2000-1
CADFILE	3041-PID-2000-1.DWG
SHEET NO	1
SHEETS	1
REV	B
3/21/2024	



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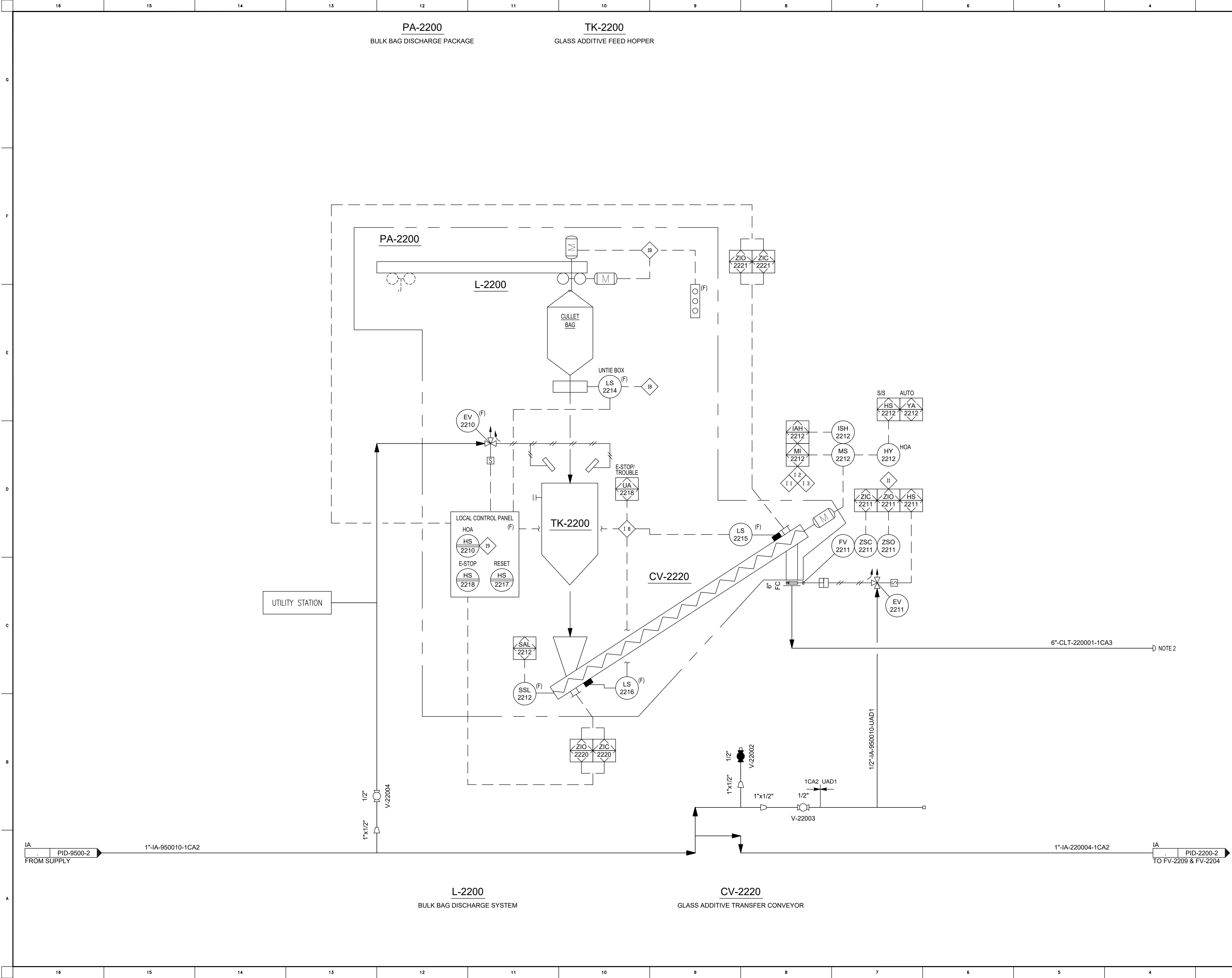
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NOTES:
1. REFER TO SHEETS 3041-PID-0001 & 3041-PID-0002 FOR LEGEND.

DRAWN: A BENNETT
DATE:
CHECKED: P PEREZ
DATE: 7/6/2020
ENGR:
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APVD:
DATE:

REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN

3041 P&ID METAL HANDLING SYSTEM	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-2100-1	REV B
CADFILE 3041-PID-2100-1.DWG	4/20/2023



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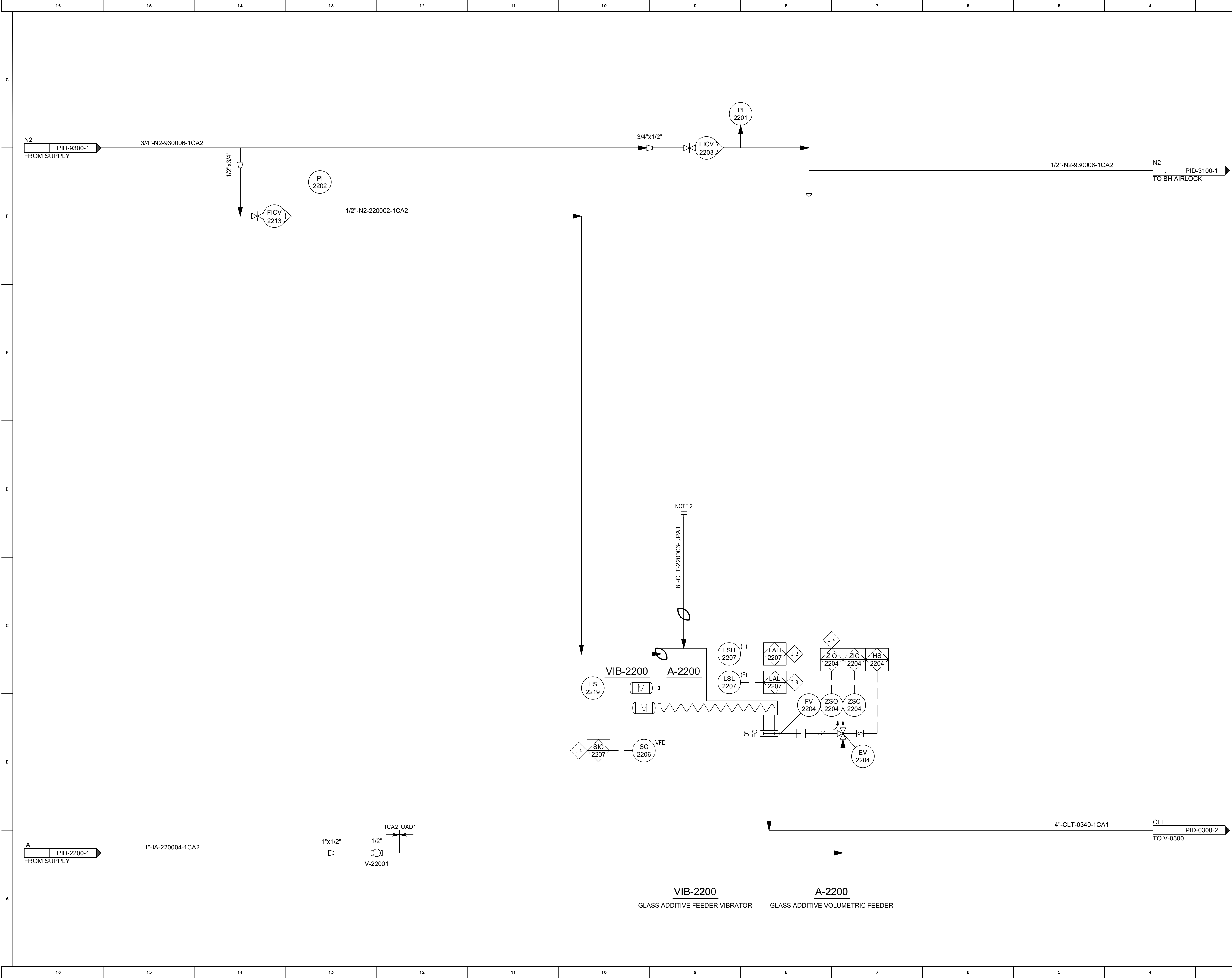
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- NOTES:**
1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 2. THIS LINE IS CURRENTLY DISCONNECTED DUE TO PLANT MODIFICATIONS.

DRAWN: A BENNETT DATE:	
CHECKED: DATE:	
ENGR: DATE:	
APVD: DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID GLASS ADDITIVES CONVEYOR	
SCALE	NONE
DWG NO	3041-PID-2200-1
CADFILE	3041-PID-2200-1.DWG
SHEET NO	1
SHEETS	2
REV	A
4/20/2023	



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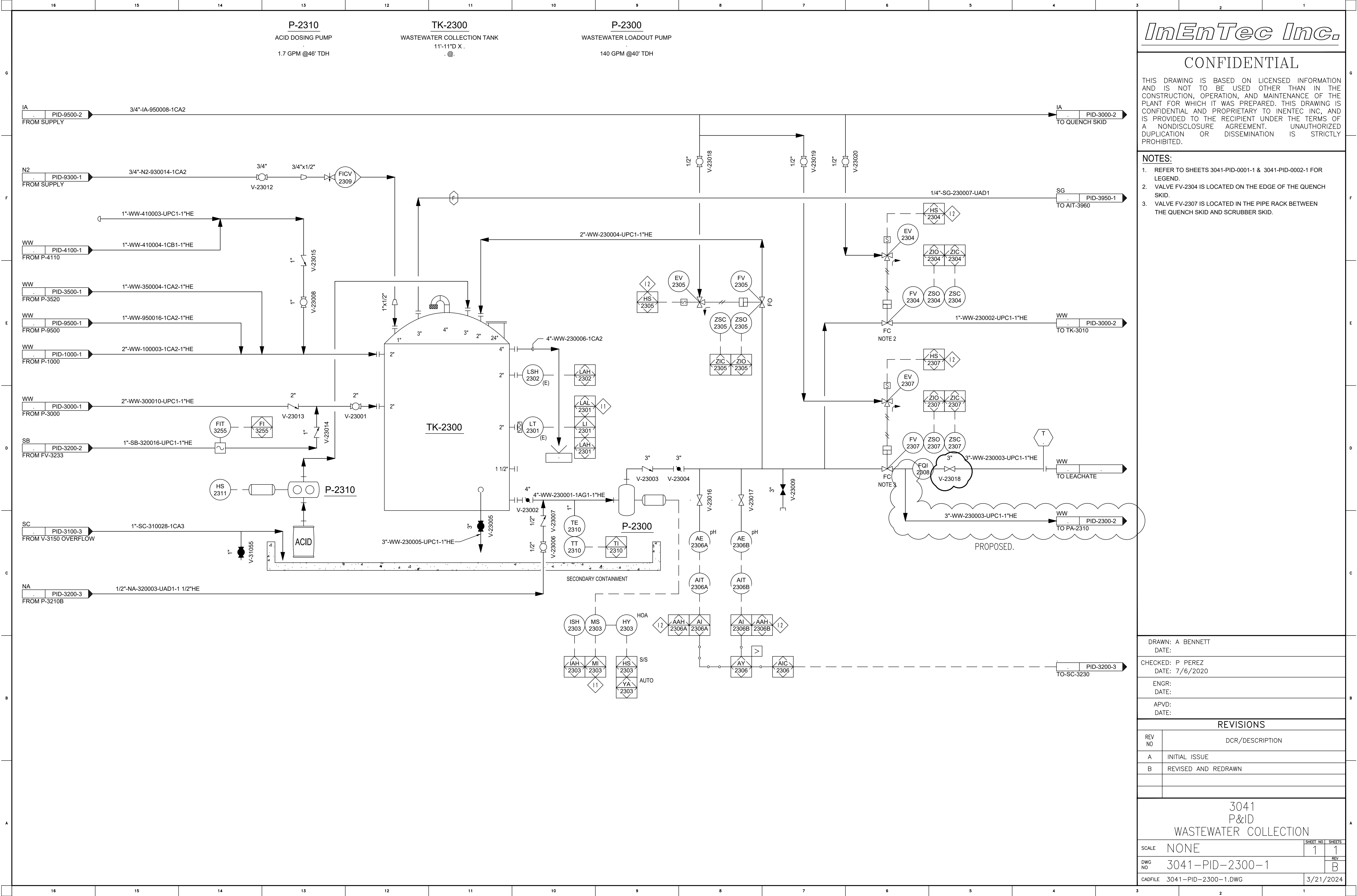
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- NOTES:
- 1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 - 2. ADDITIVE FEEDER IS CURRENTLY FILLED MANUALLY THRU FLANGED CONNECTION DUE TO PLANT MODIFICATIONS.

DRAWN: A BENNETT DATE:	
CHECKED: P PEREZ DATE: 7/6/2020	
ENGR: DATE:	
APVD: DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID GLASS ADDITIVES FEEDER		
SCALE	NONE	SHEET NO 1 SHEETS 1
DWG NO	3041-PID-2200-2	REV A
CADFILE	3041-PID-2200-2.DWG	4/20/2023



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- NOTES:
- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 - VALVE FV-2304 IS LOCATED ON THE EDGE OF THE QUENCH SKID.
 - VALVE FV-2307 IS LOCATED IN THE PIPE RACK BETWEEN THE QUENCH SKID AND SCRUBBER SKID.

DRAWN: A BENNETT
DATE:
CHECKED: P PEREZ
DATE: 7/6/2020

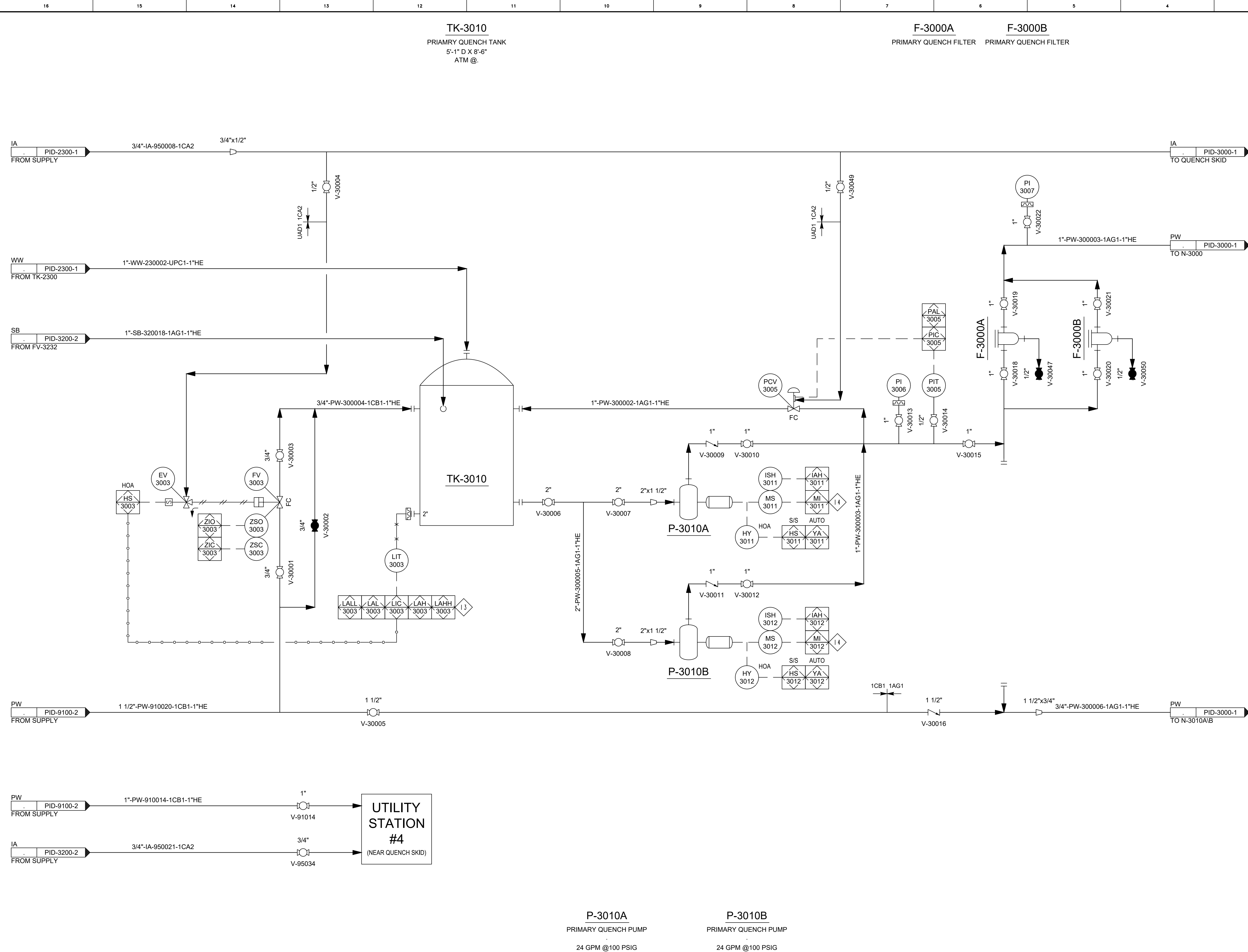
ENGR:
DATE:
APVD:
DATE:

REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN

3041 P&ID WASTEWATER COLLECTION	
SCALE NONE	SHEET NO 1 1
DWG NO 3041-PID-2300-1	REV B
CADFILE 3041-PID-2300-1.DWG	3/21/2024



CADFILE 3041-PID-3000-1.DWG 4/20/202



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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT
DATE:

CHECKED: P PEREZ
DATE: 7/6/2020

ENGR:
DATE:

APVD:
DATE:

REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041
P&ID
PRIMARY QUENCH TANK & PUMPS

SCALE	NONE	SHEET NO	1	SHEETS	1
DWG NO	3041-PID-3000-2			REV	A
CADFILE	3041-PID-3000-2.DWG			4/20/2023	

BH-3100
BAGHOUSE

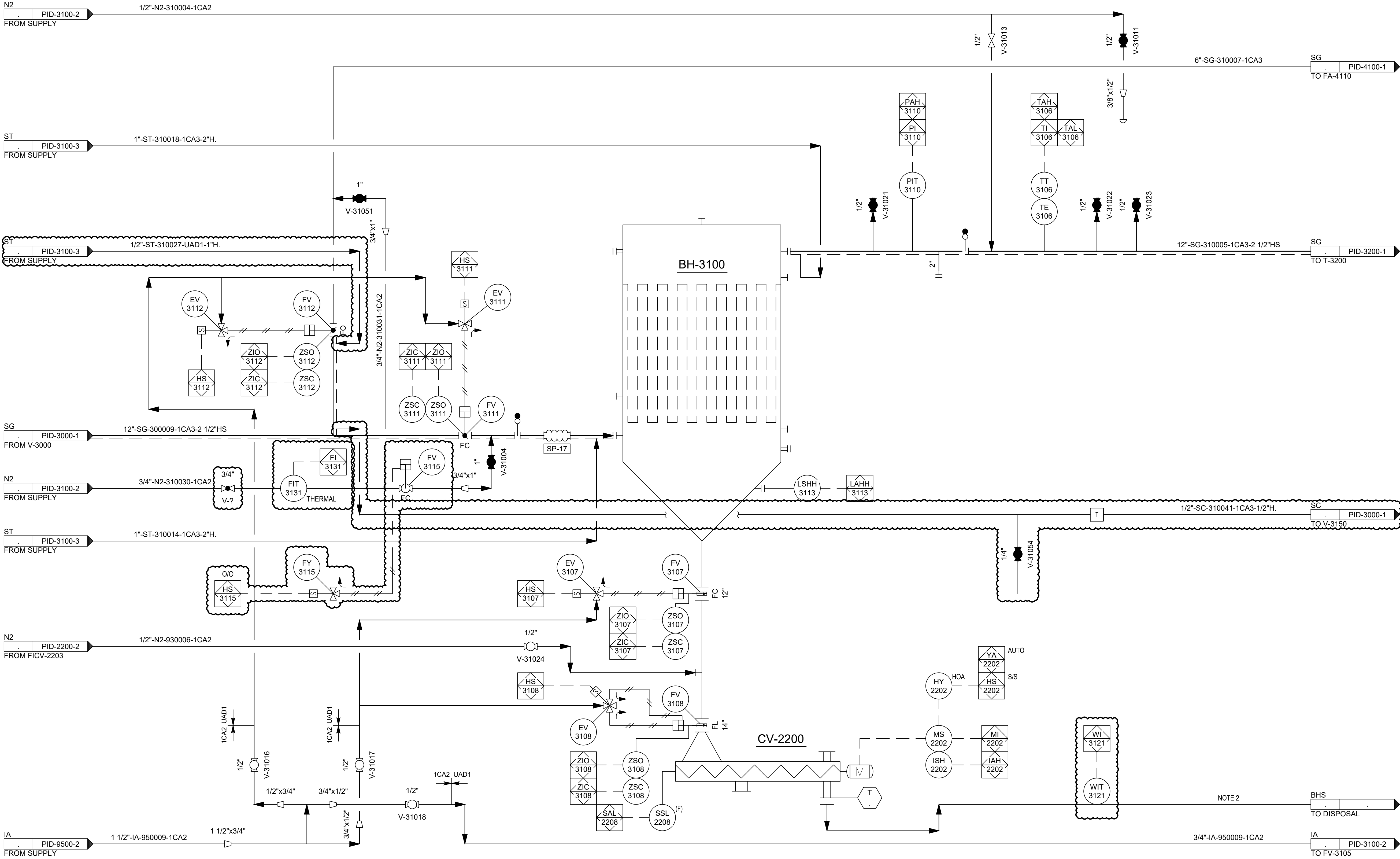
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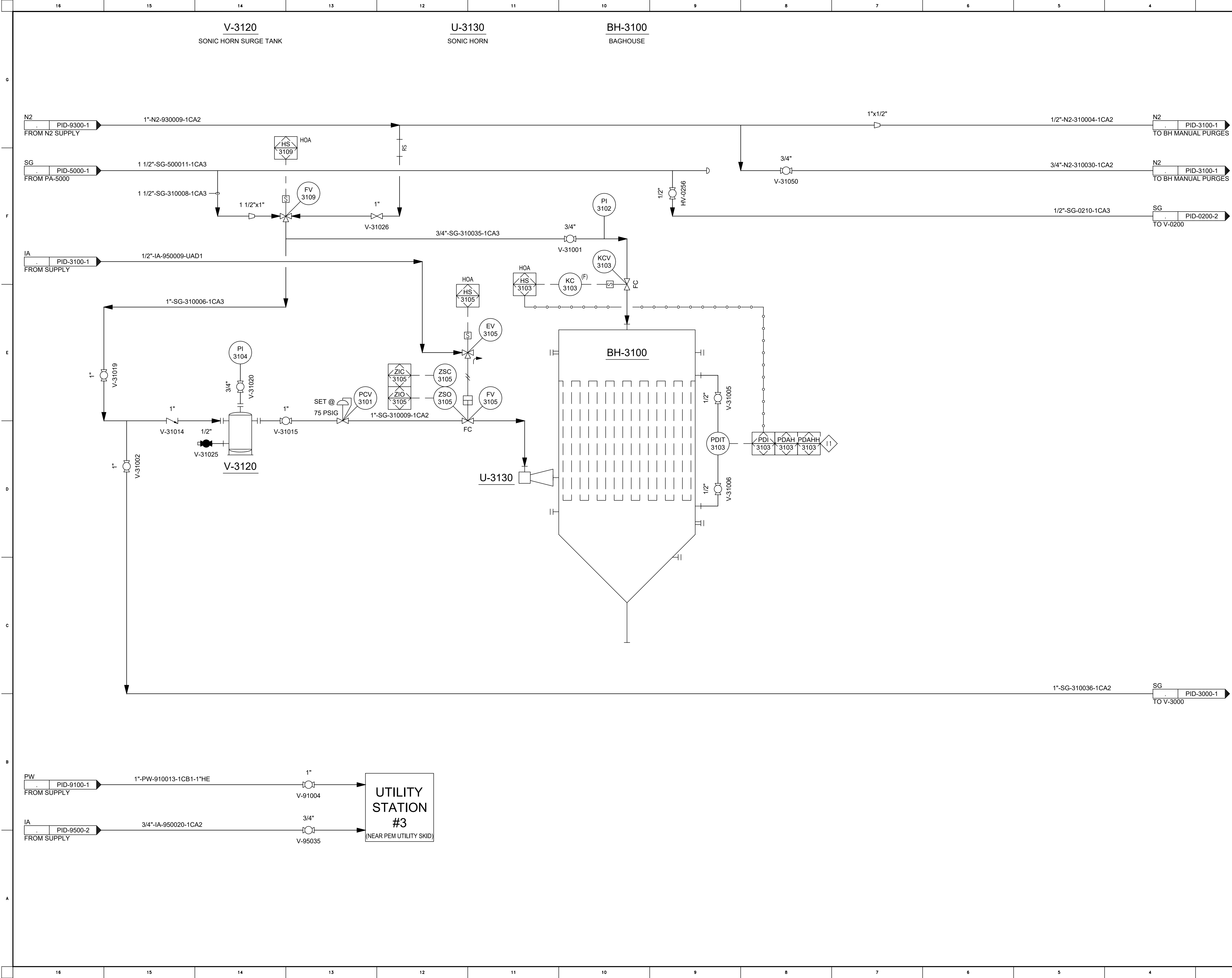
NOTES:

- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
- BAGHOUSE SOLIDS TO BE TRANSFERRED TO SUPER SACKS FOR DISPOSAL.



DRAWN: A BENNETT DATE:	
CHECKED: P PEREZ DATE: 7/6/2020	
ENGR: DATE:	
APVD: DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN

3041 P&ID BAGHOUSE	
SCALE	NONE
DWG NO	3041-PID-3100-1
CADFILE	3041-PID-3100-1.DWG
SHEET NO	1
SHEETS	1
REV	B
DATE	3/21/2024



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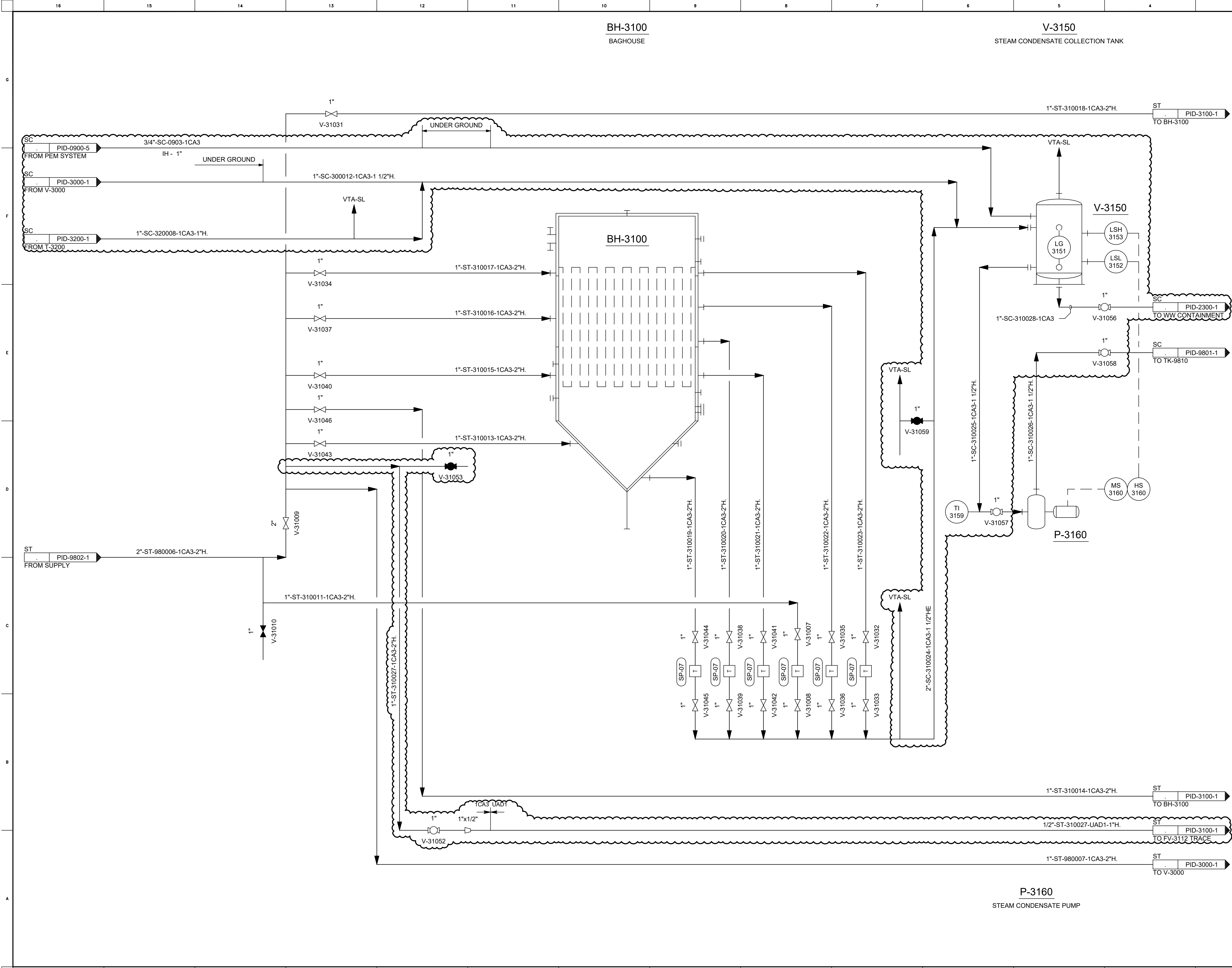
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NOTES:
1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT DATE:	
CHECKED: P PEREZ DATE: 7/6/2020	
ENGR: DATE:	
APVD: DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN

3041 P&ID BAGHOUSE	
SCALE	NONE
DWG NO	3041-PID-3100-2
CADFILE	3041-PID-3100-2.DWG
SHEET NO	1
SHEETS	1
REV	B
DATE	4/20/2023



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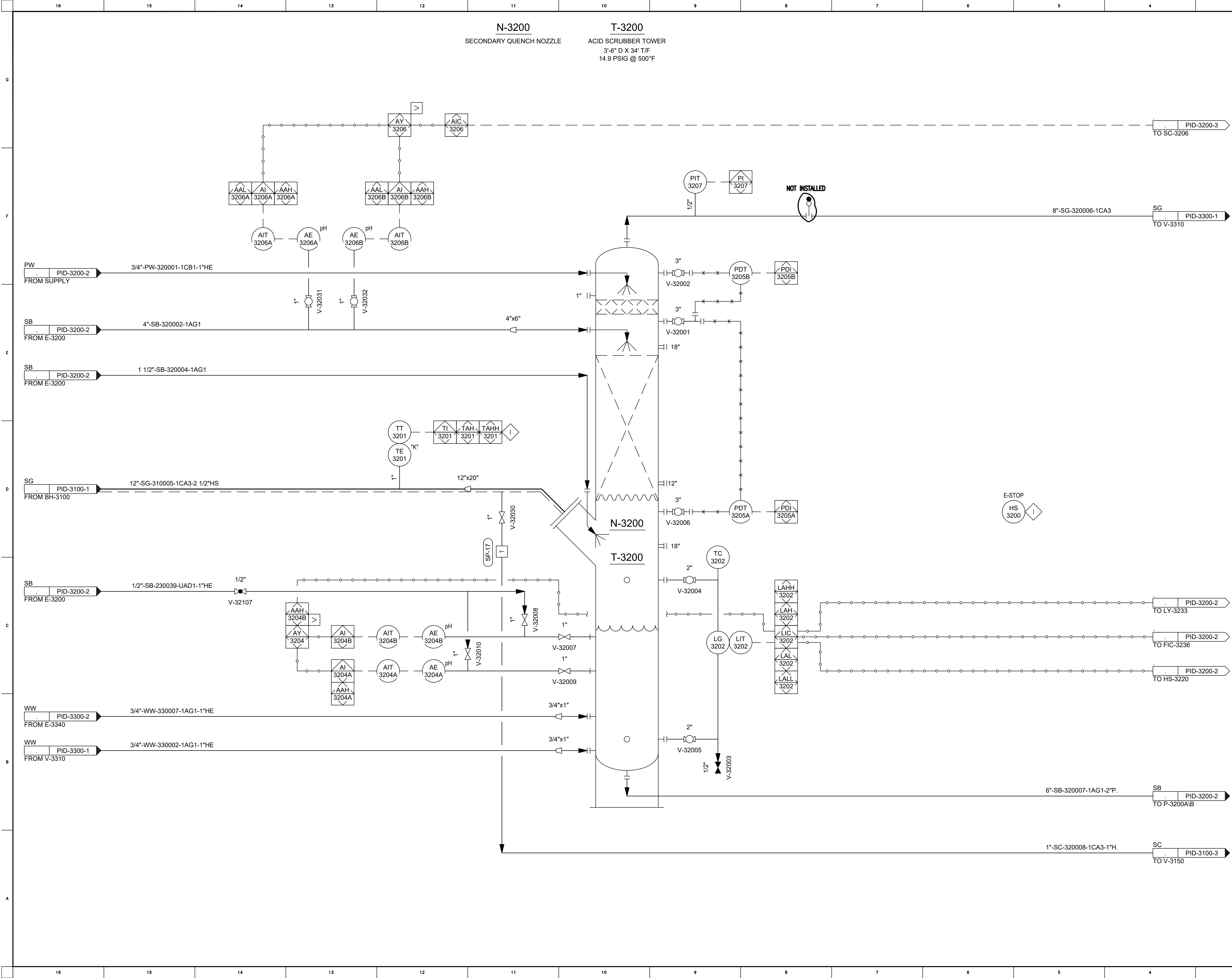
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NOTES:
1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT DATE:	
CHECKED: P PEREZ DATE: 7/6/2020	
ENGR: DATE:	
APVD: DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID BAGHOUSE HEAT TRACE	
SCALE: NONE	SHEET NO: 1 SHEETS: 1
DWG NO: 3041-PID-3100-3	REV: A
CADFILE: 3041-PID-3100-3.DWG	4/20/2023



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NOTES:
1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT
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REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

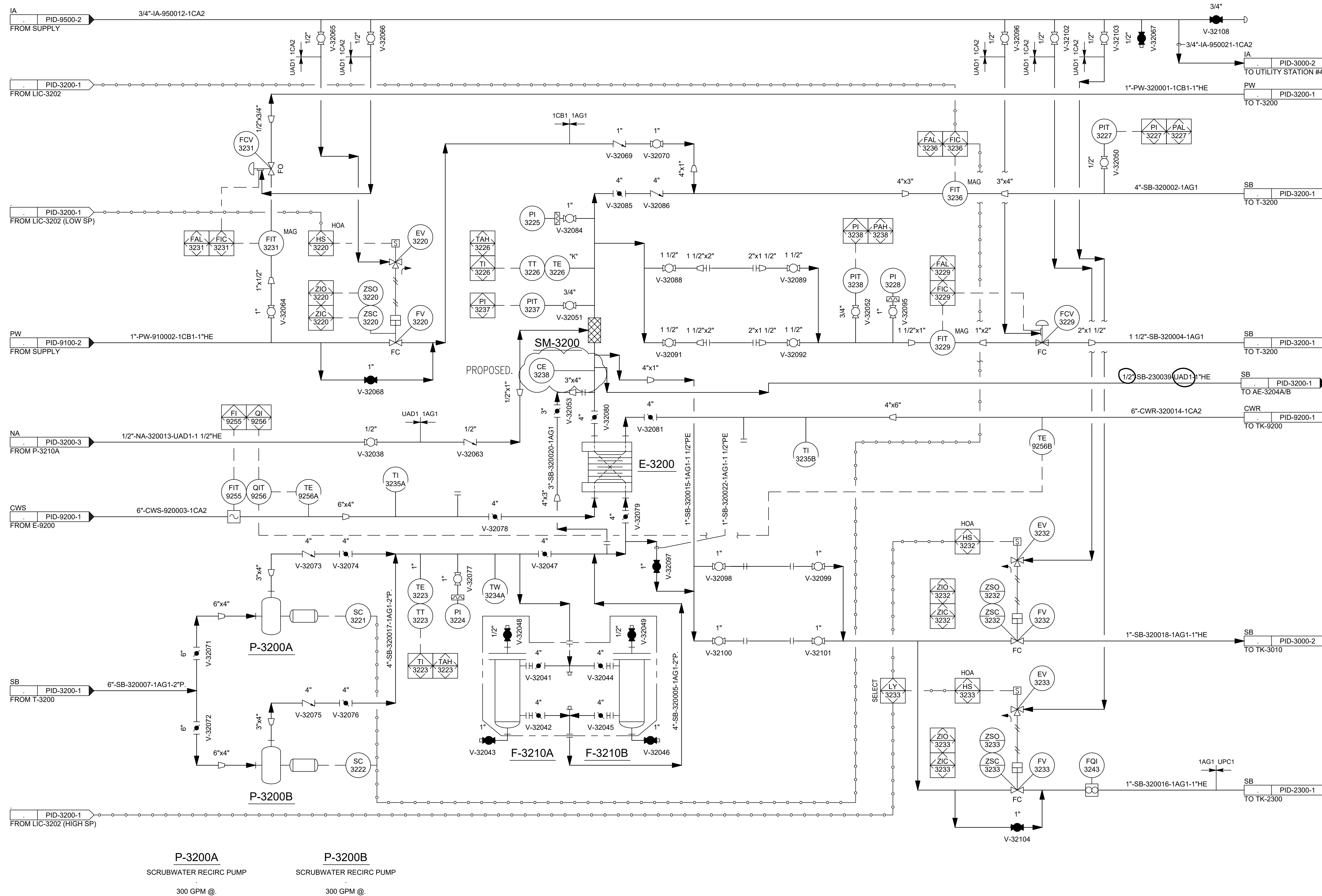
3041
P&ID
SCRUBBER

SCALE	NONE	SHEET NO	1	SHEETS	1
DWG NO	3041-PID-3200-1	REV	A		
CADFILE	3041-PID-3200-1.DWG				3/21/2024

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NOTES

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.



3041
P&ID
SCRUBBER PUMP AND PIPING

SCALE	NONE	SHEET NO	1	SHEETS	1
DWG NO	3041-PID-3200-2			REV	B
CADFILE	3041-PID-3200-2.DWG		3/21/2024		

TK-3210
CAUSTIC TANK
9'-11" D.
5250 GAL @140° F

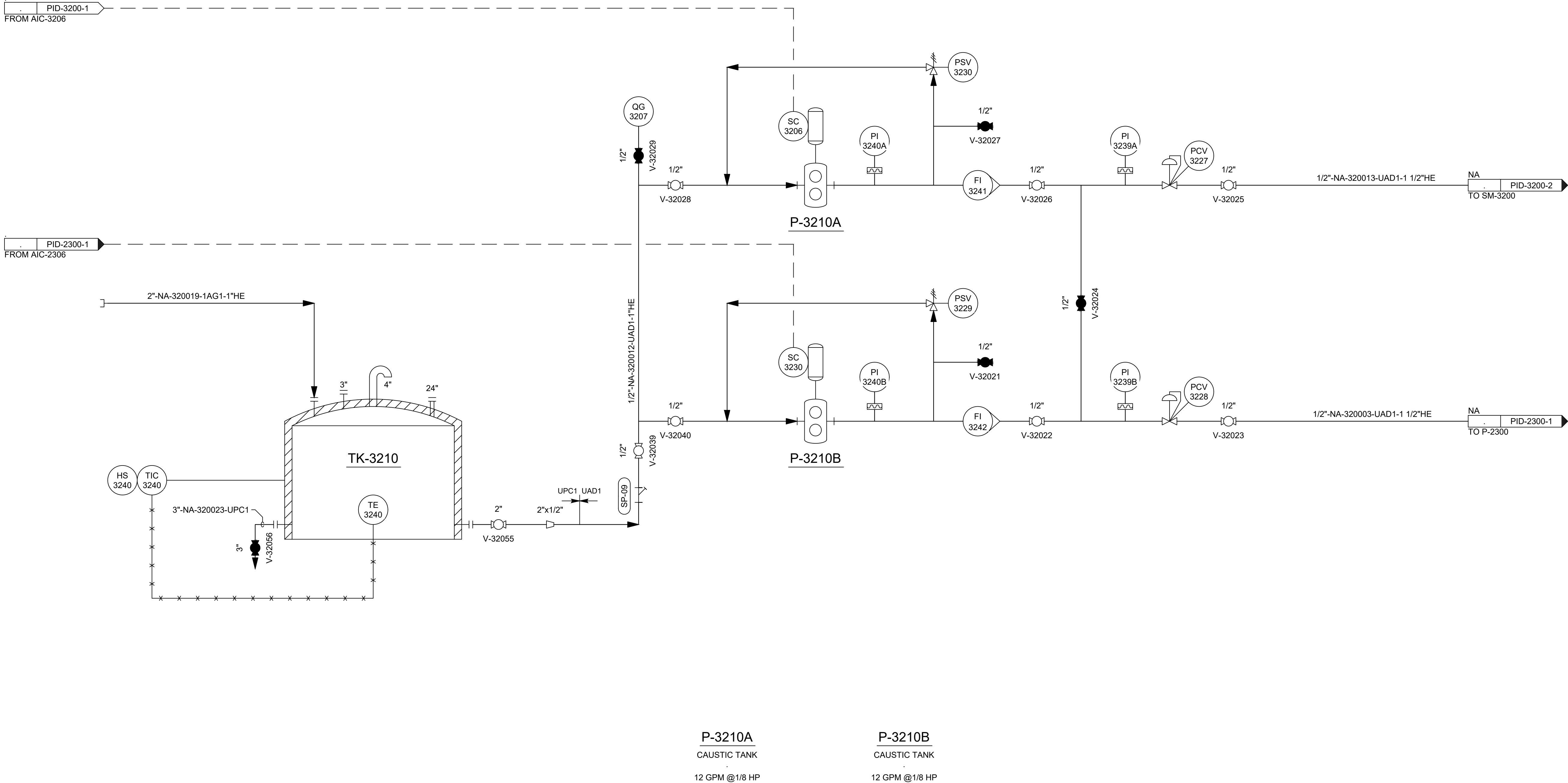
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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.



DRAWN: A BENNETT
DATE:

CHECKED: P PEREZ
DATE: 7/6/2020

ENGR:
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REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN

3041
P&ID
CAUSTIC SUPPLY

SCALE	NONE	SHEET NO	1	SHEETS	1
DWG NO	3041-PID-3200-3	REV	B		
CADFILE	3041-PID-3200-3.DWG				4/20/2023

V-3310
SYNGAS BLOWER KNOCK OUT POT
1' 8" OD X 3' 9" OAL
FV & 15 PSIG @ 500 F

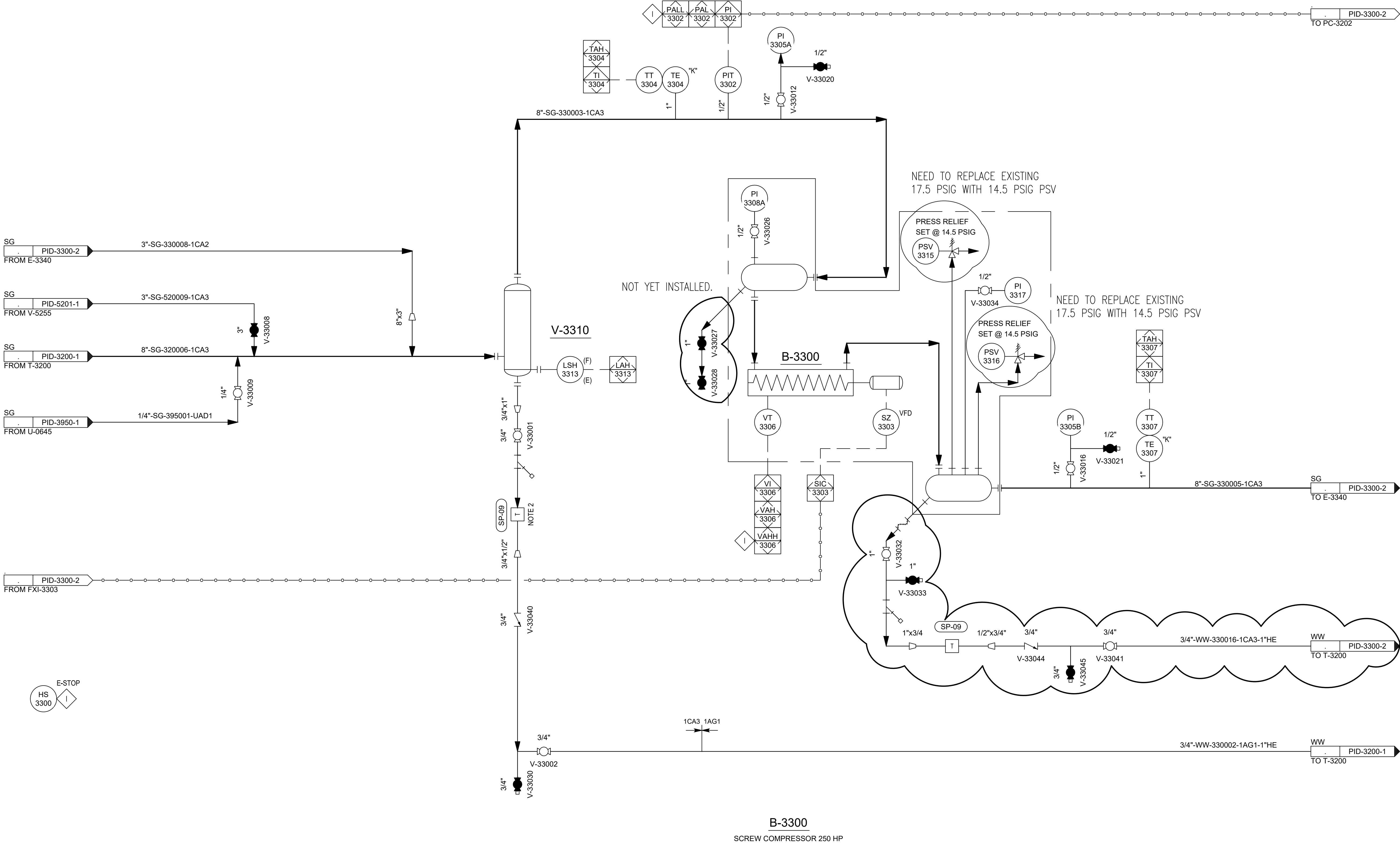
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NOTES:

- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
- TRAP MUST SAME HEIGHT OR HIGHER THAN HEIGH LEVEL ALARM ON T-3200.

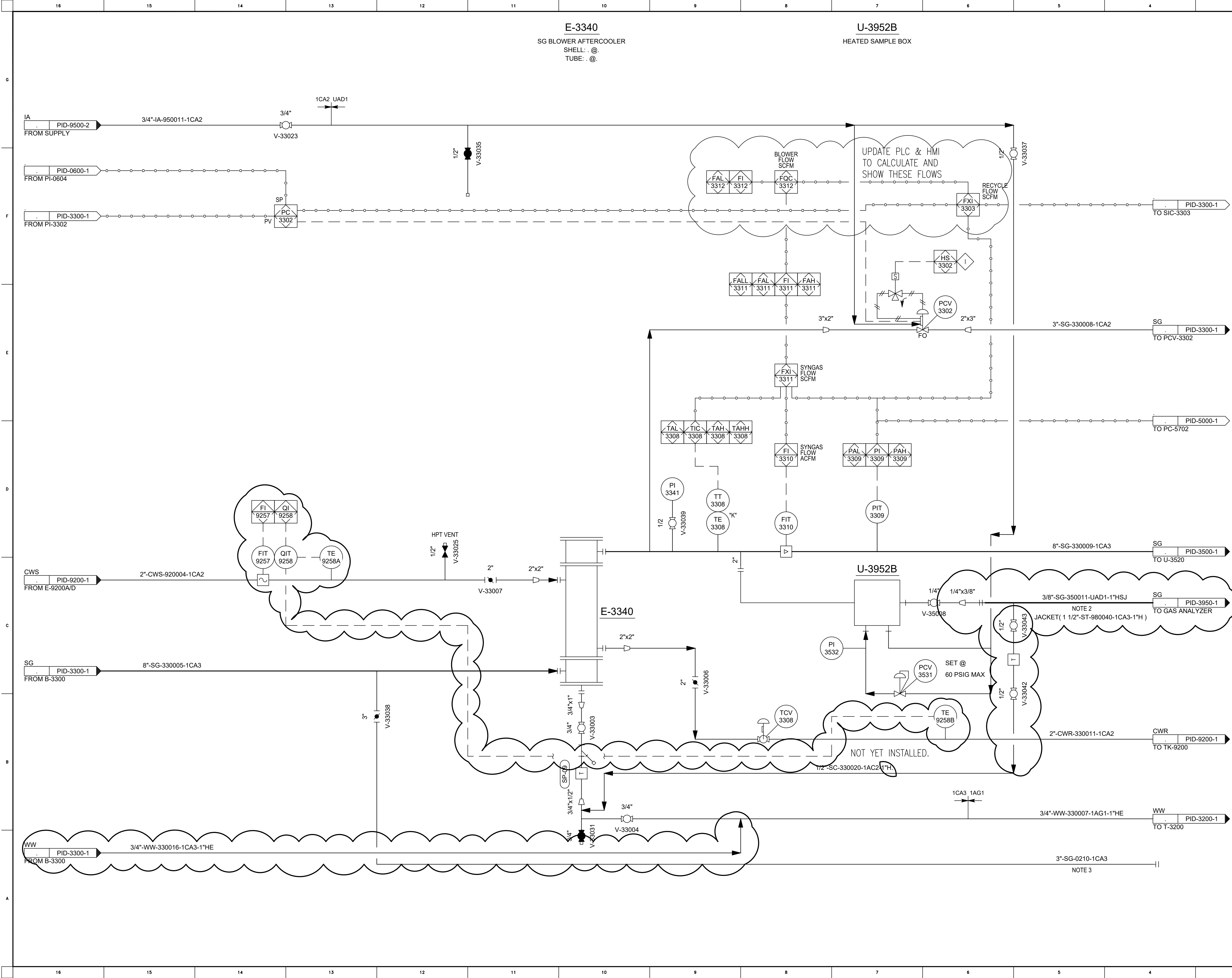


DRAWN: A BENNETT	
DATE:	
CHECKED: P PEREZ	
DATE: 7/6/2020	
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REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN
C	REVISED AND REDRAWN

3041 P&ID SYNGAS BLOWER 1	
SCALE	NONE
DWG NO	3041-PID-3300-1
CADFILE	3041-PID-3300-1.DWG

SHEET NO	SHEETS
1	1
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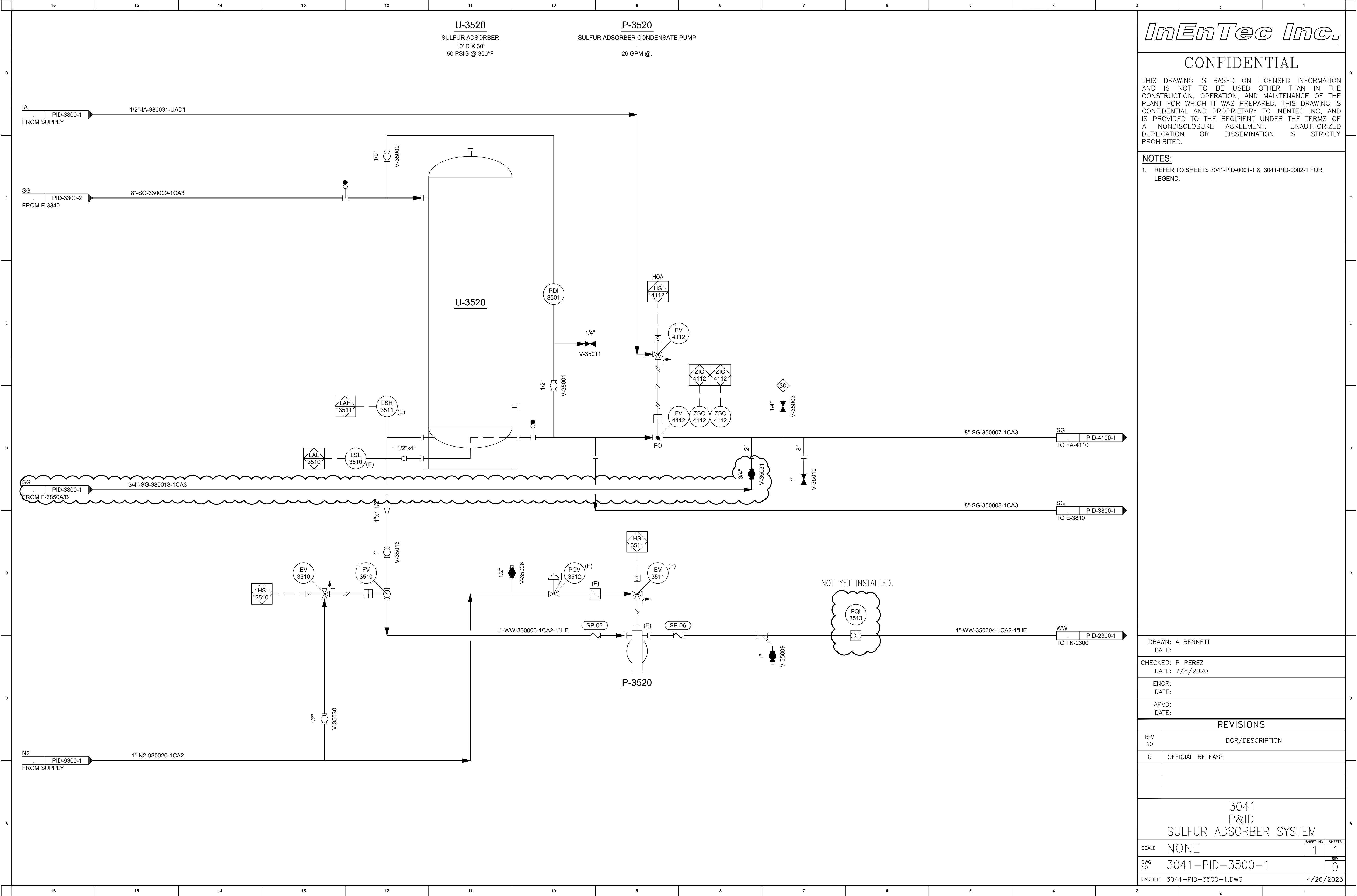
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- NOTES:
- 1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 - 2. 1-1/2" STEAM JACKETED LINE.
 - 3. CURRENTLY THIS LINE ENDS NEAR THE DDG UPPER DECK.

DRAWN: A BENNETT	
DATE:	
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APVD:	
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REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN

3041 P&ID SYNGAS BLOWER 2	
SCALE	NONE
DWG NO	3041-PID-3300-2
CADFILE	3041-PID-3300-2.DWG
SHEET NO	1
SHEETS	1
REV	B
DATE	4/20/2023



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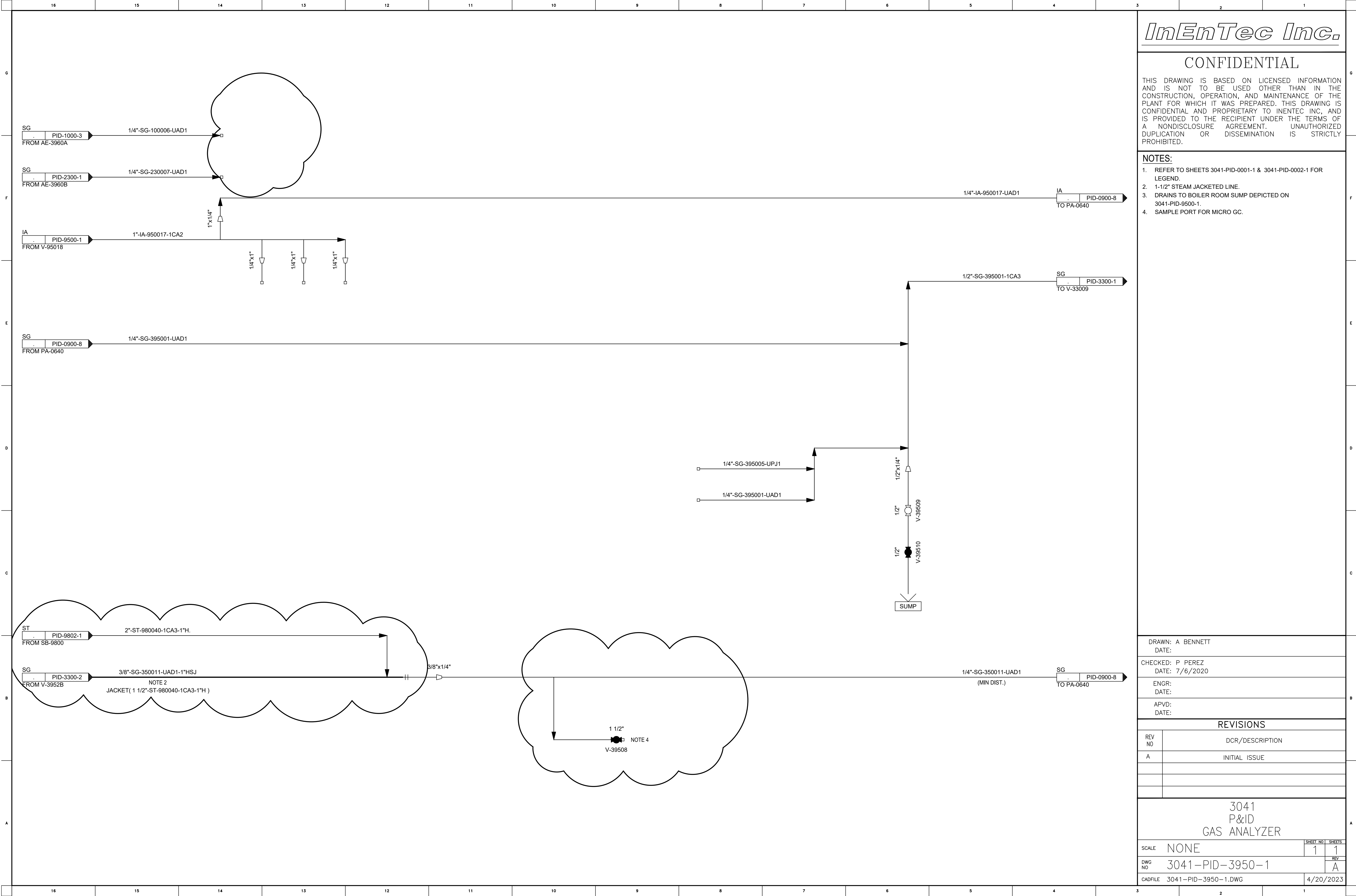
- NOTES:
- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

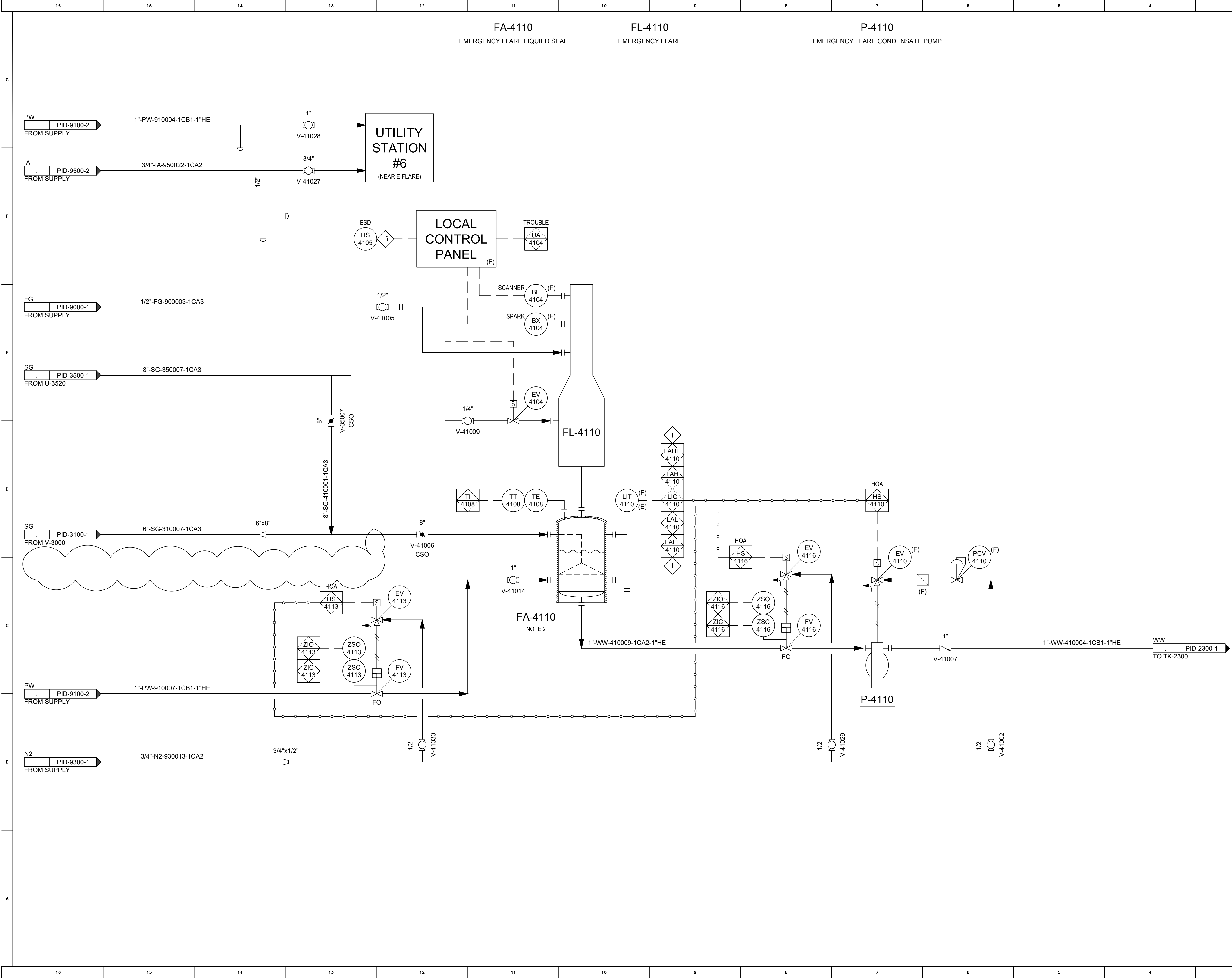
DRAWN: A BENNETT	
DATE:	
CHECKED: P PEREZ	
DATE: 7/6/2020	
ENGR:	
DATE:	
APVD:	
DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
0	OFFICIAL RELEASE

3041	
P&ID	
SULFUR ADSORBER SYSTEM	
SCALE	NONE
DWG NO	3041-PID-3500-1
CADFILE	3041-PID-3500-1.DWG
SHEET NO	1
SHEETS	1
REV	0
4/20/2023	



CADFILE 3041-PID-3800-1.DWG 4/20/2023





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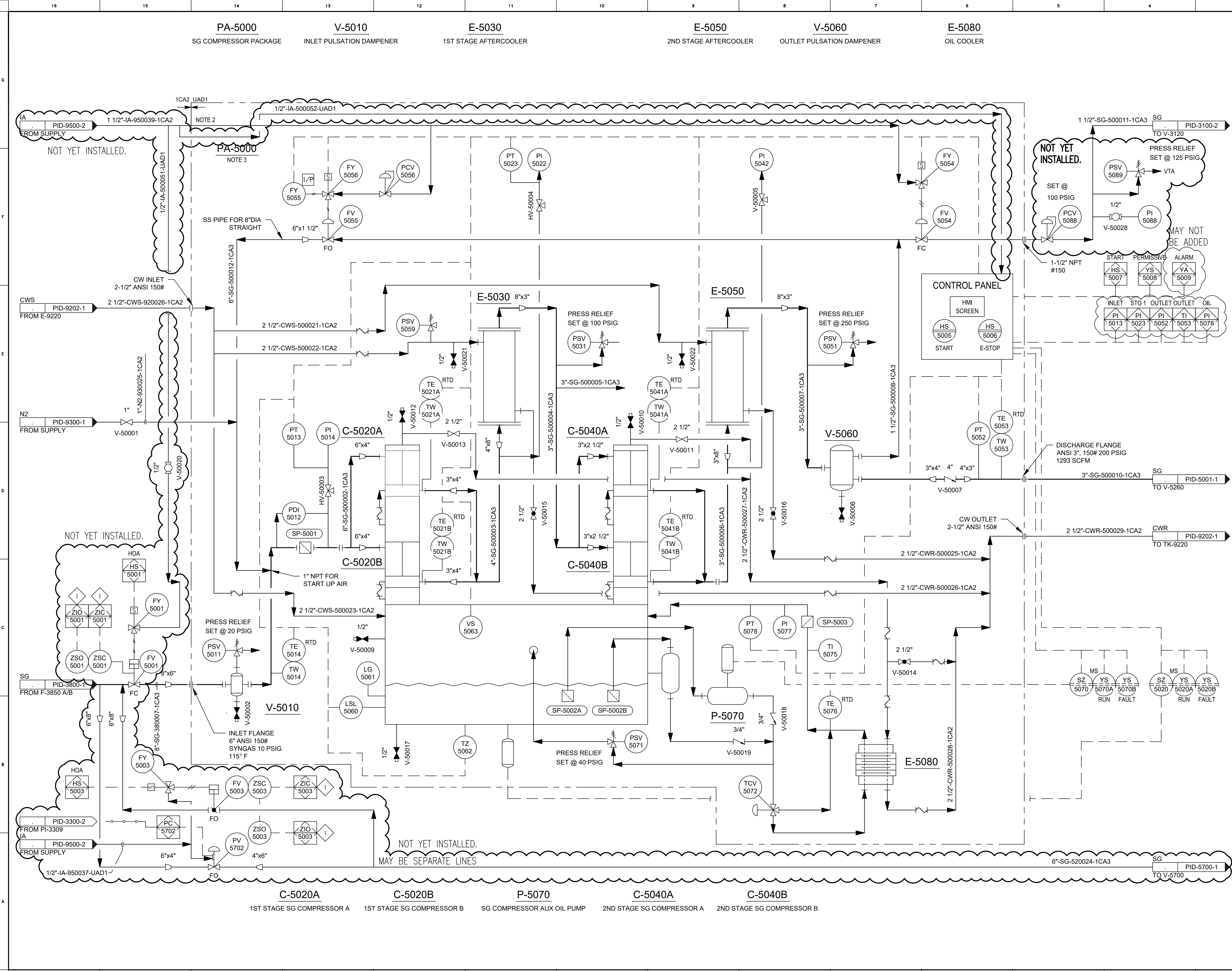
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- NOTES:
- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 - LIQUID SEAL POT IS ELECTRIC TRACED FOR FREEZE PROTECTION.

DRAWN: A BENNETT	
DATE:	
CHECKED: P PEREZ	
DATE: 7/6/2020	
ENGR:	
DATE:	
APVD:	
DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID FLARE	
SCALE	NONE
DWG NO	3041-PID-4100-1
CADFILE	3041-PID-4100-1.DWG
SHEET NO	1
SHEETS	1
REV	A
4/20/2023	



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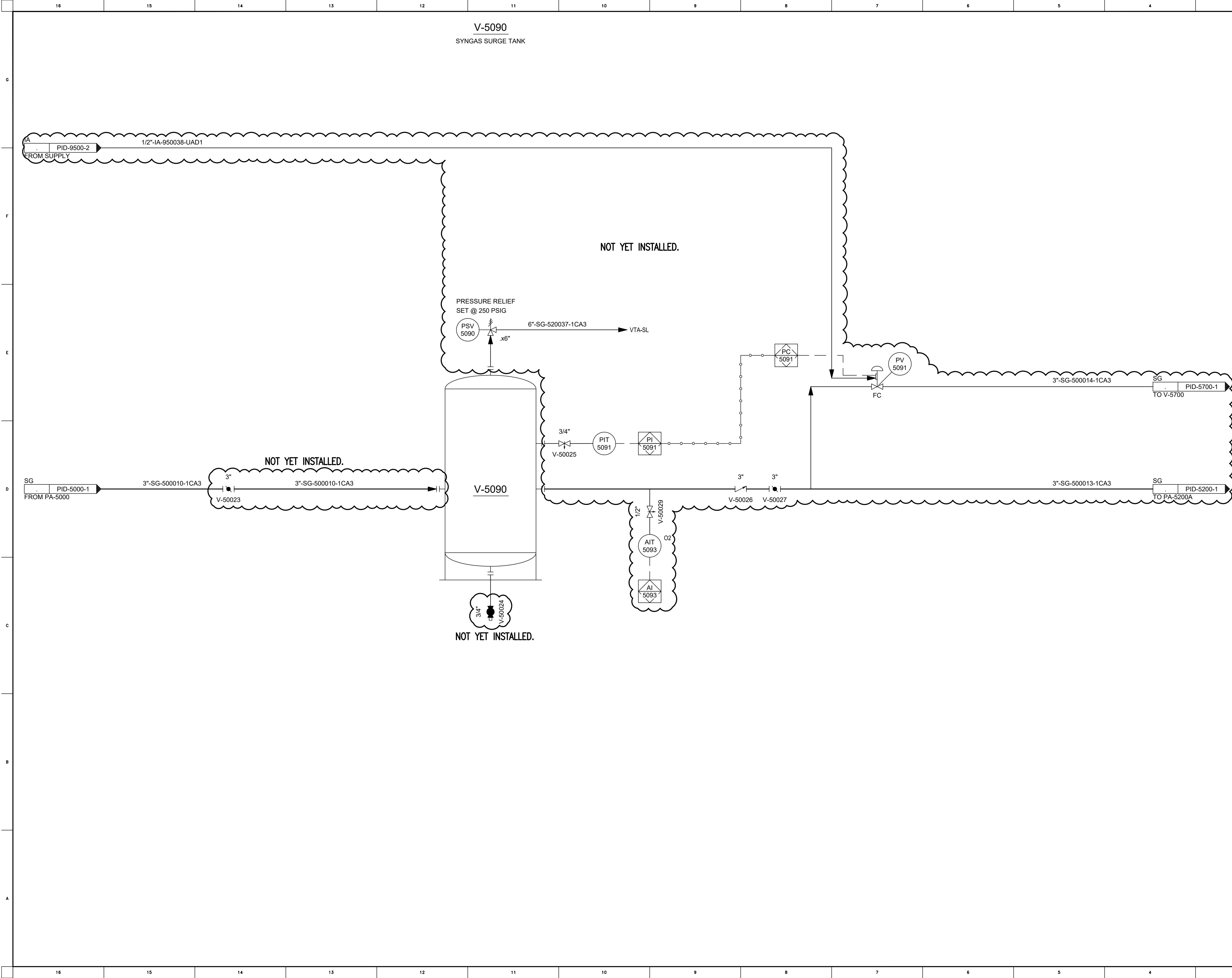
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- NOTES:
1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 2. TUBE TO EDGE OF BEDPLATE WITH 1/4" FEMALE NPT CONNECTION, INSTRUMENT AIR 80-125 PSI, SEE INSTALLATION ARRANGEMENT.
 3. PA-5000 HAS ITS OWN INDEPENDENT PLC.

DRAWN: P. SALAS	
DATE:	
CHECKED: P. PEREZ	
DATE: 6/30/2020	
ENGR:	
DATE:	
APVD:	
DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN
C	REVISED AND REDRAWN

3041	
P&ID	
SYNGAS COMPRESSOR	
SCALE	NONE
DWG NO	3041-PID-5000-1
CADFILE	3041-PID-5000-1.DWG
SHEET NO	1
SHEETS	1
REV	C
4/20/2023	



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NOTES:
1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT
DATE:

CHECKED: P PEREZ
DATE: 6/30/2020

ENGR:
DATE:

APVD:
DATE:

REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041
P&ID
SYNGAS SURGE TANK

SCALE NONE

DWG NO 3041-PID-5001-1

CADFILE 3041-PID-5001-1.DWG

SHEET NO 1

SHEETS 1

REV A

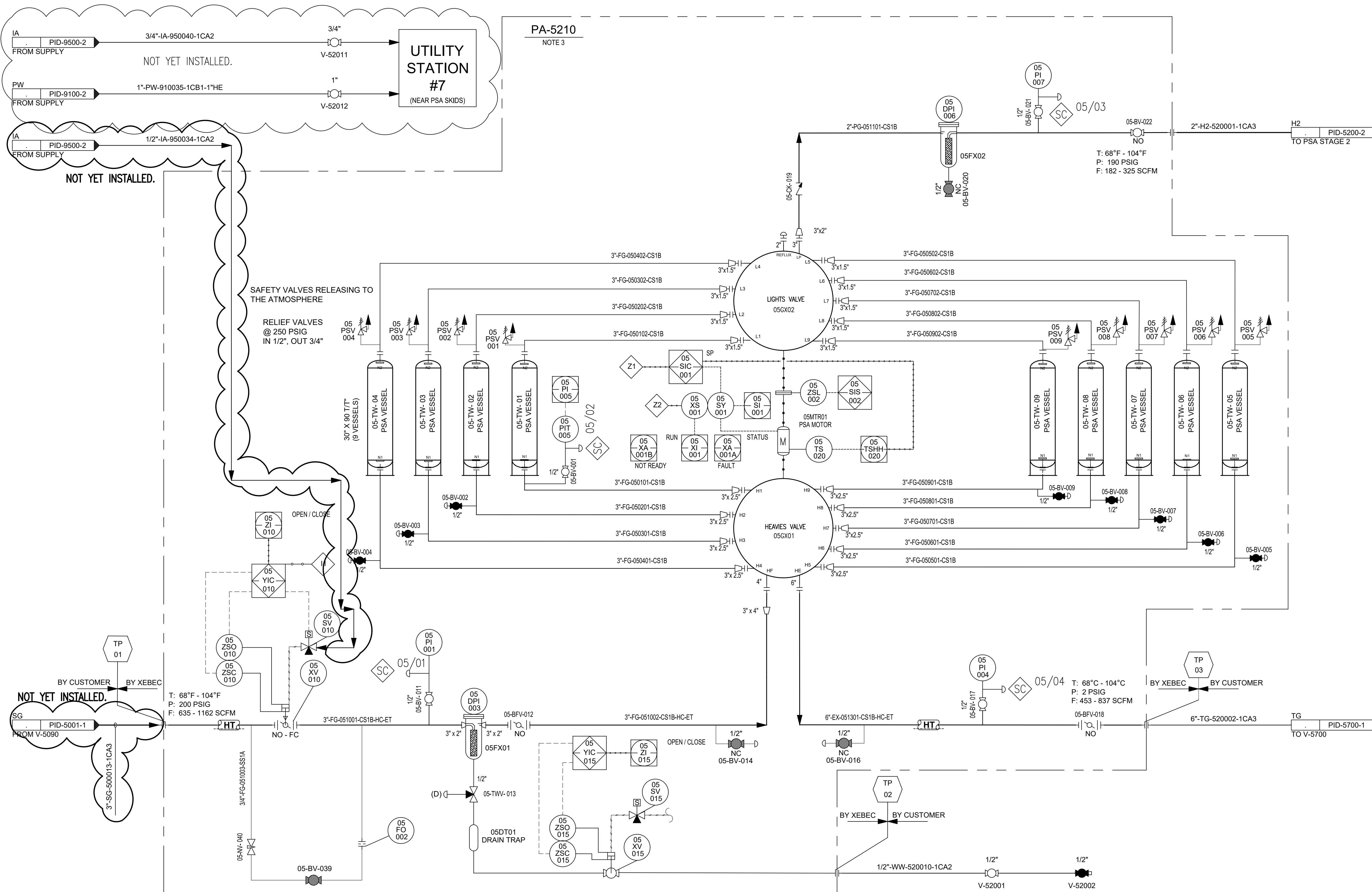
4/20/2023

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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0001-2 FOR LEGEND.
2. REFER TO XEBEC DOCUMENT 33090-05-PR-PID-001 FOR ORIGINAL EQUIPMENT PACKAGED P&ID AND 33090-05-PR-PID-000 FOR THE LEGEND.
3. PACKAGES PA-5210 AND PA-5220 ARE CONTROLLED BY AN INDEPENDENT PLC.



DRAWN: P SALAS	
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APVD:	
DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN
C	REVISED AND REDRAWN
D	REVISED AND REDRAWN

3041
P&ID
PRESSURE SWING ADSORPTION

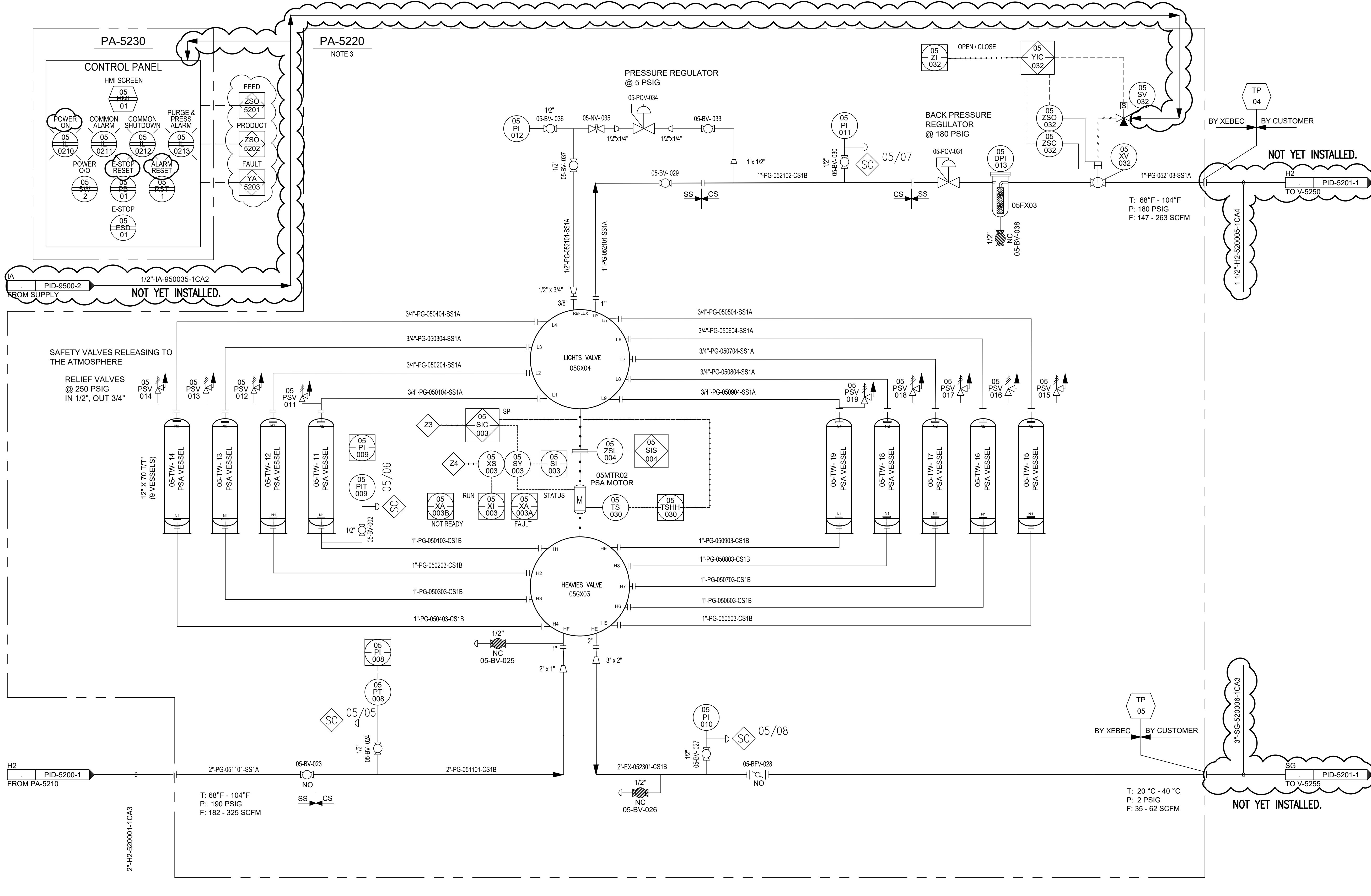
SCALE	NONE	SHEET NO	1	SHEETS	1
DWG NO	3041-PID-5200-1			REV	D
CADFILE	3041-PID-5200-1.DWG				4/20/2021

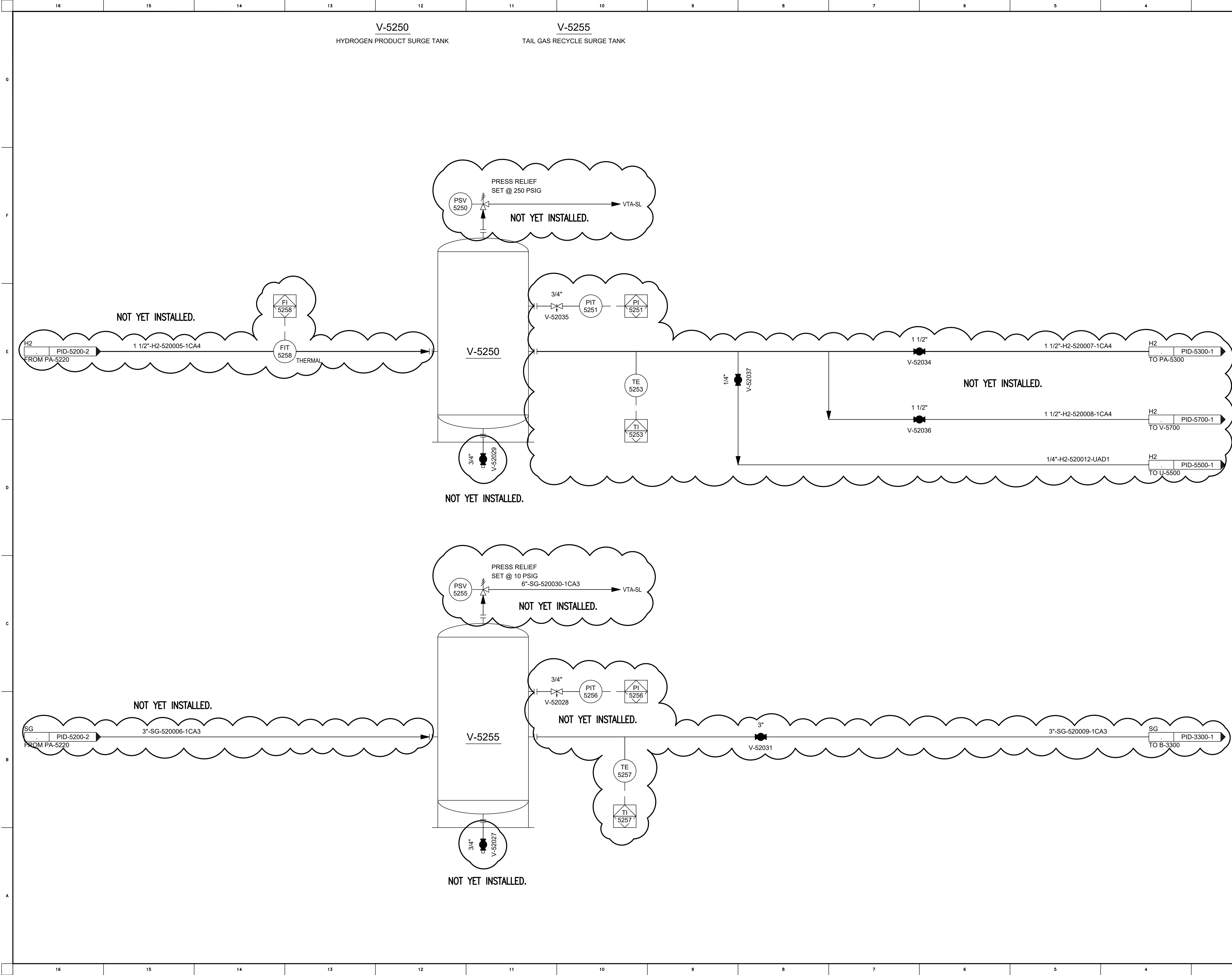
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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0001-2 FOR LEGEND.
2. REFER TO XEBEC DOCUMENT 33090-05-PR-PID-001 FOR ORIGINAL PACKAGED EQUIPMENT P&ID AND 33090-05-PR-PID-000 FOR THE LEGEND.
3. PACKAGES PA-5210 AND PA-5220 ARE CONTROLLED BY AN INDEPENDENT PLC.





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- NOTES:
- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 - ALL INSTRUMENT AIR LINES TO BE RUN IN UAD1.

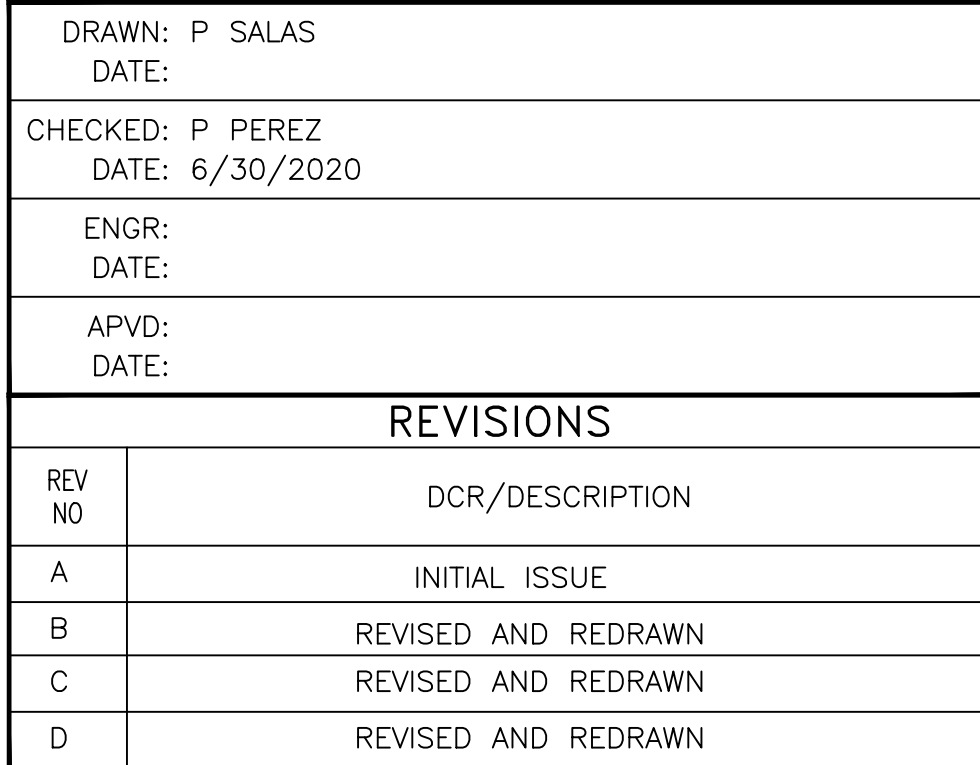
DRAWN: A BENNETT DATE:	
CHECKED: P PEREZ DATE: 6/30/2020	
ENGR: DATE:	
APVD: DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID TAIL GAS SURGE TANK	
SCALE	NONE
DWG NO	3041-PID-5201-1
CADFILE	3041-PID-5201-1.DWG
SHEET NO	1
SHEETS	1
REV	A
4/20/2023	

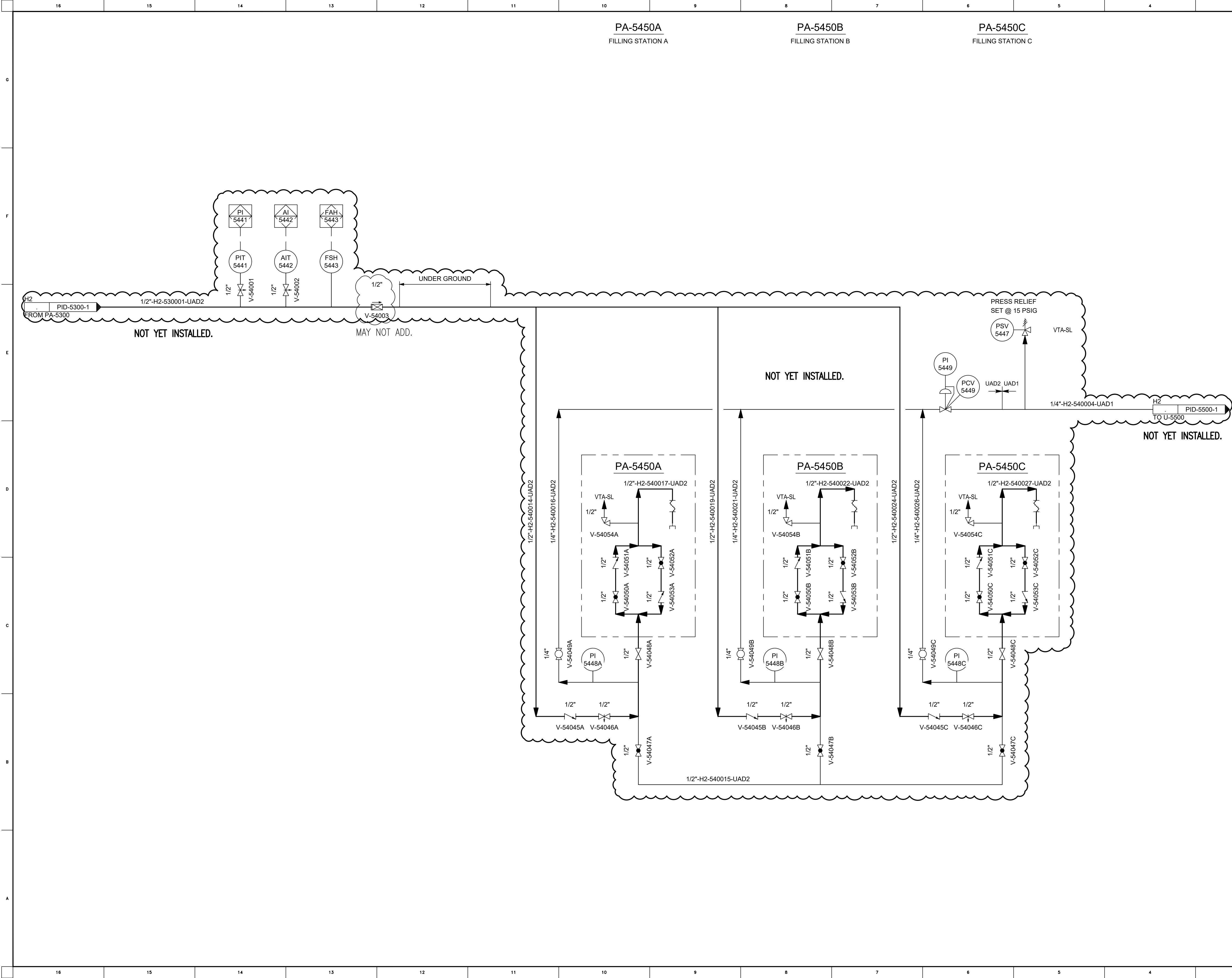
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1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0001-2 FOR LEGEND.
2. REFER TO ACS DOCUMENT PID-10177 FOR PACKAGE'S ORIGINAL P&ID.
3. PACKAGE P-5300 IS CONTROLLED BY AN INDEPENDENT PLC.



SCALE	NONE	SHEET NO	1	SHEETS	1
DWG NO	3041-PID-5300-1			REV	D
CADFILE	3041-PID-5300-1.DWG			4/20/2023	



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- NOTES:
- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

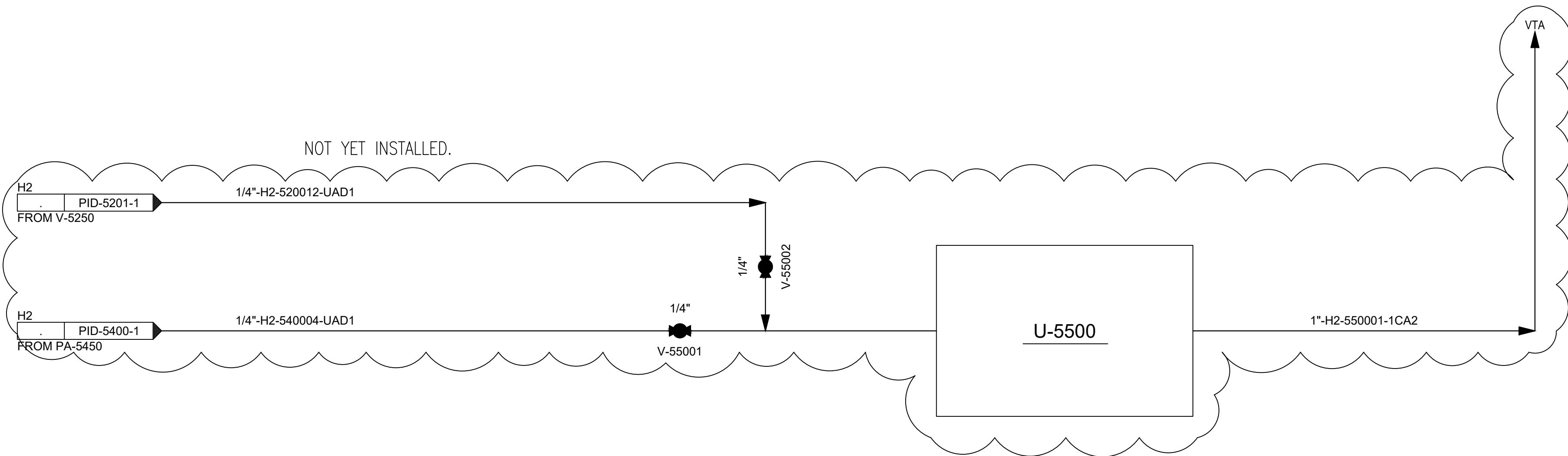
DRAWN: A BENNETT	
DATE:	
CHECKED: P PEREZ	
DATE: 6/30/2020	
ENGR:	
DATE:	
APVD:	
DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN

3041	
P&ID	
H2 TANK FILLING STATIONS	
SCALE	NONE
DWG NO	3041-PID-5400-1
CADFILE	3041-PID-5400-1.DWG
SHEET NO	1
SHEETS	1
REV	B
4/20/2023	

NOT YET INSTALLED.

Diagram of an Atmospheric Sampling Station:

- PT 5565 (Pressure Transducer) connected to PI 5565 (Pressure Indicator)
- TT 5566 (Temperature Transducer) connected to TI 5566 (Temperature Indicator)
- H₂O AT 5567 (Humidity Transducer) connected to AI 5567 (Humidity Indicator)



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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT
DATE:

CHECKED:
DATE:

ENGR:
DATE:

APVD:
DATE:

REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN
C	REVISED AND REDRAWN

3041
P&ID
H2 ANALYZER

SCALE	NONE	SHEET NO	1	SHEETS	1
DWG NO	3041-PID-5500-1	REV	A		
CADFILE	3041-PID-5500-1.DWG	4/20/2023			

The diagram shows three components connected in a sequence. The first component is a square with a triangle on top containing the text 'TAH'. Below the triangle, the number '5705' is written. The second component is a square with a triangle on top containing the text 'TI'. Below the triangle, the number '5705' is written. The third component is a circle with the text 'TT' inside, and the number '5705' written below it. Lines connect the components in a sequence from left to right.

The diagram shows three components connected in a sequence. The first component is a square with a triangle on top containing the text 'TAH'. Below the triangle, the number '5705' is written. The second component is a square with a triangle on top containing the text 'TI'. Below the triangle, the number '5705' is written. The third component is a circle with the text 'TT' inside, and the number '5705' written below it. Lines connect the components in a sequence from left to right.

TE
5705

TE
5705

8"-TG-570001-1CA2

TG

PID-5701-1
TO PA-5720

PID-5701-1
TO PA-5720

NOT YET INSTALLED?
TWO LINES?

NOT YET INSTALLED.

V-5700

NEEDS TO BE INSTALLED.

V-57005

V-57006

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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT
DATE:

CHECKED: P PEREZ
DATE: 6/30/2020

ENGR:
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APVD:
DATE:

REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN
C	REVISED AND REDRAWN

3041
P&ID
THROX WATER TRAP ARRESTOR

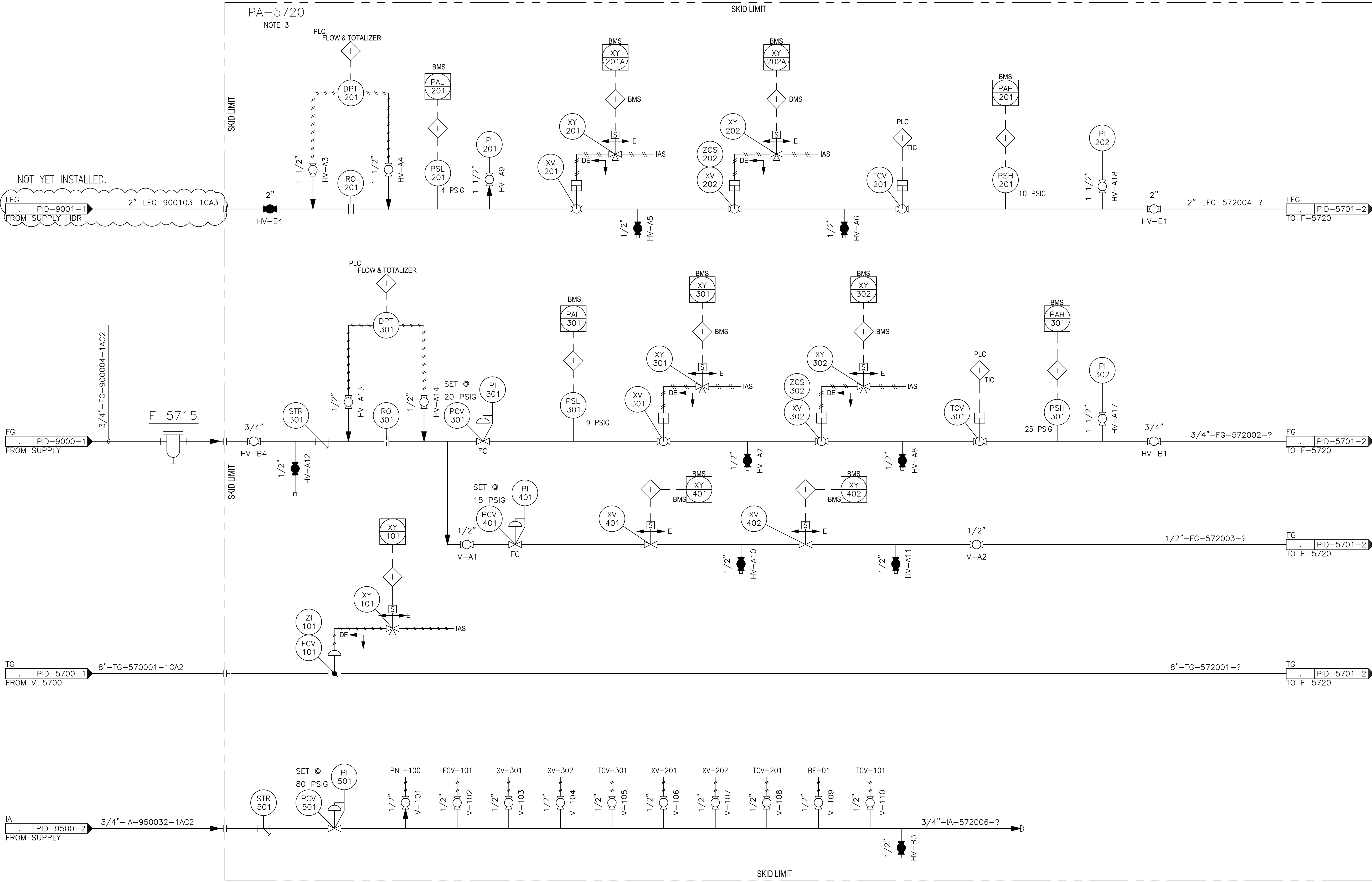
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DWG NO	3041-PID-5700-1	REV	A		
CADFILE	3041-PID-5700-1.DWG	4/20/2021			

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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. REFER TO DOCUMENTS TC-16-10-2003-101-Q01 AND TC-16-10-2003-102-Q01 FROM TULSA COMBUSTION LLC FOR ORIGINAL P&IDS FOR PACKAGE.
3. PACKAGE PA-5720 IS CONTROLLED BY AN INDEPENDENT PLC.



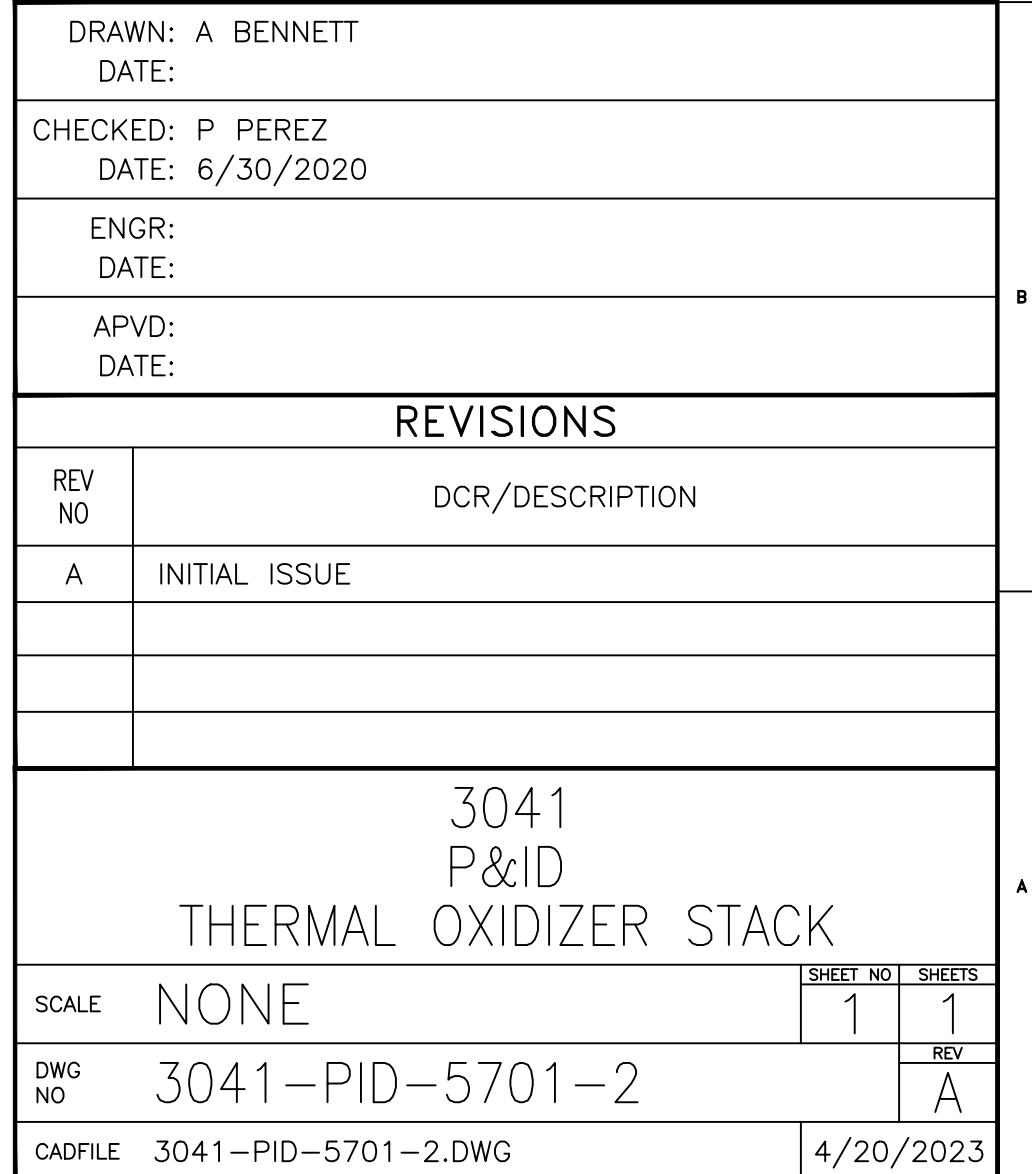
DRAWN: A BENNETT DATE:	
CHECKED: P PEREZ DATE: 6/30/2020	
ENGR: DATE:	
APVD: DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN
C	REVISED AND REDRAWN

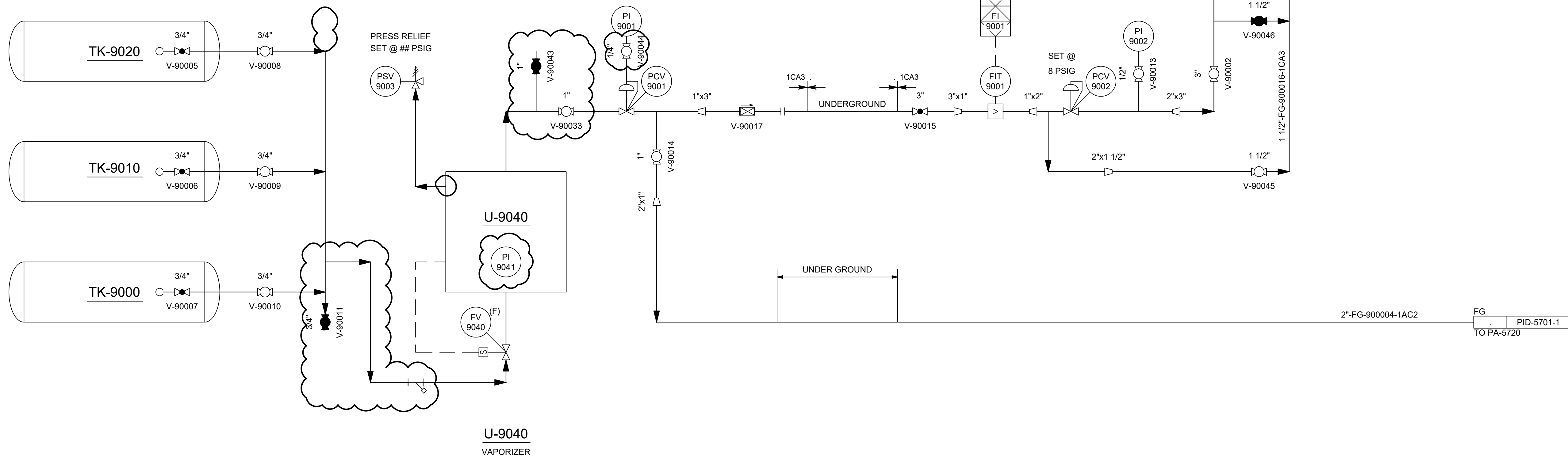
3041 P&ID THERMAL OXIDIZER HEADERS	
SCALE	NONE
DWG NO	3041-PID-5701-1
CADFILE	3041-PID-5701-1.DWG

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1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. REFER TO DOCUMENTS TC-16-10-2003-101-Q01 AND TC-16-10-2003-102-Q01 FROM TULSA COMBUSTION LLC FOR ORIGINAL P&IDS FOR PACKAGE.
3. PACKAGE PA-5720 IS CONTROLLED BY AN INDEPENDENT PLC.





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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT
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CHECKED: P PEREZ
DATE: 7/6/2020

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REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN

3041
P&ID
FUEL GAS DISTRIBUTION

SCALE	NONE	SHEET NO	1	SHEET	1
DWG NO	3041-PID-9000-1	REV	B		
CADFILE	3041-PID-9000-1.DWG	4/20/2021			

F-9075
CONDENSATE FILTER

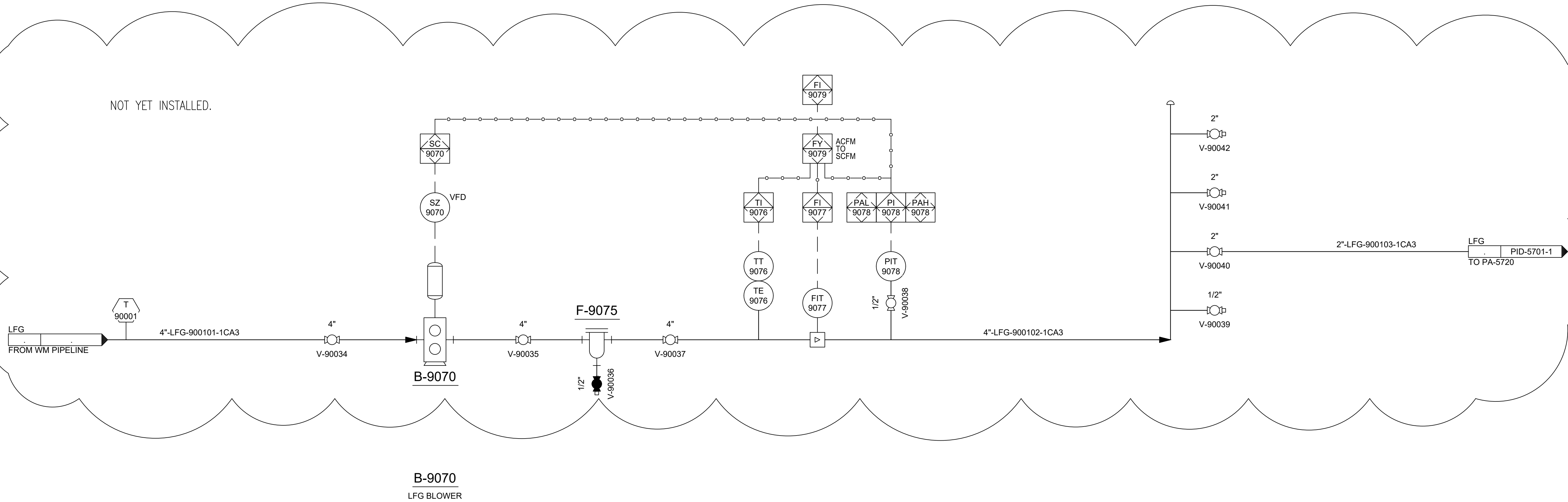
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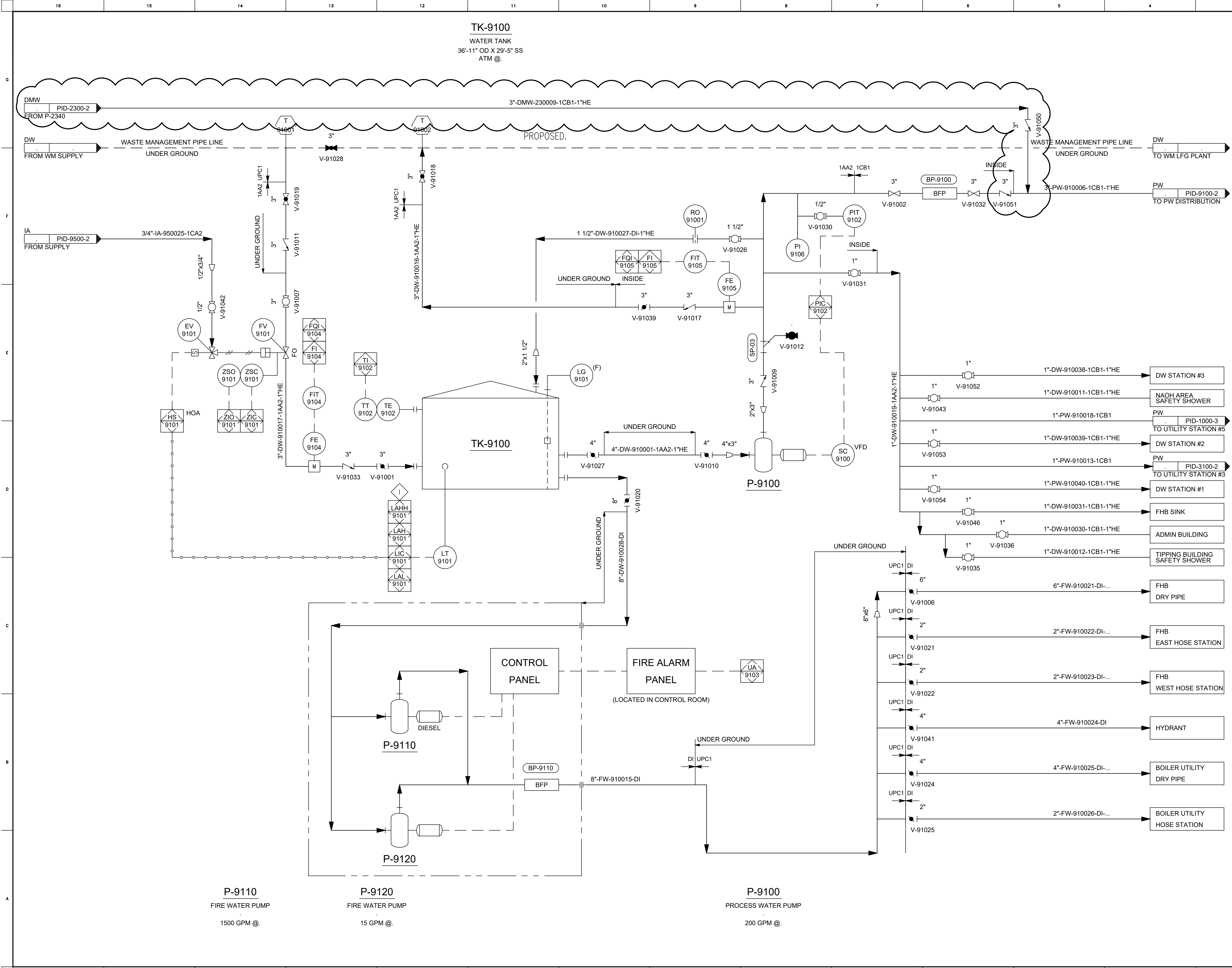
NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.



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REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID LFG SUPPLY	
SCALE	NONE
DWG NO	3041-PID-9001-1
CADFILE	3041-PID-9001-1.DWG
SHEET NO	1
SHEETS	1
REV	A
DATE	4/20/2023



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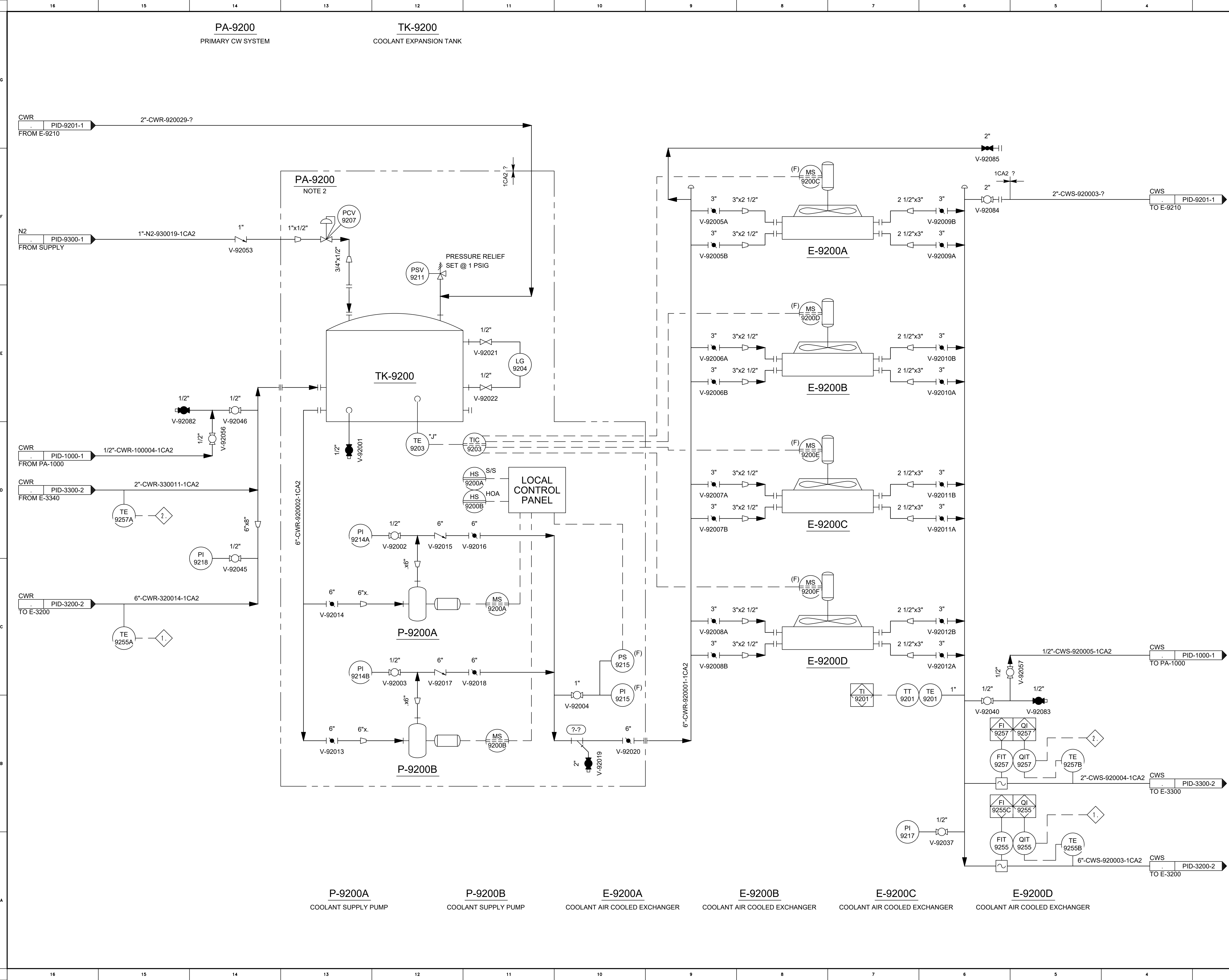
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NOTES:
1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT DATE:	
CHECKED: P PEREZ DATE:	
ENGR: DATE:	
APVD: DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID PROCESS WATER SYSTEM	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-9100-1	REV A
CADFILE 3041-PID-9100-1.DWG	3/21/2024



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- NOTES:
- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 - PA-9200 COOLING WATER SYSTEM USES GLYCOL WATER MIXTURE THAT IS TO BE FILLED MANUALLY THRU A HIGH POINT VENT.

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REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041	
P&ID	
COOLING WATER SYSTEM	
SCALE	NONE
DWG NO	3041-PID-9200-1
CADFILE	3041-PID-9200-1.DWG
SHEET NO	1
SHEETS	1
REV	A

PA-9210
PEM CW SYSTEM

TK-9210
COOLING WATER EXPANSION TANK
.X.
.@.

P-9220
PEM BACK-UP CW PUMP
20 GPM @.

E-9210
PLATE-TYPE HEAT EXCHANGER
.@.

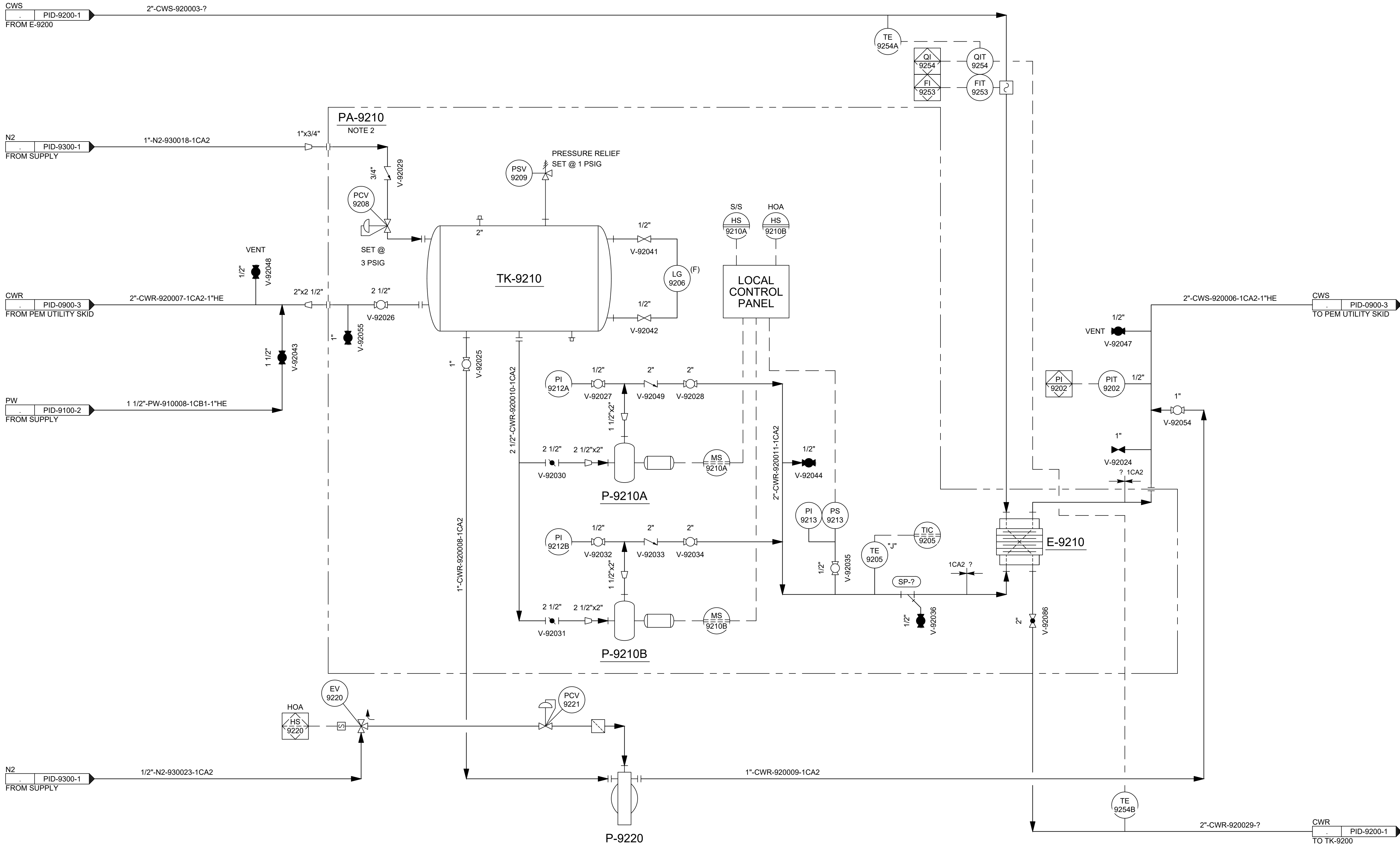
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NOTES:

- REFER TO SHEETS 3041-PID-0001 & 3041-PID-0002 FOR LEGEND.
- PA-9210 COOLING WATER SYSTEM USES HIGH PURITY WATER. FILL SYSTEM USING RO WATER MANUALLY FROM HIGH POINT VENT.



DRAWN: A BENNETT
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DATE: 7/6/2020

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APVD:
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REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041
P&ID
PEM COOLING WATER SYSTEM

SCALE NONE

DWG NO 3041-PID-9201-1

CADFILE 3041-PID-9201-1.DWG

SHEET NO	SHEETS
1	1
REV	A

3/21/2024

PA-9220
SECONDARY CW SYSTEM

TK-9230
COOLANT EXPANSION TANK

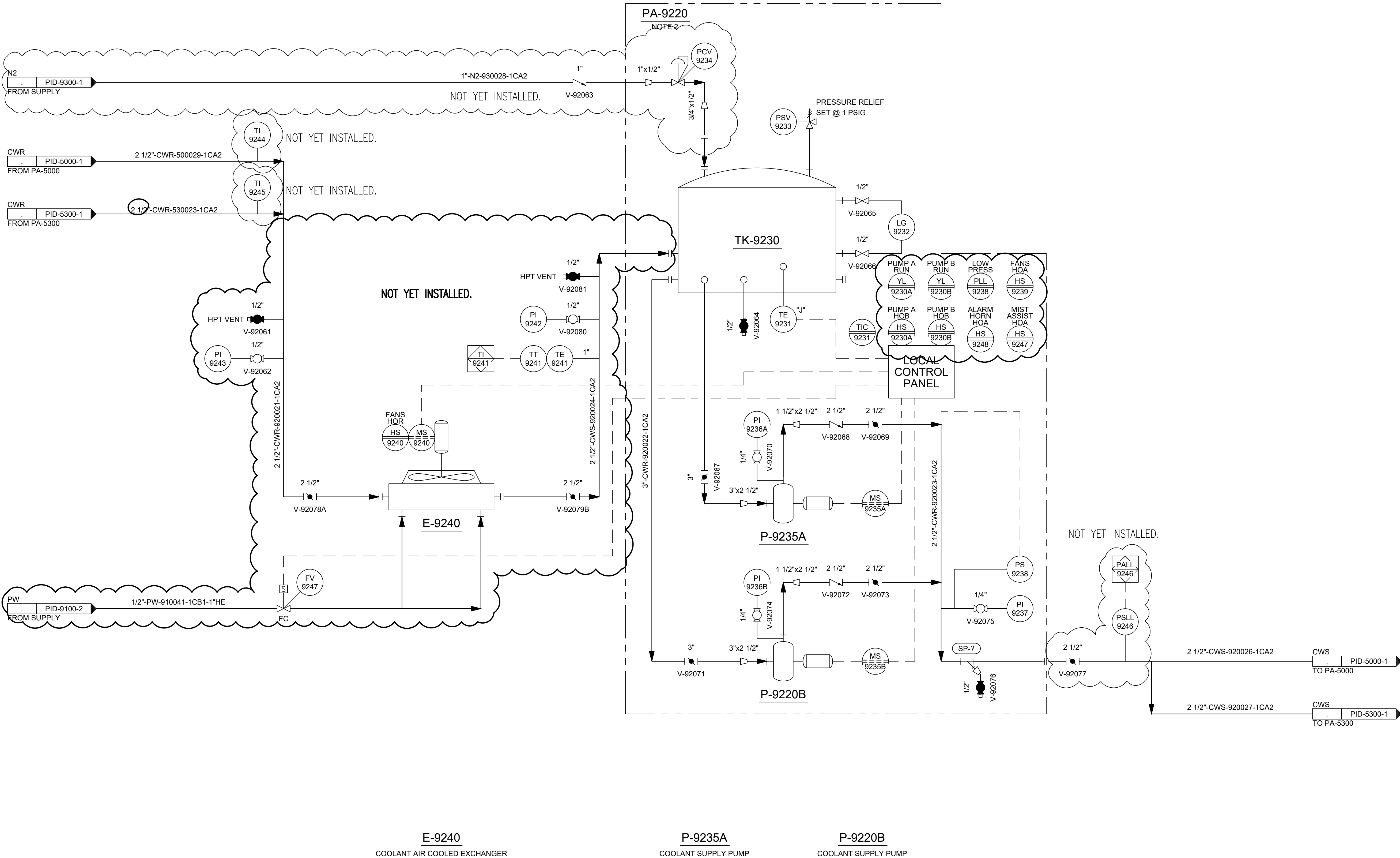
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NOTES:

- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
- PA-9220 COOLING WATER SYSTEM USES GLYCOL WATER MIXTURE THAT IS ADDED MANUALLY THRU A HIGH POINT VENT.



DRAWN: A BENNETT DATE:	
CHECKED: P PEREZ DATE: 7/6/2020	
ENGR: DATE:	
APVD: DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID H2 COOLING WATER SYSTEM	
SCALE	NONE
DWG NO	3041-PID-9202-1
CADFILE	3041-PID-9202-1.DWG
SHEET NO	1
SHEETS	1
REV	A
DATE	4/20/2023

PA-9300
NITROGEN SUPPLY SYSTEM

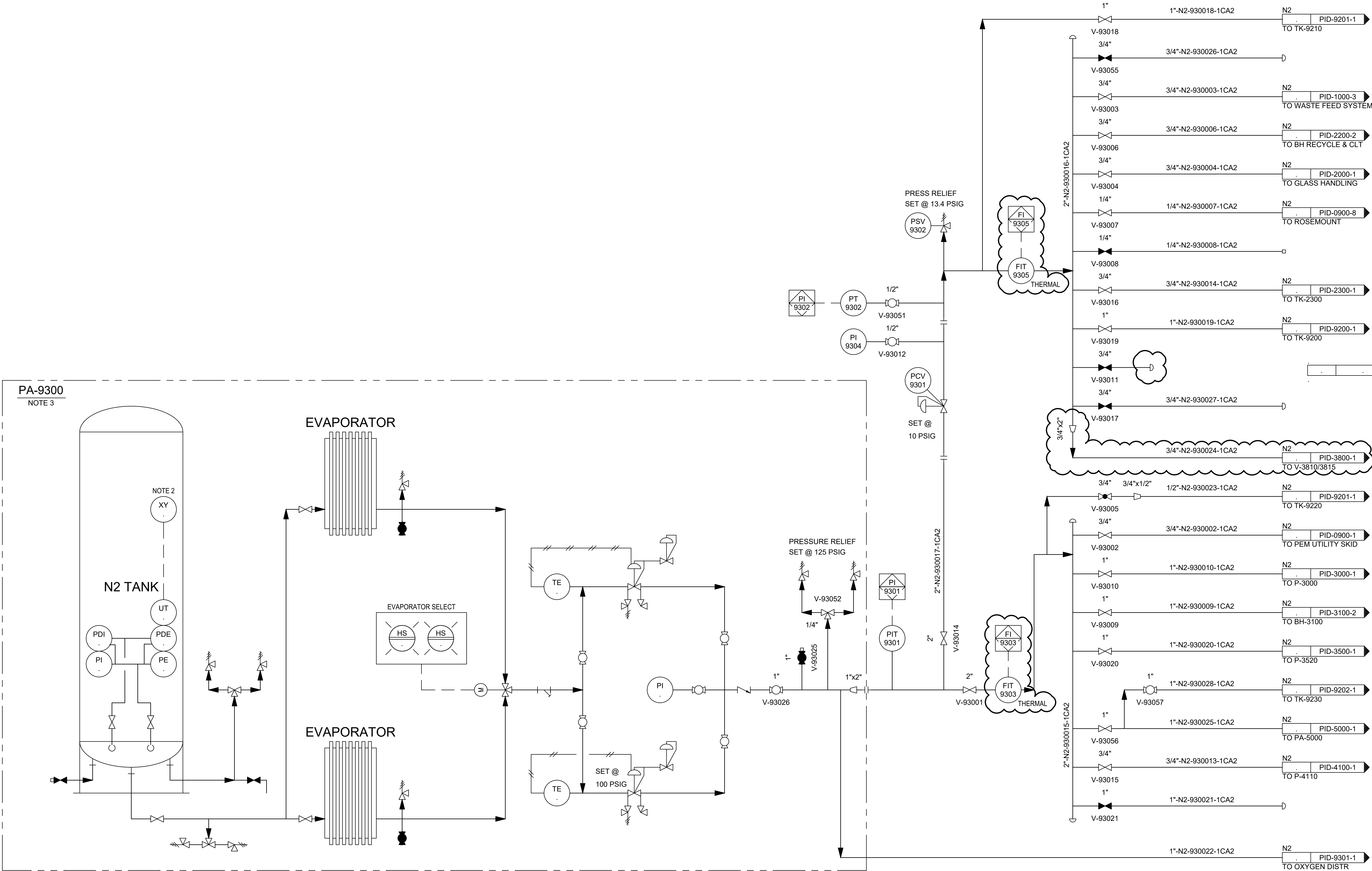
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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. ANTENNA TO TRANSMIT DATA TO NITROGEN VENDOR.
3. PA-9300 IS SUPPLIED AND OWNED BY NITROGEN VENDOR.



DRAWN: A BENNETT	
DATE:	
CHECKED: P PEREZ	
DATE: 7/6/2020	
ENGR:	
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APVD:	
DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

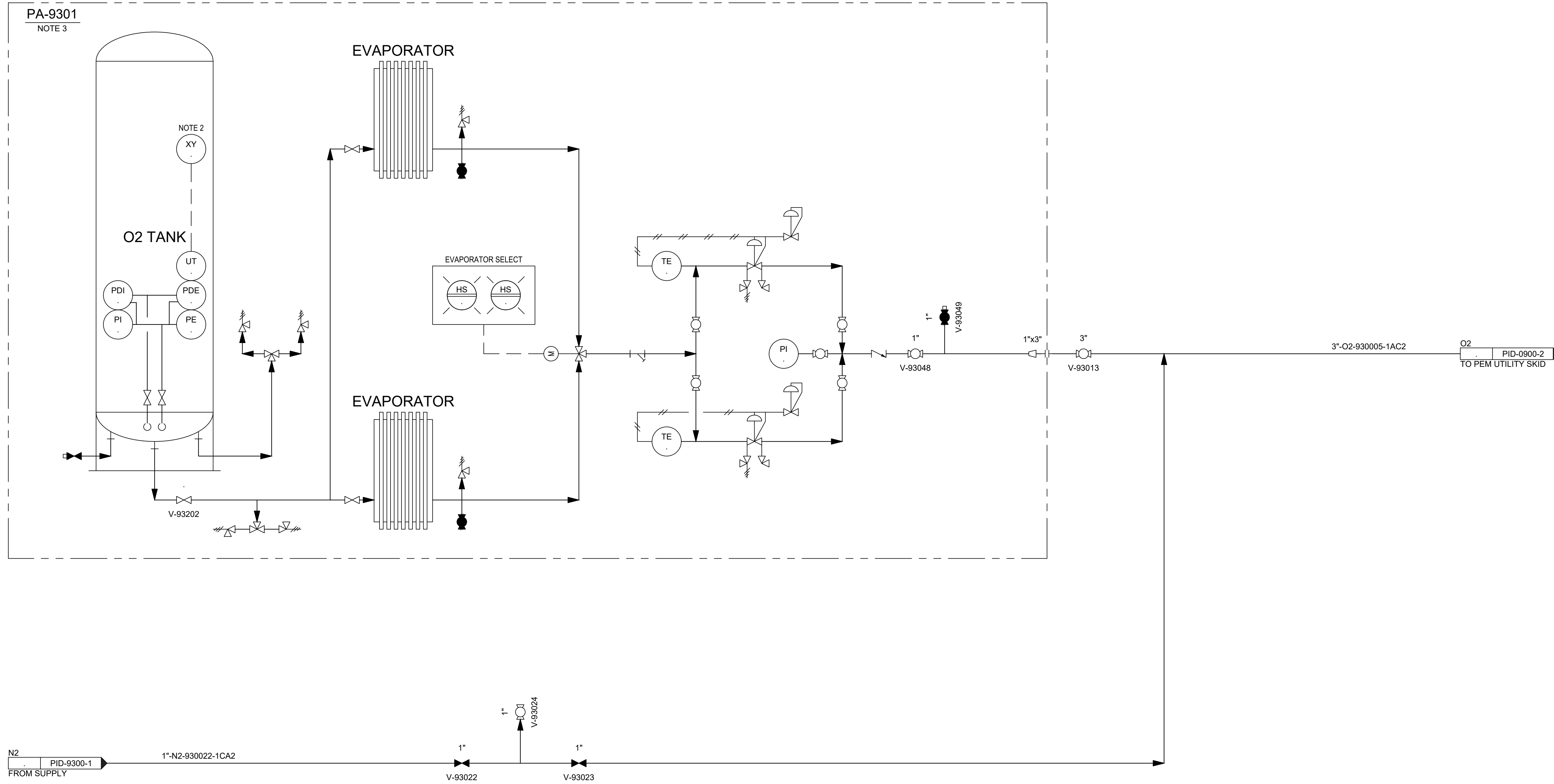
3041 P&ID NITROGEN DISTRIBUTION	
SCALE	NONE
DWG NO	3041-PID-9300-1
CADFILE	3041-PID-9300-1.DWG
SHEET NO	1
SHEETS	1
REV	A
DATE	9/27/2023

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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. ANTENNA TO TRANSMIT DATA TO OXYGEN VENDOR.
3. PA-9301 IS SUPPLIED AND OWNED BY OXYGEN VENDOR.



DRAWN: A BENNETT
DATE:

CHECKED: P PEREZ
DATE: 7/6/2020

ENGR:
DATE:

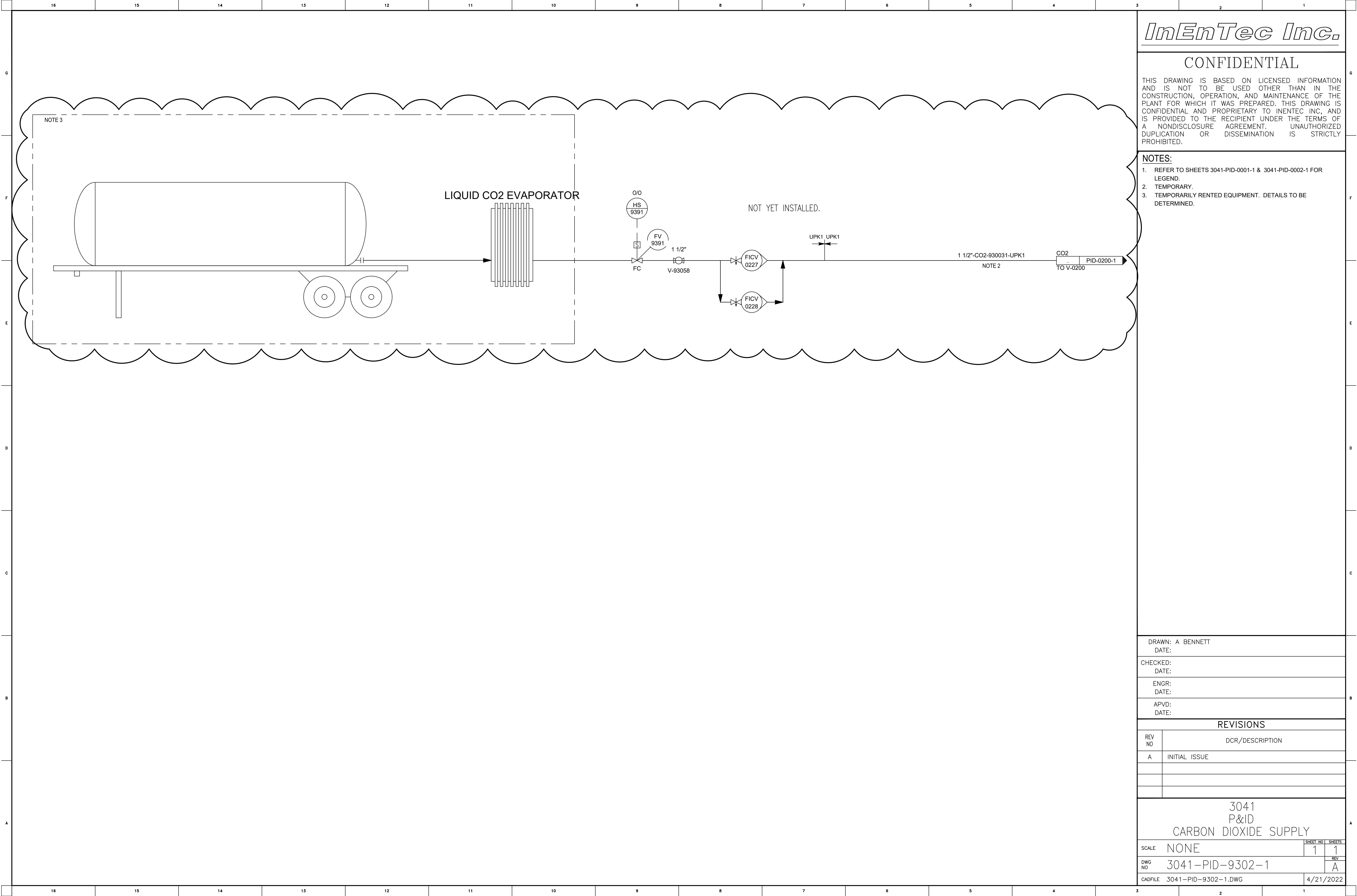
APVD:
DATE:

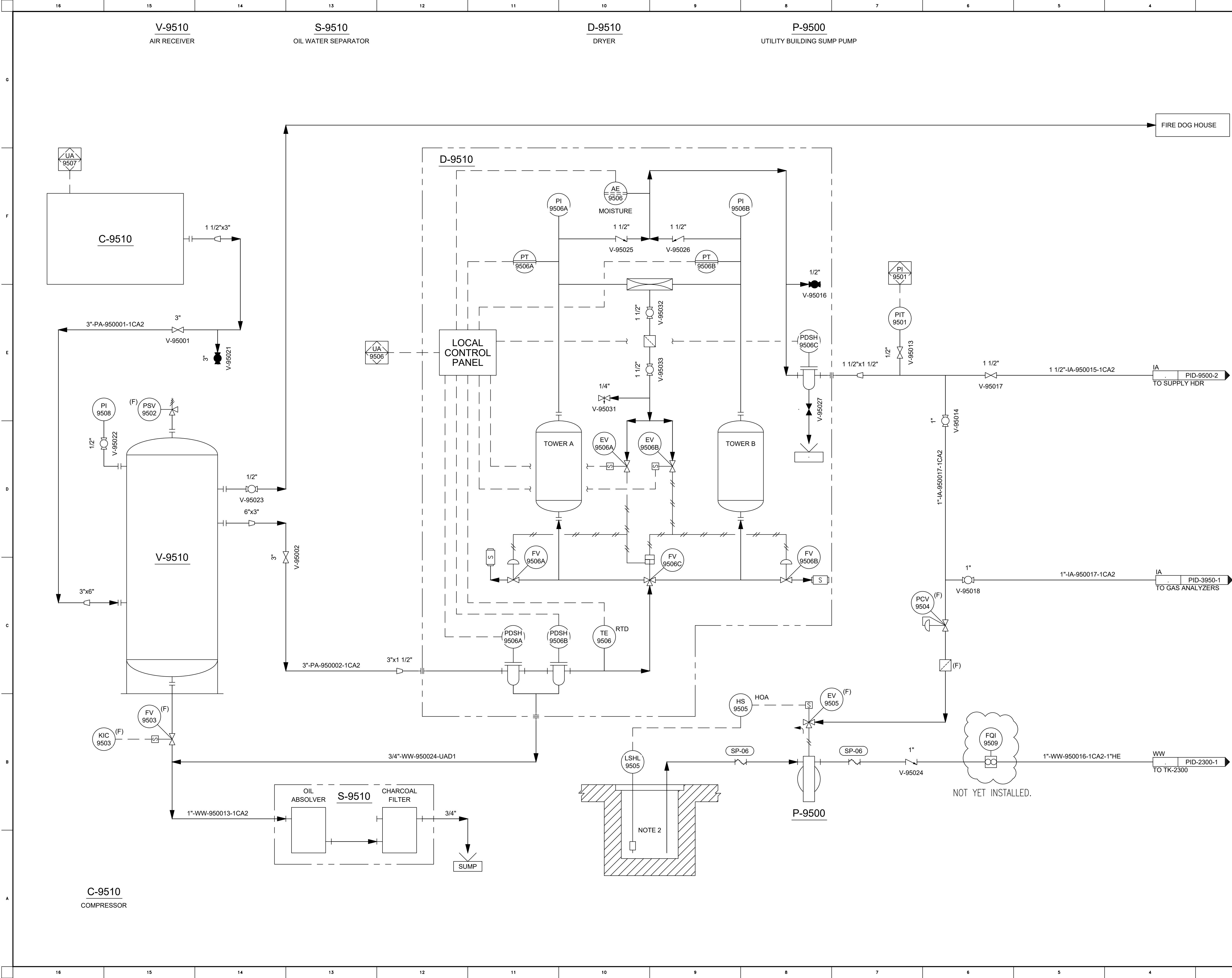
REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041
P&ID
OXYGEN DISTRIBUTION

SCALE	NONE	SHEET NO	1	SHEET	1
DWG NO	3041-PID-9301-1	REV	A		
CADFILE	3041-PID-9301-1.DWG	9/27/2021			





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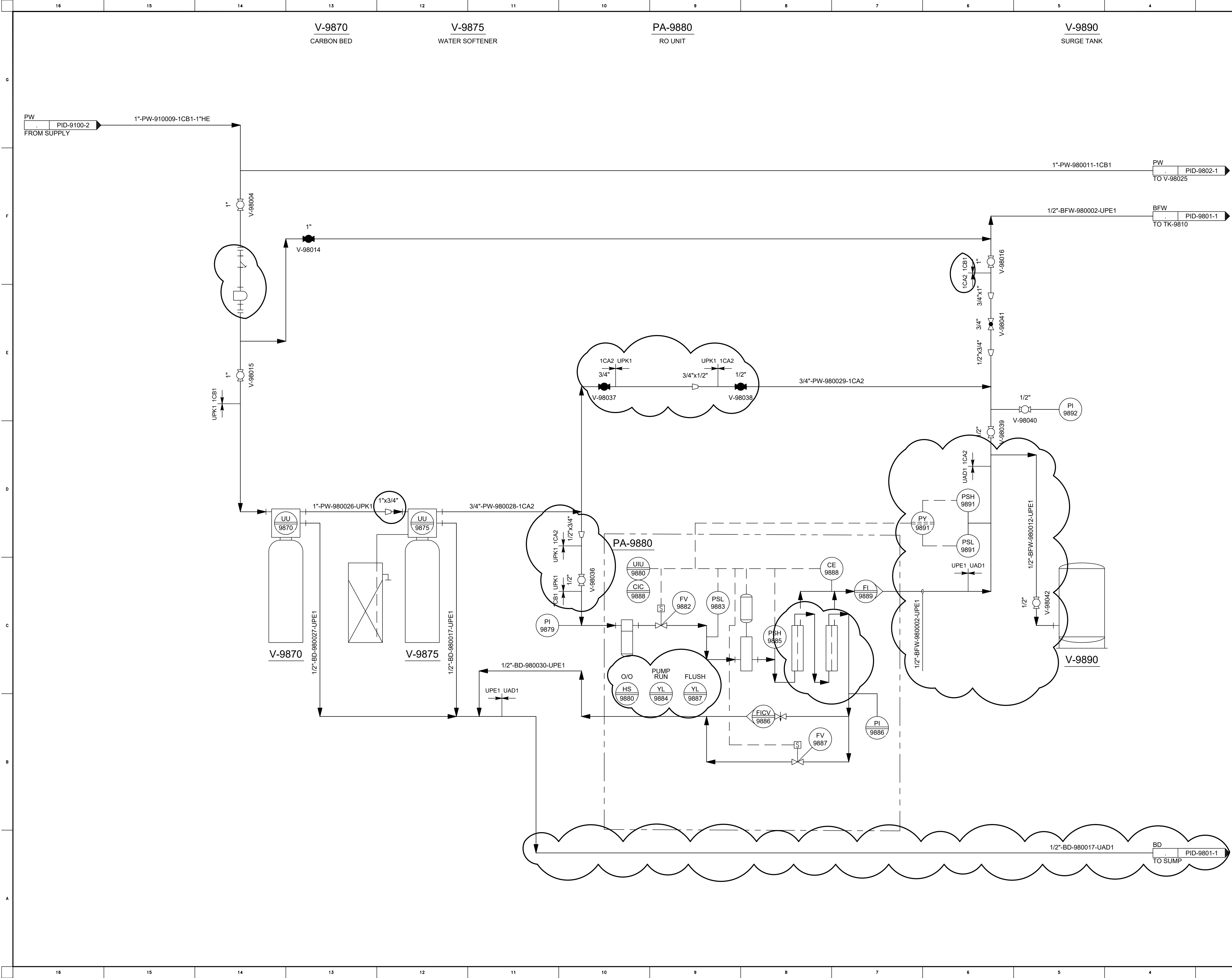
CONFIDENTIAL

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- NOTES:
- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 - BOILER ROOM SUMP.

DRAWN: A BENNETT	
DATE:	
CHECKED: P PEREZ	
DATE: 7/6/2020	
ENGR:	
DATE:	
APVD:	
DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID INSTRUMENT AIR SYSTEM	
SCALE	NONE
DWG NO	3041-PID-9500-1
CADFILE	3041-PID-9500-1.DWG
SHEET NO	1
SHEETS	1
REV	A
4/20/2023	



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- NOTES:
- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 - SUMP DRAINS TO BOILER ROOM SUMP SHOWN ON 3041-PID-9500-1.

DRAWN: A BENNETT
DATE:
CHECKED: P PEREZ
DATE: 7/6/2020
ENGR:
DATE:
APVD:
DATE:

REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID BFW SYSTEM		
SCALE	NONE	SHEET NO 1 SHEETS 1
DWG NO	3041-PID-9800-1	REV A
CADFILE	3041-PID-9800-1.DWG	3/20/2024

TK-9810
BOILER FEED WATER TANK

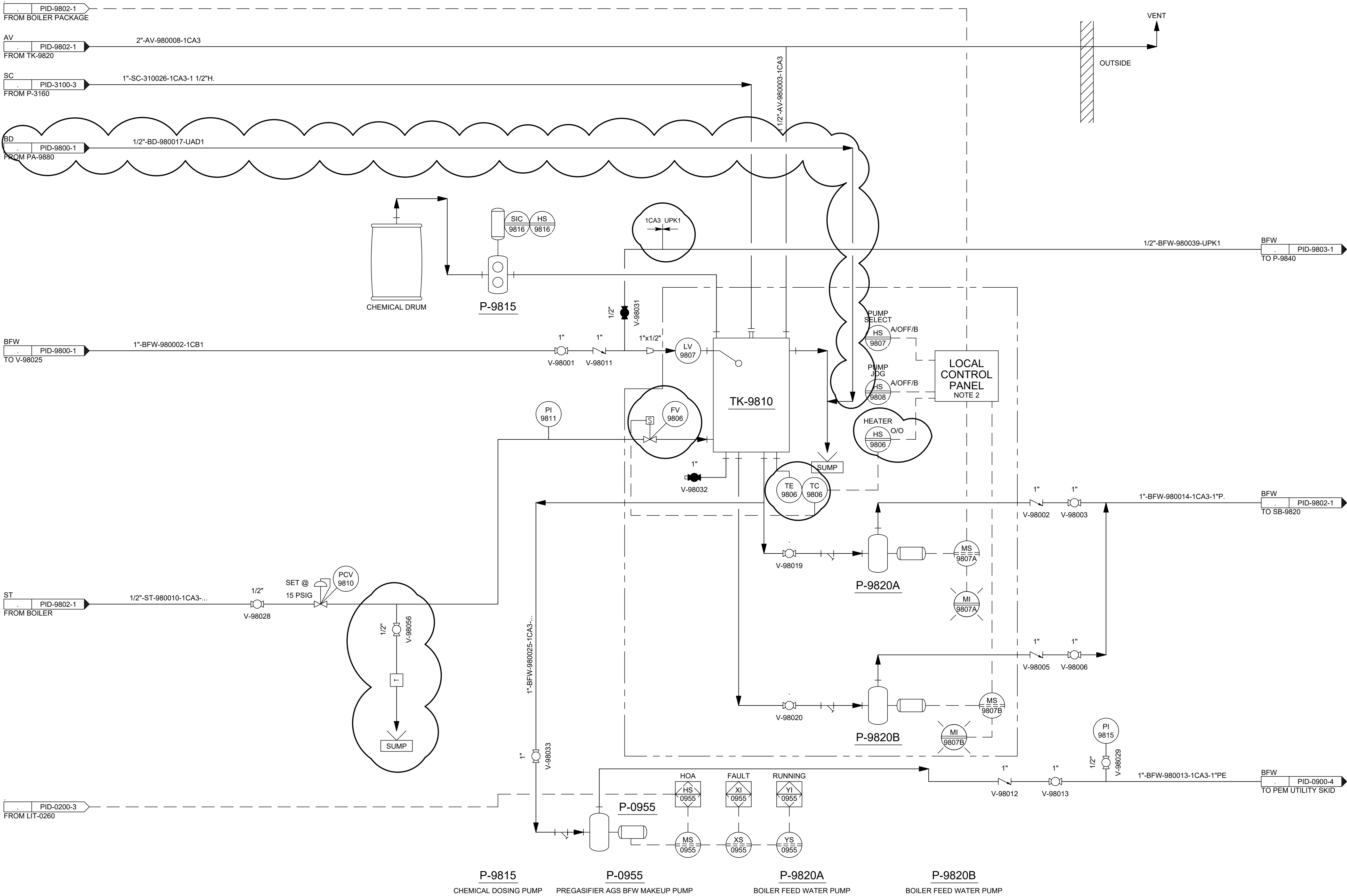
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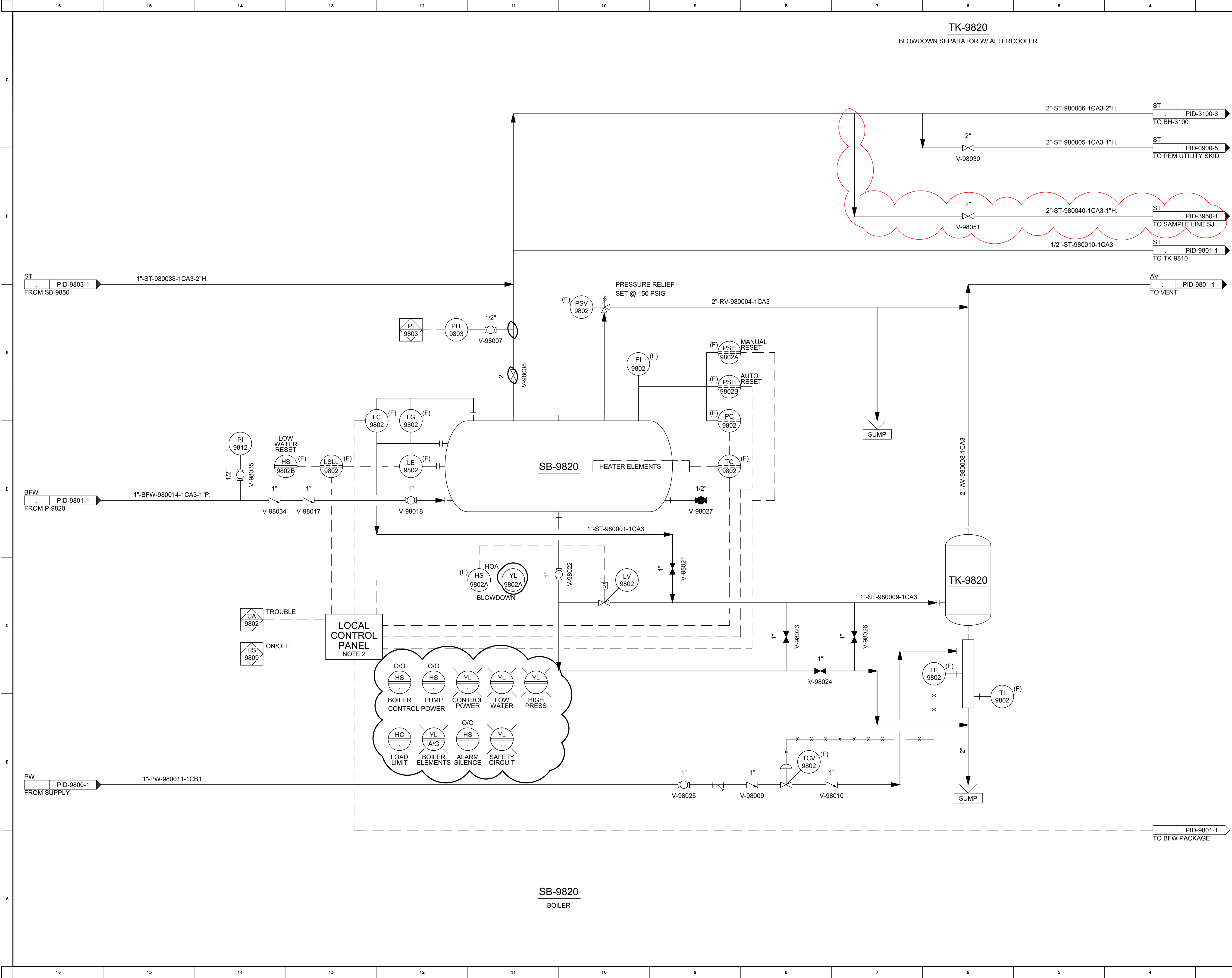
NOTES:

- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
- FOR CONTROL SYSTEM DIAGRAM SEE LATTNER DWG. WD-10-4998-J(2 OF 2).
- SUMP DRAINS TO BOILER ROOM SUMP SHOWN ON 3041-PID-9500-1.



DRAWN: A BENNETT DATE:	
CHECKED: P PEREZ DATE: 7/6/2020	
ENGR: DATE:	
APVD: DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID BFW PUMPS	
SCALE	NONE
DWG NO	3041-PID-9801-1
CADFILE	3041-PID-9801-1.DWG
SHEET NO	1
SHEETS	1
REV	A
3/19/2024	



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- NOTES:
- 1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 - 2. SEVEN STEP SEQUENCER CONTROLS 7 BANKS OF 30K HEATERS. SEE LATTNER DWG. WD-10-4988-J(1 OF 2).
 - 3. SUMP DRAINS TO BOILER ROOM SUMP SHOWN ON 3041-PID-9500-1.

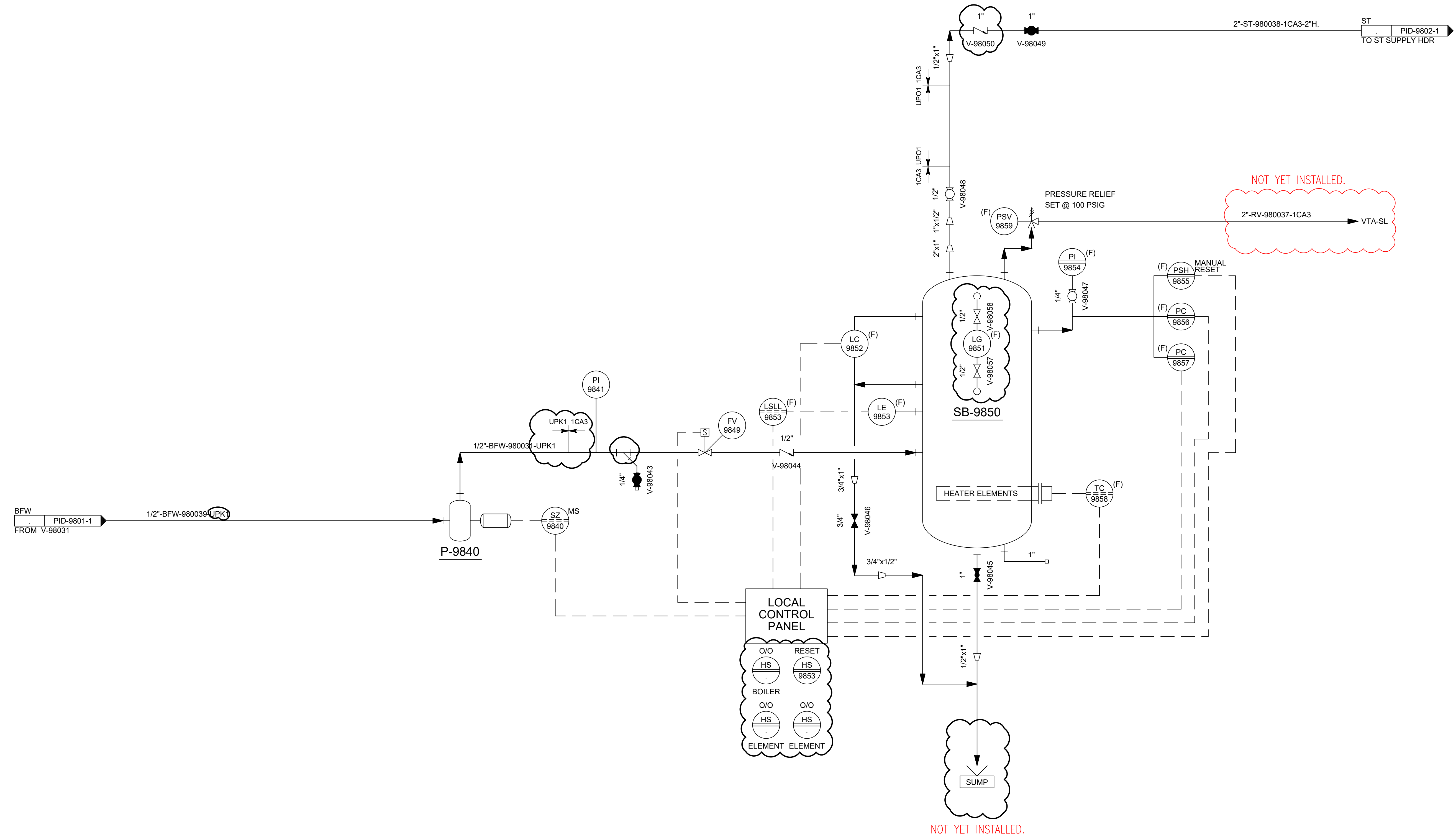
DRAWN: A BENNETT DATE:	
CHECKED: P PEREZ DATE: 7/6/2020	
ENGR: DATE:	
APVD: DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID STEAM SYSTEM	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-9802-1	REV A
CADFILE 3041-PID-9802-1.DWG	4/20/2023

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- NOTES:
- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 - SUMP DRAINS TO BOILER ROOM SUMP SHOWN ON 3041-PID-9500-1.



P-9840
HORIZONTAL CENTRIFUGAL PUMP

SB-9850
SECONDARY BOILER

DRAWN: A BENNETT DATE:	
CHECKED: P PEREZ DATE: 7/6/2020	
ENGR: DATE:	
APVD: DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID STEAM SYSTEM 2	
SCALE	NONE
DWG NO	3041-PID-9803-1
CADFILE	3041-PID-9803-1.DWG
SHEET NO	1
SHEETS	1
REV	A
3/19/2024	